

CA FINAL · PAPER 6

Advanced Financial Management **Formula Book**

~ every chapter, every formula, built up step-by-step ~

14 Chapters

Every Derivation

Print & Revise

Made for: **Lokeshwaran** — a one-copy, handwritten-style formula companion for quick last-mile revision.

Chapters follow ICAI's current Advanced Financial Management structure (Modules 1-3, 14 chapters) as best known to me — live syllabus look-up wasn't reachable while building this, so please do a quick one-time check against your own study material's index and flag anything that doesn't match.

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Frequently used



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Financial Policy & Corporate Strategy

How a firm chooses its growth direction, how financial policy must fund and support that choice, and the yardsticks used to check whether the strategy is actually creating value or heading toward distress.

Strategic Planning vs Financial Planning and Policy

Strategic planning is the top-management process of deciding where the firm should compete and how it will win there — choosing which businesses, products and markets to be in, and building a sustainable competitive advantage around that choice; it typically looks three to five years or more ahead and is shaped by the firm's mission, its environment, and its competitors' moves. Financial planning and policy sits one level below this: once strategy has decided what the firm wants to do, financial policy decides how it will be paid for — the mix of debt, equity and retained earnings used to raise funds, the dividend policy that governs how much profit is ploughed back versus paid out, the capital budgeting rules used to screen individual investments, and the working-capital and risk policies that keep the firm solvent while it executes. In short, strategic planning asks "what should we do and why," while financial planning asks "can we afford it, and what is the cheapest, safest way to fund it" — strategy sets the destination, and financial policy supplies and constrains the route to get there. A brilliant strategy can still fail in practice if financial policy is not aligned with it — for instance, if the funds raised are too expensive, too short-term in tenor, or simply too small for what the strategy actually demands.

Ansoff's Product-Market Growth Matrix

Ansoff's matrix is a simple 2×2 grid that maps a firm's growth options along two axes — whether the product being sold is existing or new, and whether the market being sold into is existing or new — giving four growth strategies of steadily rising risk. Market penetration (existing product, existing market) is the safest cell: selling more of what the firm already makes to the customers it already serves, through higher usage, better pricing, or winning share from competitors. Product development (new product, existing market) launches new or improved products at the firm's current, already-understood customer base, while market development (existing product, new market) takes the firm's existing, proven products into new geographies, segments or customer groups it has not served before. Diversification (new product, new market) is the riskiest cell of all — selling products the firm has never made before to customers it has never served before — and is usually only justified once it is closely related to the firm's existing capabilities, or after the other three options have been largely exhausted. The matrix's core lesson is that risk rises steadily as a firm moves away from what it already knows — its current products and current customers.

BCG Growth-Share Matrix (Stars, Cash Cows, Question Marks, Dogs)

The BCG matrix is a portfolio-planning tool for a firm running several products or business units at once – it plots each one on two axes, the growth rate of its market (high or low) and its relative market share within that market (high or low), sorting every business into one of four cells. Stars sit in high-growth, high-share markets – current or future leaders that generate substantial revenue but also demand heavy continued investment to defend that lead as the market keeps expanding. Cash cows are high-share businesses in low-growth, mature markets – since the market has stopped expanding, they need little further investment, so the surplus cash they throw off is used to fund the Stars and the more promising Question Marks elsewhere in the portfolio. Question marks (also called problem children) hold low share in high-growth markets – they consume heavy cash to try and build share, forcing a choice between investing further to turn them into Stars or divesting before they slide into Dogs. Dogs hold low share in low-growth markets, generating little cash and tying up capital for little return, and are usually harvested (milked for whatever cash is left, with minimal reinvestment) or divested outright. Read together, the matrix pushes a firm toward a self-funding portfolio – use Cash Cow money to fund Stars and the best Question Marks, while starving or exiting the Dogs.

How Corporate Strategy and Financial Policy Interact

Corporate strategy and financial policy are not separate conversations – the strategic choices made using tools like Ansoff's matrix or the BCG matrix only work in practice if financial policy is deliberately shaped to support them. Sources of finance matter first: an aggressive push into diversification or market development needs substantial funding, and the choice between fresh debt, fresh equity, or ploughed-back retained earnings decides whether that growth is affordable without over-leveraging the balance sheet or diluting existing owners' control. Dividend policy is the second lever – a firm chasing high growth typically needs to retain most of its earnings (a low payout, high retention ratio) to self-fund investment and keep its sustainable growth rate high, while a mature, cash-cow-stage business with fewer good reinvestment opportunities can comfortably pay out most of its profit as dividends instead. Risk appetite is the third link – a firm's willingness to take on financial risk through leverage must be consistent with its willingness to take on strategic risk through new products and markets; staying conservatively geared in normal times builds spare capacity that can be drawn on to fund a bold strategic move later, whereas a firm that is already highly leveraged has far less room to pursue such bets without risking financial distress. Financial policy, in other words, is what turns a strategic ambition into something the firm can actually afford and survive executing.

1.1 Sustainable Growth Rate (SGR)

★★ Frequently Asked

$$g = b \times ROE$$

Where:

- b retention ratio — the fraction of net profit the firm keeps back instead of paying as dividend; $b = 1 - \text{Dividend Payout Ratio}$
- ROE Return on Equity — net profit available to equity shareholders divided by shareholders' equity (net worth)
- g Sustainable Growth Rate — the rate at which the firm can grow its equity base (and, if margins/leverage stay unchanged, its sales and assets too) using only internally generated funds, with no fresh equity issue

Build-up logic:

1. Ignoring any fresh share issue, a firm's equity base grows only through profits it ploughs back — so the rupee amount added to equity each year is simply the retention ratio times net profit: $b \times \text{Net Profit}$.
2. Express this addition as a percentage of the equity it is being added to: growth rate of equity = $(b \times \text{Net Profit}) / \text{Equity}$.
3. But Net Profit divided by Equity is exactly the definition of ROE — substitute it in, and the equity base grows at exactly $b \times ROE$ per year, with no outside financing needed at all.
4. If the firm also holds its profit margin, asset turnover and debt-equity ratio steady — i.e. keeps its whole operating and financial policy unchanged — then sales, assets and equity must all grow at this same rate, which is why $b \times ROE$ is called the firm's "sustainable" growth rate: the fastest it can grow on retained earnings alone, without altering policy or issuing new shares.
5. Because ROE itself splits, by the DuPont identity, into Net Profit Margin $\times$ Asset Turnover $\times$ Equity Multiplier, sustainable growth really has four levers — retain more profit, improve margins, use assets more efficiently, or take on more leverage — and pulling any one of them raises g without needing outside equity.

 *Tip: This simple form is exact when ROE is measured on opening equity. If ROE is measured on closing equity (the more common ratio-analysis convention), the more precise version is $g = \frac{b \times ROE}{1 - b \times ROE}$ — watch which convention a question's data implies. Also remember: if the firm's desired growth exceeds g , it must raise fresh equity, add leverage, cut dividends, or improve ROE — it cannot outgrow g on unchanged policy alone.*

1.2 Economic Value Added (EVA)

★ MUST KNOW

$$EVA = NOPAT - (WACC \times \text{Capital Employed})$$

Where:

NOPAT	Net Operating Profit After Tax – $EBIT \times (1 - \text{tax rate})$; pure operating profit, before financing charges
WACC	weighted average cost of capital – the blended cost of every long-term source of funds (debt and equity), weighted by its share in the capital structure
Capital Employed	total long-term funds invested in the business's operations – equity plus long-term debt, equivalently $\text{Total Assets} - \text{Current Liabilities}$

Build-up logic:

1. Ordinary accounting profit only subtracts explicit costs – interest on debt, tax – and never charges anything for the cost of equity, even though equity investors also demand a return for the risk they bear.
2. So a firm can report a perfectly healthy accounting profit while still destroying shareholder value, if that profit does not even cover what investors could have earned elsewhere for the same risk – EVA exists to catch exactly this.
3. Start from NOPAT, the pure operating profit after tax and before any financing charges, because EVA wants to judge how well the capital employed performed, independent of how that capital happens to be financed.
4. Now charge this NOPAT for the full cost of ALL capital employed – both debt and equity – at the blended WACC rate; this "capital charge," $WACC \times \text{Capital Employed}$, is the minimum return investors require just to compensate them for the money tied up in the business.
5. Whatever NOPAT is left after covering this capital charge is the true economic value created for shareholders over and above what investors demanded – a positive EVA means the firm earned more than its cost of capital; a negative EVA means it was accounting-profitable yet still destroying shareholder wealth.

 *Tip: A firm can show steady, positive accounting profit for years and still carry a negative EVA throughout – that gap is the entire reason EVA is used as a performance yardstick instead of plain profit.*

1.3 Market Value Added (MVA)

★ Good to Know

$$MVA = \text{Market Value of the Firm} - \text{Total Capital Invested}$$

Where:

Market Value of the Firm	what the firm is worth to the market today – market value of equity (share price × number of shares), plus market value of debt where debt is separately valued
Total Capital Invested	the book value of all capital shareholders and lenders have put into the firm over its life – total Capital Employed (paid-up capital + reserves + long-term borrowings)

Build-up logic:

1. Book value shows what shareholders and lenders originally put into the firm, plus retained profits – a backward-looking, historical-cost figure.
2. Market value shows what the firm is actually worth today, based on everything the market currently knows or believes about its future prospects – a forward-looking figure.
3. If the firm has used the capital invested in it well – profitable projects, sound strategy – the market should be willing to pay more for the whole firm than the raw capital that was put in; if it has destroyed value instead, market value falls below capital invested.
4. So simply subtract what was originally invested (book value of capital) from what the market says the firm is worth today – the difference is the total wealth created, or destroyed, for investors over the firm's entire life.
5. Unlike EVA, which measures value created in just one period, MVA is a cumulative, all-time scorecard – in fact, MVA is best understood as the market's own estimate of the present value of every future EVA the firm is expected to generate.

💡 *Tip: EVA and MVA can disagree in the short run – a firm can post positive EVA this year but still show weak or falling MVA if the market doubts that performance will be sustained.*

1.4 Altman's Z-Score

★ Good to Know

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

Where:

- X_1 Working Capital / Total Assets – short-term liquidity cushion relative to firm size
- X_2 Retained Earnings / Total Assets – cumulative profitability and maturity; young firms naturally score low here
- X_3 EBIT / Total Assets – operating earning power of the asset base, independent of tax and financing
- X_4 Market Value of Equity / Book Value of Total Liabilities – the solvency cushion: how far assets can fall in value before liabilities exceed them
- X_5 Sales / Total Assets – asset turnover, how efficiently the asset base is used to generate sales

Build-up logic:

1. A firm heading toward bankruptcy rarely fails for just one reason – it typically shows weakness across several fronts at once: poor liquidity, poor profitability (both current and accumulated), poor efficiency, and a thin solvency cushion – so no single ratio can reliably flag distress alone.
2. Altman picked one ratio to represent each danger zone: X_1 for short-term liquidity, X_2 for accumulated profitability and track record, X_3 for current operating earning power, X_4 for the market's own solvency cushion, and X_5 for efficiency in generating sales from assets.
3. These five ratios don't all matter equally in separating firms that actually went bankrupt from those that survived, so each is multiplied by a weight – 1.2, 1.4, 3.3, 0.6 and 1.0 – fitted statistically from real failed-versus-healthy company data, with X_3 (operating earning power) getting by far the heaviest weight since it proved the single strongest predictor of failure.
4. Adding up these five weighted ratios collapses many separate pieces of financial-health information into one number – the higher the score, the healthier the firm looks across all five dimensions at once.
5. Because the model was calibrated against real historical bankruptcy outcomes, Altman was also able to identify the actual score cut-offs that best separated failed firms from survivors – giving the three interpretation zones used to read the final score: Safe ($Z > 2.99$), Grey ($1.81 \leq Z \leq 2.99$), and Distress ($Z < 1.81$).

 *Tip: This is the original model for publicly-traded manufacturing firms – note that X_4 uses the MARKET value of equity, not book value, so it needs the share price, not just the balance sheet.*

Example: $X_1 = 0.15$, $X_2 = 0.20$, $X_3 = 0.10$, $X_4 = 1.50$, $X_5 = 1.20$. $Z = 1.2(0.15) + 1.4(0.20) + 3.3(0.10) + 0.6(1.50) + 1.0(1.20) = 0.18 + 0.28 + 0.33 + 0.90 + 1.20 = 2.89$ – falls in the Grey Zone, so caution is warranted even though bankruptcy is not imminent.

2

Risk Management

How to measure the risk hiding inside a project's uncertain cash flows, build that risk into the accept-or-reject decision, and quantify a portfolio's worst-case loss using Value at Risk.

2.1 Expected Value, Variance & Standard Deviation of a Probability Distribution

★ Good to Know

$$E(X) = \sum_{i=1}^n p_i X_i$$

$$\sigma = \sqrt{\sum_{i=1}^n p_i (X_i - E(X))^2}$$

Where:

- p_i probability of the i^{th} possible outcome or economic state (e.g. recession, normal, boom)
- X_i value of the variable in that state – a project's cash flow or NPV, or a security's return
- $E(X)$ expected value – the probability-weighted average outcome
- σ standard deviation of X – the usual risk measure quoted; σ^2 is its variance

Build-up logic:

1. A project's or security's future outcome is not one fixed number – it depends on which economic state actually plays out, and each possible state carries some probability.
2. To get one usable "central" figure, weight every possible outcome by how likely it is and add them up – this probability-weighted average is the expected value, the single best estimate of what will happen on average.
3. Risk is about how far actual outcomes might swing away from this expected value, so measure each outcome's deviation from the mean, $(X_i - E(X))$.
4. Simply averaging these deviations would always give zero, since positive and negative swings cancel out – so square each deviation first (this also penalises bigger swings more heavily), then take the probability-weighted average of the squared deviations to get variance.
5. Variance sits in awkward squared units, so take its square root to bring the risk figure back to the same unit as the original outcome – that is the standard deviation, the number actually compared across projects or securities.

✍ Tip: If a question gives equally-likely scenarios without stating probabilities, just use $p_i = 1/n$ for each. This same year-by-year logic, applied to a project's annual cash flows, is exactly the building block behind Hillier's model (2.3–2.4) below.

2.2 Coefficient of Variation (CV)

★ Good to Know

$$CV = \frac{\sigma}{E(X)}$$

Where:

σ standard deviation of the project's/security's returns (or NPV), from 2.1

$E(X)$ expected value (mean) of the same distribution, from 2.1

Build-up logic:

1. Standard deviation measures risk in absolute terms – but ₹10,000 of SD means something very different for a project with an expected NPV of ₹20,000 than for one with an expected NPV of ₹2,00,000.
2. To compare risk fairly across projects or securities of very different sizes, risk needs to be expressed relative to the return it is attached to, not as a raw rupee number.
3. Dividing SD by the expected value does exactly this – it rescales risk into "risk per unit of expected return," a pure ratio that strips out the effect of scale.
4. A lower CV means less risk is being taken per unit of expected return, so when two projects have different expected NPVs, ranking them by CV – not raw SD – gives the fairer comparison, and the lower-CV project is generally the relatively safer choice for the return it offers.

 *Tip: Never rank mutually exclusive projects of very different sizes purely on SD – a bigger project almost always shows a bigger absolute SD. CV is the fair comparison; lower CV is better.*

2.3 Expected NPV & Standard Deviation of NPV — Independent Cash Flows (Hillier's Model)

★ MUST KNOW

$$E(NPV) = \sum_{t=1}^n \frac{E(CF_t)}{(1 + R_f)^t} - I_0$$

$$\sigma(NPV) = \sqrt{\sum_{t=1}^n \frac{\sigma_t^2}{(1 + R_f)^{2t}}}$$

Where:

$E(CF_t)$ expected cash flow in year t , computed from that year's own probability distribution (as in 2.1)

σ_t standard deviation of the cash flow in year t (also from 2.1)

R_f risk-free rate — used to discount here because risk is measured separately through $\sigma(NPV)$, not folded into the rate

I_0 initial investment (outflow at time 0, assumed certain)

n project life in years

$E(NPV)$ expected net present value of the project

$\sigma(NPV)$ standard deviation of the project's NPV — its overall risk across the whole project life

Build-up logic:

1. When a project's future cash flows are uncertain, both its expected value and its risk need to be measured — not just one number, as under a plain certain-cash-flow NPV.
2. First find expected NPV exactly like an ordinary NPV, but using each year's expected cash flow — and discount at the risk-free rate, not a risk-adjusted rate, because risk is about to be captured separately via $\sigma(NPV)$; using a risk-adjusted rate here would double-count the risk.
3. For risk: each year's cash flow has its own SD, σ_t , which also needs bringing to a present-value basis — discounting scales the cash flow itself, so it scales variance by the discount factor SQUARED, hence dividing σ_t^2 by $(1 + R_f)^{2t}$.
4. If the cash flow in one year is statistically independent of every other year's cash flow — a bad year for CF_1 says nothing about CF_2 — a key statistical result applies: the variance of a sum of independent variables is simply the sum of their individual variances, with no covariance cross-terms to add.
5. So total variance of NPV is just the sum of each year's discounted variance, and its square root gives $\sigma(NPV)$ — this square-root-of-sum-of-squares shape is exactly what happens when independent risks partly offset each other over time (some years' surprises are good, some bad, and they don't move together), leaving total risk smaller than a plain sum of the individual years' SDs would suggest.

 *Tip: Independence is the default assumption in Hillier's model unless the question explicitly says cash flows move together across years (e.g. driven by the same demand trend) — that is the perfectly-correlated case in 2.4.*

2.4 Standard Deviation of NPV — Perfectly Correlated Cash Flows

★★ Frequently Asked

$$\sigma(NPV) = \sum_{t=1}^n \frac{\sigma_t}{(1 + R_f)^t}$$

Where:

σ_t standard deviation of year t 's cash flow (same meaning as in 2.3)

R_f risk-free rate

$\sigma(NPV)$ standard deviation of the project's overall NPV, now under the perfectly-correlated assumption — $E(NPV)$ itself is still computed exactly as in 2.3, since expected value is always additive regardless of correlation

Build-up logic:

1. Sometimes a project's cash flows across years are not independent at all — they are driven by the same underlying factor (if the product is a hit, every year runs high; if it flops, every year runs low), which is perfect positive correlation, a correlation coefficient of +1 between every pair of years.
2. When two uncertain quantities are perfectly positively correlated, a clean statistical result applies: the standard deviation of their SUM equals the plain sum of their individual standard deviations — unlike the independent case, where variances (not SDs) are the things that add up.
3. So here, discount each year's own σ_t by $(1 + R_f)^t$ to bring it to present-value terms, exactly as before, but then simply ADD these discounted SDs directly across all n years — no squaring, summing of squares, and square-rooting this time.
4. This gives a noticeably larger $\sigma(NPV)$ than the independent case for the very same set of yearly σ_t 's, because a plain sum of positive numbers is always at least as large as the square root of the sum of their squares — strictly larger whenever more than one year is involved.
5. This makes complete economic sense: when cash flows are independent, a bad year can be cushioned by a good year elsewhere, so risk partly cancels out over time — exactly like diversification across different securities. When cash flows are perfectly correlated instead, a bad outcome in one year means a bad outcome in EVERY year at once — nothing cancels out, so the full risk of every year adds up directly, making total project risk much higher for the same individual yearly risks.

Example: A 3-year project has an undiscounted $\sigma_t = ₹10,000$ in each year. Discounted at $R_f = 10\%$, this comes to $\sigma_1 \approx ₹9,091$ (year 1), $\sigma_2 \approx ₹8,264$ (year 2) and $\sigma_3 \approx ₹7,514$ (year 3). Independent case: $\sigma(NPV) = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2} \approx ₹14,401$. Perfectly correlated case: $\sigma(NPV) = \sigma_1 + \sigma_2 + \sigma_3 \approx ₹24,869$ — about 73% higher, for the exact same underlying yearly risk.

2.5 Risk-Adjusted Discount Rate (RADR) Method

★ MUST KNOW

$$NPV = \sum_{t=1}^n \frac{CF_t}{(1 + RADR)^t} - I_0, \quad RADR = R_f + \text{Risk Premium}$$

Where:

CF_t	expected (risky) cash flow in year t
$RADR$	risk-adjusted discount rate — the discount rate loaded with an extra premium to compensate for this project's risk
R_f	risk-free rate
Risk Premium	extra return demanded specifically for bearing this project's risk — larger for riskier projects
I_0	initial investment

Build-up logic:

1. A plain NPV calculation assumes cash flows are certain — but real project cash flows are only estimates, and riskier projects deserve a bigger penalty than safer ones when deciding whether to accept them.
2. The most intuitive way to build risk into NPV is to make the "price of waiting" itself steeper for riskier projects — use a higher discount rate for riskier projects and a lower one for safer projects, instead of applying the firm's plain cost of capital to every project regardless of risk.
3. Build this adjusted rate up from the risk-free rate — the return needed even with zero risk — plus a risk premium, the extra return demanded specifically for this project's risk; riskier projects or divisions get a larger premium added on top.
4. Discount every future cash flow at this higher rate, $RADR$, instead of the plain cost of capital — since a higher discount rate shrinks the present value of every future rupee more aggressively, riskier projects get penalised automatically, without touching the cash flow estimates themselves.
5. This keeps the mechanics of ordinary NPV completely unchanged — still discount and subtract the initial investment — only the discount rate is adjusted upward for risk, which is exactly why RADR is the most widely used, easiest-to-apply way of building risk into a capital budgeting decision.

💡 *Tip: RADR discounts the RAW expected cash flows at a higher rate. Don't confuse this with the Certainty Equivalent method (2.6), which instead shrinks the cash flows themselves and discounts at the risk-free rate.*

2.6 Certainty Equivalent (CE) Method

★★ Frequently Asked

$$NPV = \sum_{t=1}^n \frac{\alpha_t \times CF_t}{(1 + R_f)^t} - I_0$$

Where:

CF_t risky (expected) cash flow in year t

α_t certainty-equivalent coefficient for year t , between 0 and 1 — converts the risky cash flow into an equivalent certain amount; a smaller α_t means the cash flow is treated as riskier

R_f risk-free rate — used because the cash flows have already been converted to their certain, risk-free-equivalent form

I_0 initial investment

Build-up logic:

1. Instead of loading the risk penalty onto the discount rate the way RADR does, an alternative is to load it directly onto the cash flow itself — ask "what smaller, certain amount would I consider exactly as good as this larger, risky expected cash flow?"
2. That smaller certain amount is the certainty equivalent, found by multiplying the risky expected cash flow by a coefficient α_t that shrinks more for riskier or more distant cash flows.
3. Once every year's cash flow has been converted into its risk-free-equivalent amount, there is no risk left in these numbers to account for — so they can now be discounted at the ordinary risk-free rate, R_f , exactly like any certain cash flow.
4. This keeps the discount rate "clean" — always R_f , project after project — and instead makes the risk adjustment explicit and visible year by year through the α_t coefficients, which is why CE is considered more theoretically precise than RADR: it lets risk vary period by period instead of forcing one flat risk premium onto every year of the project's life.
5. Summing these present values of certainty-equivalent cash flows and subtracting the initial investment gives NPV exactly as usual — accept if positive, reject if negative.

 *Tip: α_t usually gets smaller as t increases, since more distant cash flows are harder to forecast reliably and deserve a bigger haircut. If a question gives risk premiums or a risk-adjusted rate instead of α_t coefficients, use RADR instead.*

RADR vs Certainty Equivalent — Why the Two Can Disagree

RADR applies one constant risk-adjusted rate to every year of the project, and compounding that single higher rate implicitly assumes risk keeps growing at the same fixed proportional pace year after year — since a bigger discount factor bites progressively harder on more distant cash flows. Certainty Equivalent makes no such assumption: it lets the analyst assign a separately judged coefficient to each year, so it can reflect any pattern of risk over time — rising, falling, or uneven — not only one that grows at a fixed compound rate. Because the two methods build their risk adjustment into the cash flows so differently, they will typically agree on NPV and on the accept/reject verdict only when a project's real risk profile happens to match RADR's built-in assumption of steadily compounding risk; whenever risk instead rises unevenly, or is concentrated in a particular year (say, a single large cash flow far in the future, or an early year that is riskier than the years after it), the two methods can rank projects differently, and can occasionally even flip the accept/reject conclusion. This is exactly why CE is generally regarded as the more theoretically sound method — it separates the time-value adjustment (done via R_f) from the risk adjustment (done via α_t) instead of bundling both into a single rate — even though RADR remains the far more common method in practice for its sheer simplicity.

Sensitivity Analysis

Sensitivity analysis tests how sensitive a project's NPV is to a change in just one key underlying variable at a time — sales volume, selling price, variable cost, or useful life — while every other variable is held at its base-case value. For each variable in turn, the analyst usually finds the percentage change that would swing NPV all the way to zero; a variable that needs only a small change to wipe out NPV is highly sensitive and needs the closest monitoring, while one that would need a huge swing to hurt NPV is comparatively safe. The technique is simple to build — just recompute NPV, one variable at a time, for a small change up and down — which is exactly why it is usually the first step in risk analysis. Its real limitation is that it changes only one variable at a time and never captures what happens when several variables move together, as they usually do in real life (a recession, for instance, tends to hit sales volume and price at once) — so it identifies which single variables a project is most exposed to, but does not by itself measure the project's overall risk.

Scenario Analysis

Scenario analysis fixes sensitivity analysis's one-variable-at-a-time limitation by changing several related variables together, consistently, under a small number of realistic combined scenarios — typically a Pessimistic case (low sales, high costs), a Most Likely case (the base case), and an Optimistic case (high sales, low costs). NPV is recomputed in full under each scenario, giving a small range of possible outcomes, and if probabilities are attached to each scenario, an expected NPV and standard deviation can be worked out exactly as in the probability-distribution approach earlier in this chapter. Because variables move together in economically consistent combinations — a downturn scenario logically drags down sales, price and collections all at once, not just one of them in isolation — scenario analysis gives a far more realistic feel for a project's genuine downside and upside than sensitivity analysis's isolated single-variable swings. It is still limited, though, to only the handful of discrete scenarios chosen, unlike a full simulation, which can generate a virtually continuous range of outcomes.

2.7 Value at Risk (VaR) — Parametric Method

★ MUST KNOW

$$\begin{aligned} VaR_1 &= Z \times \sigma \times V \\ VaR_t &= Z \times \sigma \times V \times \sqrt{t} = VaR_1 \times \sqrt{t} \end{aligned}$$

Where:

- Z standard normal Z-value for the chosen confidence level (e.g. 1.645 for 95%, 2.33 for 99%, one-tailed)
- σ standard deviation (volatility) of the portfolio's/asset's returns, usually a daily figure
- V value of the portfolio or position being measured, in currency terms
- t the holding period/horizon, in the same time unit as σ (e.g. $t = 10$ days for a 10-day VaR, when σ is daily)
- VaR Value at Risk — the maximum loss expected over the given horizon, at the chosen confidence level, under normal market conditions

Build-up logic:

1. VaR answers one specific question: "what is the worst loss I can expect on this portfolio over the next day (or week), such that I am, say, 95% or 99% confident actual losses won't exceed it?"
2. The parametric (variance-covariance) method assumes returns follow a normal distribution — under this assumption, a given confidence level corresponds to a fixed number of standard deviations away from the mean, the Z-value (e.g. the worst 5% of one-tailed outcomes lie beyond 1.645 SDs below the mean).
3. So the potential loss, as a fraction of portfolio value, is simply Z standard deviations of return, $Z \times \sigma$; multiply by the portfolio's actual rupee value, V , to turn this percentage loss into a real currency figure.
4. This gives VaR for the SAME period over which σ was measured — a 1-day VaR if σ is daily volatility — but a longer holding period is often needed, say 10 days.
5. Since variance is additive over independent, identically distributed periods (the same idea used for Hillier's independent-cash-flow case in 2.3), variance over t days equals t times the 1-day variance, so standard deviation scales by $\sqrt{t}$ — simply multiply the 1-day VaR by $\sqrt{t}$ to scale it up to a t -day VaR.

 *Tip: Common one-tailed Z-values — 90% confidence → 1.28; 95% → 1.645; 99% → 2.33. Always check whether the question wants a one-tailed or two-tailed Z before picking the value.*

Example: A portfolio worth ₹50,00,000 has a daily return SD of 2%. At 99% confidence ($Z = 2.33$), 1-day $VaR = 2.33 \times 0.02 \times ₹50,00,000 = ₹2,33,000$. For a 10-day horizon, $VaR = ₹2,33,000 \times \sqrt{10} \approx ₹2,33,000 \times 3.162 \approx ₹7,36,811$.

Historical Simulation and Monte Carlo Simulation (VaR Approaches)

The parametric method is quick, but it leans entirely on the assumption that returns are normally distributed — real market returns often have "fatter tails" (more extreme moves) than a normal curve predicts, which can make parametric VaR understate real risk. Historical simulation sidesteps this assumption completely: it takes the portfolio's actual historical returns over some past window (say, the last 500 trading days), applies today's portfolio weights to each of those historical days to build a hypothetical profit or loss for every day, ranks these hypothetical outcomes from worst to best, and simply reads off the loss at the required percentile (the 5th-worst percentile for 95% confidence, say) directly from this ranked list — no assumption about the shape of the return distribution is needed at all. Monte Carlo simulation goes a step further: rather than relying only on what actually happened historically, a computer randomly generates thousands of hypothetical future price paths for the portfolio based on assumed statistical properties — mean, volatility, correlations — of the underlying risk factors, and VaR is again read off the resulting distribution of simulated outcomes. Monte Carlo is the most flexible of the three approaches, able to handle complex portfolios and non-linear instruments like options under any assumed distribution, but it is also the most computationally demanding, and its accuracy depends heavily on how well the assumed model and inputs reflect reality.

3

Security Analysis

How to judge a security from the outside in – economy-to-company fundamentals, price charts and indicators, and whether markets are too efficient for any of this to matter.

Fundamental Analysis: Economy-Industry-Company (EIC) Framework

Fundamental analysis tries to find a security's "true" intrinsic value by working top-down through three layers. Economy analysis studies the big picture first – GDP growth, inflation, interest rates, fiscal/monetary policy, and the stage of the business cycle – because no industry or company can escape a struggling economy. Industry analysis then narrows down to the specific sector's life-cycle stage, competitive structure, government regulation, and demand outlook, since even in a healthy economy some industries do far better than others. Finally, company analysis studies the specific firm's financial statements, management quality, competitive position, and growth prospects to arrive at an intrinsic value, which is then compared with the current market price to decide buy, hold, or sell. The logic runs strictly top-down because a great company in a dying industry, or a great industry in a collapsing economy, rarely delivers good returns.

Dow Theory

Dow Theory is the founding theory of technical analysis – it holds that the market's own price movement, not news or fundamentals, reveals everything needed about the trend, and that this movement has three components running at once, like waves of different sizes on the sea. The *primary trend* is the big multi-month-to-multi-year move (a bull or bear market); the *secondary (intermediate) trend* is a shorter correction or rally against the primary trend, usually lasting a few weeks to a few months; and *minor trends* are the small day-to-day wiggles, mostly just noise, that the theory largely ignores. Its key tenets: the market discounts everything (all known information is already in the price); market averages must confirm each other before a move is trusted; a trend remains in force until a clear reversal signal appears; and volume must confirm price (rising prices on rising volume signal a healthy trend). Because minor trends are noise, Dow theorists focus almost entirely on correctly reading the primary trend.

Elliott Wave Principle

Elliott Wave theory holds that market prices move in repeating crowd-psychology patterns rather than randomly. In the direction of the larger trend, price advances in 5 numbered "impulse" waves: waves 1, 3 and 5 move with the trend while waves 2 and 4 are smaller pullbacks against it (wave 3 is usually the longest and strongest of the five). Once this impulse move is complete, the market corrects the whole advance in 3 lettered "corrective" waves, A, B and C, moving against the main trend. Each of these waves can itself be broken down into the same 5-3 pattern at a smaller scale — a "fractal" structure — which is why the theory is applied across timeframes from minutes to decades. In practice, it is used to judge where the current wave count stands and what should logically follow next, though labelling the waves correctly in real time is famously subjective.

3.1 Simple Moving Average (SMA)

★ Good to Know

$$SMA_n = \frac{P_1 + P_2 + \dots + P_n}{n}$$

Where:

$P_1, P_2, \dots, P_n$ closing prices of the security over each of the last n periods being averaged

n number of periods in the average — the chosen window length (e.g. a 5-day, 10-day, or 200-day moving average)

Build-up logic:

1. A single day's closing price is noisy — random short-term swings make it hard to see the real underlying trend.
2. To smooth this out, look at a window of the most recent n days together instead of any one day alone.
3. Add up these n prices and divide by n — a plain average — to get one representative "trend" value for that window.
4. As each new day passes, drop the oldest price from the window and bring in the newest one, then recompute — this is the "moving" part of moving average.
5. Plotting this rolling average day after day gives a smoothed line that filters out day-to-day noise and reveals the underlying direction of the trend.

Tip: A shorter-period MA crossing above a longer-period MA is read as bullish (a "golden cross"); crossing below is bearish (a "death cross").

3.2 Exponential Moving Average (EMA)

★★ Frequently Asked

$$EMA_t = (P_t \times \alpha) + (EMA_{t-1} \times (1 - \alpha)), \quad \alpha = \frac{2}{n + 1}$$

Where:

- P_t closing price for the current (today's) period
- EMA_{t-1} the EMA value already calculated for the previous period
- α smoothing constant — decides how much weight today's price gets versus the carried-forward history
- n number of periods chosen for the average (e.g. a 12-day EMA uses $n = 12$)

Build-up logic:

1. A simple moving average treats every one of the n days as equally important — but a big price move yesterday is usually more relevant to today's trend than one from three weeks ago.
2. So instead of a flat average, give more weight to recent prices and let the weight taper off for older ones — this is what "exponential" weighting means.
3. The easiest way to build this: today's EMA is mostly yesterday's EMA (carrying forward all the already-smoothed history) plus a small nudge from today's actual price.
4. That nudge is controlled by the smoothing constant α , set as $2/(n + 1)$ so that an " n -day EMA" reacts with roughly the sensitivity of an n -day window while still weighting recent data more heavily.
5. Because each day's EMA already embeds all prior days' smoothed history, only yesterday's EMA and today's price are needed to update it — no need to store all n old prices as a simple moving average would.

 *Tip: A larger n gives a smaller α , so the EMA reacts more slowly and lags more; a smaller n gives a larger α , so it hugs the current price more closely.*

3.3 Relative Strength Index (RSI)

★★ Frequently Asked

$$RS = \frac{\text{Average Gain}}{\text{Average Loss}}$$
$$RSI = 100 - \frac{100}{1 + RS}$$

Where:

Average Gain average of the positive price changes (up-moves) over the last n periods

Average Loss average of the negative price changes (down-moves), taken as a positive magnitude, over the same n periods

RS Relative Strength — the ratio of average gain to average loss

n number of periods used in the calculation (commonly 14 days)

Build-up logic:

1. The goal is one clean number showing whether a stock's recent price action has been dominated by up-moves or down-moves — a "strength" gauge.
2. Look back over the last n days and split each day's price change into either a gain (price up) or a loss (price down, recorded as a positive amount).
3. Average the gains and average the losses separately over those n days, then take their ratio, RS . On its own, RS can range from zero to a very large number, which is awkward to interpret or compare.
4. To convert this into a clean 0–100 scale, apply the RSI transformation, $100 - 100/(1 + RS)$: when gains dominate (RS large), RSI approaches 100; when losses dominate (RS near zero), RSI approaches 0.
5. Extreme readings are then read as signals — an RSI above roughly 70 usually flags an "overbought" stock due for a correction, while below roughly 30 flags "oversold," due for a bounce.

 *Tip: RSI is always strictly between 0 and 100 — if a computed value falls outside this range, recheck the average gain/loss figures.*

Example: Over the last 14 days, Average Gain = ₹2.50 and Average Loss = ₹1.25. $RS = 2.50/1.25 = 2$, so $RSI = 100 - 100/(1 + 2) = 100 - 33.33 \approx 66.67$ — approaching overbought territory.

3.4 Rate of Change (ROC)

★ Good to Know

$$ROC = \frac{P_t - P_{t-n}}{P_{t-n}} \times 100$$

Where:

- P_t closing price today (the current period)
 P_{t-n} closing price n periods ago
 n the look-back period used for comparison

Build-up logic:

1. Momentum simply asks: how much has price actually moved over a fixed look-back window, compared with where it started?
2. Take today's price and subtract the price from n periods back — this gives the absolute change over that window.
3. Express this change as a percentage of the starting price, P_{t-n} , so the indicator is comparable across stocks trading at very different price levels, not just a rupee number.
4. Multiply by 100 to state it as a clean percentage.
5. A rising ROC line shows accelerating upward momentum; a falling or negative ROC shows momentum fading or reversing — sometimes even before the price itself turns down.

 *Tip: ROC oscillates around zero — a crossover from negative to positive is often read as a bullish momentum signal, and vice versa.*

Moving Average Convergence Divergence (MACD)

MACD is a momentum-cum-trend-following indicator built entirely from EMAs. The MACD line is the difference between a faster and a slower EMA: $MACD\ Line = EMA_{12} - EMA_{26}$. A signal line is then created by taking a 9-day EMA of the MACD line itself: $Signal\ Line = EMA_9(MACD\ Line)$. When the MACD line crosses above the signal line, it hints at rising bullish momentum; when it crosses below, momentum is turning bearish. The gap between the two lines — often plotted as a histogram — shows how strongly the trend is accelerating or fading, which is why MACD is used both to spot trend direction and to time entries/exits around the crossovers.

Efficient Market Hypothesis (EMH) and Its Three Forms

EMH holds that a security's market price at any time already reflects all available information, so consistently "beating the market" through analysis is not possible — fresh information gets absorbed into price almost instantly. It comes in three forms, based on how much information is assumed to already be in the price. The *weak form* says price reflects all past price and volume data, so technical analysis (charts, moving averages, RSI, and the like) cannot generate extra returns — only genuinely new information can move price from here. The *semi-strong form* says price reflects all publicly available information too (financial statements, news, announcements), so even fundamental analysis of public data cannot beat the market — only access to private, non-public information could. The *strong form* goes furthest, saying price reflects all information, public and private (insider) alike, so even insider trading could not consistently earn abnormal profits — a form most researchers consider unrealistic in practice. So the stronger the form of EMH that actually holds, the less useful both technical analysis (undermined even by the weak form) and fundamental analysis (undermined by the semi-strong form and above) become.

Putting a fair rupee value on a bond or a share – coupons and redemption value for bonds; dividends, free cash flows and earnings for equity.

4.1 Bond Valuation

★ MUST KNOW

$$V_0 = C \times \left[\frac{1 - (1 + k_d)^{-n}}{k_d} \right] + \frac{RV}{(1 + k_d)^n}$$

Where:

V_0 current value (price) of the bond

C annual coupon (interest) amount = coupon rate $\times$ face value

k_d required rate of return / market discount rate for a bond of this risk (the yield investors currently demand)

n number of years remaining to maturity

RV redemption value received at maturity (often equals face value, but can be at a premium or discount to it)

Build-up logic:

1. A bond is just a legal promise to pay two things: a fixed coupon every year until maturity, and one lump-sum redemption amount when it matures.
2. Money in the future is worth less than money today, so these future rupee amounts cannot simply be added up – each must be discounted back to today at the rate investors currently demand for this bond's risk, k_d .
3. The coupons form a level annual annuity of C for n years, so their combined present value is just C multiplied by the standard annuity discounting factor for n years at k_d – the bracketed term.
4. The redemption value RV is a single lump sum received only once, at the end of year n , so its present value is simply RV discounted back n years at k_d .
5. Adding these two present values together gives the bond's fair value today – what a rational investor should be willing to pay for it.
6. This is exactly why bond price and yield move in opposite directions: raise k_d in the formula and every discount factor shrinks, pulling V_0 down, and vice versa.

 *Tip: If the coupon rate equals k_d , the bond trades at par (V_0 = face value); if the coupon rate is higher than k_d , it trades at a premium; if lower, at a discount.*

4.2 Current Yield

★ Good to Know

$$\text{Current Yield} = \frac{\text{Annual Coupon}}{\text{Market Price}} \times 100$$

Where:

Annual Coupon the fixed yearly interest amount received (coupon rate × face value)

Market Price the price at which the bond currently trades in the market

Build-up logic:

1. An investor buying a bond today at its current market price wants a quick sense of the annual cash return they're getting right now, purely from coupon income.
2. The rupee amount of that return each year is fixed and already known — the annual coupon.
3. To express this as a rate of return comparable across bonds, put the annual coupon over what the investor is actually paying today, the market price, not the face value — because that is the true amount invested.
4. Multiply by 100 to state it as a percentage.
5. Notice this ignores any capital gain or loss from buying below or above face value and holding to maturity — that is exactly why current yield is a quick, rough measure, not the same thing as YTM.

 *Tip: Current yield ignores the capital gain/loss at redemption — it overstates true return for a bond bought at a discount and understates it for one bought at a premium.*

4.3 Yield to Maturity (YTM)

★ MUST KNOW

$$YTM \approx \frac{C + \frac{RV - P_0}{n}}{\frac{RV + P_0}{2}}$$

Where:

- C annual coupon amount
 RV redemption value at maturity
 P_0 current market price of the bond
 n number of years remaining to maturity

Build-up logic:

1. YTM is really the bond's own internal rate of return – the single rate that makes the present value of all coupons plus redemption exactly equal today's price. Solving this exactly needs trial and error, which isn't practical for quick use, so a good approximation is built instead.
2. Start with the annual coupon C – the obvious, recurring part of the investor's return.
3. If the bond is bought at a discount (P_0 below RV), the investor also earns $(RV - P_0)$ as a capital gain spread evenly over the n years to maturity; if bought at a premium, this term is a loss. Add this average annual gain/loss to the coupon to get total average annual return.
4. Now express this average annual return as a percentage of the amount actually tied up. But the investment isn't static – it starts at P_0 and ends at RV – so use their average, $(RV + P_0)/2$, as a fair stand-in for the "average" money invested over the bond's life.
5. Dividing total average annual return by this average investment gives a good approximate yield – close to, though not exactly equal to, the true IRR-based YTM.

 *Tip: Whenever a question gives P_0 , RV , C and n and asks to estimate YTM, reach for this formula directly – attempt trial-and-error interpolation only if the exact IRR-based YTM is specifically demanded.*

Example: A ₹1,000 face-value, 10% coupon bond ($C = ₹100$) with 5 years to maturity, $RV = ₹1,000$, trading at $P_0 = ₹950$: $YTM \approx \frac{100 + (1000 - 950)/5}{(1000 + 950)/2} = \frac{110}{975} \approx 11.28\%$.

4.4 Yield to Call (YTC)

★★ Frequently Asked

$$YTC \approx \frac{C + \frac{CP - P_0}{n_c}}{\frac{CP + P_0}{2}}$$

Where:

C annual coupon amount

CP call price – the amount the issuer pays to redeem the bond early, often at a slight premium to face value

P_0 current market price of the bond

n_c number of years remaining until the (first) call date

Build-up logic:

1. Many bonds give the issuer the right to redeem the bond early at a call price once it becomes economical to do so – typically once interest rates have fallen enough for the issuer to refinance more cheaply.
2. If a bond is likely to be called, the investor's real holding period is shorter than the stated maturity, and the amount received at the end is the call price, not the redemption/face value.
3. So the YTC formula is built exactly like the YTM approximation, simply substituting the call price CP for RV and the years-to-call n_c for n – same logic, different assumed end point.
4. This gives an estimated yield assuming the bond is called at the earliest opportunity – the more realistic, usually more conservative, figure to use for a premium bond trading above its call price.

 *Tip: For a premium bond likely to be called, check both YTM and YTC and use the lower of the two ("yield to worst") as the more prudent estimate.*

4.5 Realized Yield

★ Good to Know

$$\text{Realized Yield} = \left(\frac{TV}{P_0} \right)^{1/n} - 1$$

Where:

TV terminal value — total cash in hand at the end of the holding period: every coupon compounded forward to that date at the actual/assumed reinvestment rate, plus the redemption or sale proceeds received then

P_0 original purchase price of the bond

n number of years in the holding period

Build-up logic:

1. YTM is a promise, not a guarantee — it is only truly earned if every coupon received along the way is reinvested at that same YTM rate. In real life, rates move, so coupons often get reinvested at a different rate — this is "reinvestment risk."
2. Realized yield measures what the investor actually earned, given the reinvestment rate that actually applied, or one the question specifically assumes.
3. To find it, first work out the terminal value: compound every coupon forward from the year it was received to the end of the holding period at the reinvestment rate, then add the redemption or sale proceeds received at that final date.
4. Now there is one lump sum in hand at the end, TV , against one lump sum paid out at the start, P_0 — exactly like a simple fixed-deposit problem. The compound annual rate that grows P_0 into TV over n years is $(TV/P_0)^{1/n} - 1$.
5. This realized yield equals the original YTM only if the reinvestment rate assumption happens to exactly equal YTM — any difference between the two shows up directly as a gap between realized yield and quoted YTM.

 *Tip: If a question says coupons are reinvested at a given rate, it wants realized yield — compound each coupon forward to the terminal date at that rate, don't discount them.*

4.6 Macaulay Duration

★ MUST KNOW

$$D = \frac{\sum_{t=1}^n t \times \frac{CF_t}{(1+y)^t}}{\sum_{t=1}^n \frac{CF_t}{(1+y)^t}} = \frac{\sum_{t=1}^n t \times PV(CF_t)}{P_0}$$

Where:

- t the time period (year) in which a cash flow is received
- CF_t the cash flow (coupon, or coupon plus redemption in the final year) received at time t
- y the bond's yield to maturity, used as the discount rate
- $PV(CF_t)$ present value of the cash flow at time t , i.e. $CF_t / (1+y)^t$
- P_0 the bond's current price – the sum of the present values of all its cash flows

Build-up logic:

1. A bond doesn't pay all its cash back at once – coupons trickle in over many years, and one big redemption arrives at the end. Maturity (n years) alone doesn't say when, on average, the investor's money actually comes back.
2. To find this true average timing, first find the present value of each individual cash flow – exactly what gets computed when pricing the bond – which shows how much each future payment is really worth today.
3. Weight each cash flow's time period t by how much of the bond's total value that cash flow represents today: multiply each t by its $PV(CF_t)$, so cash flows worth more in present-value terms count more heavily.
4. Add up all these weighted times, then divide by the total present value of all cash flows (which is simply the bond price, P_0) – this turns the weighted sum back into a proper weighted average, measured in years.
5. The result, Macaulay Duration, is the weighted-average time an investor waits to receive the bond's cash flows – always less than or equal to the bond's maturity n , equal only for a zero-coupon bond, where the entire cash flow (and so its full weight) sits at time n .

 Tip: Duration is always $\leq$ maturity for a coupon bond, and exactly equals maturity only for a zero-coupon bond.

4.7 Modified Duration (MD)

★ Good to Know

$$MD = \frac{D}{1 + y}$$

Where:

D Macaulay Duration, in years

y yield to maturity (per period)

Build-up logic:

1. Macaulay Duration is measured in years — it shows the average timing of cash flows, but on its own it doesn't directly say how sensitive the bond's price is to a change in yield.
2. To convert this "timing" measure into a "price sensitivity" measure, adjust for the fact that discounting is compounded — a small change in yield acts on the $(1 + y)$ discount factor, not directly on the raw cash flows.
3. Dividing Macaulay Duration by $(1 + y)$ rescales it into exactly the right units to be multiplied directly against a change in yield to get a percentage price change — this adjusted figure is Modified Duration.
4. Modified Duration is therefore always a little smaller than Macaulay Duration, since it's divided by a number greater than 1, and it is this version — not Macaulay Duration directly — that is actually used to estimate price changes.

 *Tip: Always convert Macaulay Duration to Modified Duration before estimating a % price change — using Macaulay Duration directly in that formula is a common exam slip.*

4.8 Price-Yield Relationship (Duration-based)

★★ Frequently Asked

$$\% \Delta P \approx -MD \times \Delta y$$

Where:

$\% \Delta P$ approximate percentage change in the bond's price

MD Modified Duration

Δy change in yield, in decimal form (e.g. a 50 bps rise is entered as +0.005)

Build-up logic:

1. Modified Duration is already rescaled to link directly to price sensitivity — the whole point of building it was to answer "if yield changes a little, how much does price move?"
2. Bond prices and yields always move in opposite directions (higher yield means more discounting, so lower price), so the relationship must carry a negative sign.
3. For a small yield change, the price-yield curve behaves almost like a straight line locally, so the percentage price change is roughly proportional to the yield change — and Modified Duration is exactly that proportionality constant, the "slope."
4. Multiplying Modified Duration by the change in yield, Δy , and flipping the sign gives the approximate percentage price change.
5. This is only an approximation because the true price-yield relationship is curved (convex), not straight — the approximation gets less accurate for larger yield swings, which is exactly the gap the convexity adjustment corrects for.

 *Tip: Rising yield means falling price and vice versa — always sanity-check the sign of the answer against this before finalizing it.*

4.9 Bond Convexity

★ Good to Know

$$\text{Convexity} = \frac{1}{P_0(1+y)^2} \sum_{t=1}^n \frac{t(t+1)CF_t}{(1+y)^t}$$
$$\% \Delta P \approx (-MD \times \Delta y) + \left(\frac{1}{2} \times \text{Convexity} \times (\Delta y)^2 \right)$$

Where:

P_0	current bond price
y	yield to maturity
CF_t	cash flow received at time t
n	number of periods to maturity
MD	Modified Duration
Δy	change in yield (decimal form)

Build-up logic:

1. The duration-based $\% \Delta P$ formula assumes the price-yield relationship is a straight line – but it's actually curved (convex), bowing below the straight-line estimate when yield falls and above it when yield rises. For small yield changes this barely matters, but for larger swings the straight-line estimate drifts off.
2. Convexity measures exactly how curved this relationship is – it uses the same weighted-average-time idea as duration, but weights each cash flow's timing by $t(t+1)$ instead of just t , which is what picks up the curvature duration's straight-line version misses.
3. Because bond prices are convex to yield, the true price is always a little higher than what plain duration alone predicts, whichever way yield moves – so the convexity adjustment term is always added, never subtracted, and $(\Delta y)^2$ keeps it positive regardless of direction.
4. Combining both effects gives a far more accurate estimate: the linear duration term captures the first-order (straight-line) effect, and the $\frac{1}{2} \times \text{Convexity} \times (\Delta y)^2$ term corrects for the curvature – the higher the convexity, the more this correction matters, especially for large yield swings.
5. In practice, a bond with higher convexity is more attractive to hold for the same duration, since it gains more when yields fall and loses less when yields rise than a lower-convexity bond would.

 *Tip: The convexity adjustment term is always positive, regardless of whether yield rose or fell – only the duration term's sign flips with the direction of Δy .*

4.10 Zero-Coupon Bond Valuation

★★ Frequently Asked

$$V_0 = \frac{RV}{(1 + k_d)^n}$$

Where:

V_0 price of the zero-coupon bond today

RV redemption/maturity value, often the face value

k_d required yield/discount rate

n years to maturity

Build-up logic:

1. A zero-coupon (deep discount) bond pays no periodic interest at all — the investor's entire return comes from the gap between what they pay today and what they receive at maturity.
2. The general bond valuation formula still applies, just with the coupon C set to zero — the entire "present value of coupon annuity" term simply disappears.
3. What's left is only the present value of the single lump sum RV received n years from now, discounted at the required yield k_d .
4. This is exactly why such bonds are always issued and traded well below their face/redemption value — the investor's entire return is baked into this discount rather than paid out separately as interest.

 Tip: The implicit yield on a zero-coupon bond can also be found by rearranging the same formula:

$$k_d = \left(\frac{RV}{V_0} \right)^{1/n} - 1.$$

Bonds vs Equity Valuation — the Big Picture

Both bond and share valuation rest on the same principle: value equals the present value of the future cash flows a security is expected to pay, discounted at a rate reflecting its risk. For bonds, the cash flows (coupons and redemption) and the discount rate (YTM) are contractually fixed and relatively certain, so valuation is largely mechanical, as seen above. For equity shares, both the future cash flows — dividends, earnings, or free cash flows — and the appropriate discount rate (cost of equity) must be estimated rather than read off a contract, which makes equity valuation inherently more judgment-heavy. That is exactly why several alternative equity models exist (dividend-based, cash-flow-based, earnings-based) instead of one universal formula — the rest of this chapter works through each of them.

4.11 Dividend Discount Model — Zero Growth

★ Good to Know

$$P_0 = \frac{D}{K_e}$$

Where:

P_0 current value (price) of the share

D constant annual dividend expected to be paid forever, with no growth

K_e cost of equity — the required rate of return of shareholders

Build-up logic:

1. A share's value should equal the present value of every future dividend it will ever pay, discounted at the return shareholders require, K_e — the same present-value logic used for bonds.
2. In the simplest possible case, assume the company pays the exact same dividend D every year forever, with no growth at all — a level, never-ending payment stream.
3. A level cash flow received forever is a perpetuity, and the present value of any perpetuity is simply the periodic amount divided by the discount rate: D / K_e .
4. So under the zero-growth assumption, price collapses to this one clean ratio — it is really just the standard perpetuity formula applied to dividends.

 *Tip: This is just the special case of the Gordon Growth Model (next) with $g = 0$ — no need to memorize it separately once Gordon's model is known.*

4.12 Gordon Growth Model (Constant Growth DDM)

★ MUST KNOW

$$P_0 = \frac{D_1}{K_e - g}$$

Where:

P_0 current value of the share

D_1 expected dividend at the end of year 1, i.e. $D_0(1 + g)$

K_e cost of equity / required rate of return

g constant expected annual growth rate of dividends, assumed to continue forever

Build-up logic:

1. In the zero-growth model dividends never grow – but real companies usually grow, so their dividends usually grow too, at some constant long-run rate g assumed to hold forever.
2. So instead of every future dividend being the same, each year's dividend equals last year's grown by $(1 + g)$: $D_1, D_1(1 + g), D_1(1 + g)^2$, and so on forever – a growing perpetuity.
3. Discounting each of these infinitely many growing cash flows back to today at K_e and adding them up is an infinite geometric series; using the standard sum of an infinite geometric series, the whole thing collapses neatly to $D_1 / (K_e - g)$, with no need to sum infinite terms by hand.
4. This only works, and only makes economic sense, if K_e is greater than g – if growth equalled or exceeded the discount rate, value would be infinite or negative, which is meaningless, so the model implicitly assumes growth eventually moderates below K_e .
5. This one clean formula – a perpetuity growing at a constant rate – is the workhorse valuation model for stable, mature, dividend-paying companies.

💡 Tip: The formula needs D_1 (next year's dividend), not D_0 (the dividend just paid) – if given D_0 , compute $D_1 = D_0(1 + g)$ first. This is the single most common slip with this formula.

4.13 Growth Rate ($g = b \times ROE$)

★★ Frequently Asked

$$g = b \times ROE$$

Where:

b retention ratio – the proportion of earnings retained (not paid out) by the company, $= 1 - \text{payout ratio}$
 ROE Return on Equity – the rate of return the company earns on shareholders' funds ($\text{Net Profit} \div \text{Net Worth}$)

Build-up logic:

1. Every growth-based DDM formula needs a growth rate g as an input, but companies rarely announce "our dividends will grow at exactly X% forever" – g usually has to be estimated from the company's own fundamentals.
2. A company can only grow future earnings, and so future dividends, by reinvesting part of today's profit back into the business – the portion paid out as dividends leaves the business and can't fund growth.
3. So the fraction of profit retained and reinvested, the retention ratio b , is the "fuel" for growth – the more a company retains relative to paying out, the more capital it has to grow with, all else equal.
4. But how efficiently that reinvested rupee turns into future profit depends on the return earned on it, ROE – a company retaining a lot of profit but earning a poor ROE won't grow much either.
5. Multiplying the two together – fraction retained times the return earned on what's retained – gives the sustainable growth rate: $g = b \times ROE$, how fast earnings (and dividends, at a constant payout ratio) can grow purely from reinvested profits.

 *Tip: If a question gives the payout ratio instead of the retention ratio, remember $b = 1 - \text{payout ratio}$ before applying this formula.*

4.14 Two-Stage Dividend Growth Model

★ MUST KNOW

$$P_0 = \sum_{t=1}^n \frac{D_0(1+g_1)^t}{(1+K_e)^t} + \frac{1}{(1+K_e)^n} \times \frac{D_{n+1}}{K_e - g_2}$$

Where:

D_0 dividend just paid (base year)

g_1 high/initial growth rate during the first (explicit) stage, lasting n years

K_e cost of equity

n number of years in the high-growth (first) stage

D_{n+1} dividend expected in year $n + 1$, i.e. $D_n(1 + g_2)$ – the first dividend of the stable-growth stage

g_2 stable, long-run growth rate assumed to hold forever after year n

Build-up logic:

1. The single constant-growth (Gordon) model assumes one growth rate forever – unrealistic for a currently fast-growing company, since no firm can keep outgrowing the economy indefinitely.
2. A more realistic story: let the company grow fast at g_1 for a limited, explicit forecast period of n years (stage 1), then assume it settles into a stable, sustainable growth rate g_2 forever after (stage 2), once competition and scale have caught up.
3. Value stage 1 by explicitly forecasting and discounting each individual dividend for years 1 to n back to today at K_e – direct present-value work, one year at a time, since growth here isn't constant enough for a shortcut.
4. Value stage 2 using the Gordon growth formula, but applied as of the end of year n , since from year n onward growth is finally constant at g_2 – this gives a lump-sum "terminal value" sitting at time n , equal to $D_{n+1}/(K_e - g_2)$.
5. This terminal value is a future amount as of year n , not today, so it must itself be discounted back to today over n years, exactly like any other single future cash flow.
6. Adding the present value of the explicit stage-1 dividends to the present value of the stage-2 terminal value gives the share's value today – blending a realistic near-term forecast with a simplified long-term assumption.

Example: $D_0 = ₹2$, growing at 20% for 3 years then 6% forever, $K_e = 14\%$: discount D_1 – D_3 individually, find $D_4/(K_e - g_2)$ as the year-3 terminal value and discount that back 3 years too; adding both parts gives $P_0 \approx ₹37.6$.

4.15 H-Model

★ Good to Know

$$P_0 = \frac{D_0(1 + g_2)}{K_e - g_2} + \frac{D_0 \times H \times (g_1 - g_2)}{K_e - g_2}$$

Where:

D_0 dividend just paid (current year)

g_1 current, initial (high) growth rate

g_2 eventual stable, long-run growth rate

K_e cost of equity

H half of the number of years it takes growth to decline linearly from g_1 to g_2 , i.e. $H = N/2$ where N is that declining period

Build-up logic:

1. The two-stage model assumes growth suddenly drops from high to stable overnight at year n – unrealistic, since in practice a company's growth advantage usually fades gradually as competition catches up.
2. The H-Model instead assumes growth declines in a straight line from g_1 down to g_2 over N years, then stays at g_2 forever – a smoother, arguably more realistic transition.
3. The value splits into two clean pieces: the first term, $D_0(1 + g_2)/(K_e - g_2)$, is what the share would be worth if growth were already at the stable rate g_2 right now – the Gordon-model baseline.
4. The second term adds a premium for the extra growth earned during the transition, above and beyond g_2 . This excess growth's impact is captured, on average, as if it applied for H years (the half-life of the declining period), then valued like a perpetuity discounted at $(K_e - g_2)$.
5. Adding the two pieces approximates the value that precisely modelling every single year's changing growth rate would give – but through one neat closed-form formula instead of a year-by-year forecast.

 *Tip: H is the declining period divided by 2, not the full period – a common slip is plugging in N instead of $N/2$ for H .*

4.16 FCFE-based Equity Valuation

★★ Frequently Asked

$$P_0 = \frac{FCFE_1}{K_e - g}$$

Where:

P_0 value per share (equity value)

$FCFE_1$ expected Free Cash Flow to Equity next year – the cash left over for equity shareholders after operating expenses, interest, tax, reinvestment (capex plus working capital) needs, and net debt repayments

K_e cost of equity

g constant expected growth rate of FCFE

Build-up logic:

1. Dividends are only the cash a company chooses to actually pay out – but the cash it could pay to shareholders, after covering everything else including debt obligations, is FCFE, a fuller measure of what equity holders are truly entitled to.
2. Some companies pay out far less than their FCFE, retaining cash, or occasionally more, funded by debt or reserves – so valuing purely on actual dividends can understate or distort true equity value when payout policy doesn't match cash-generating capacity.
3. FCFE-based valuation simply substitutes FCFE wherever the DDM would use dividends – the entire present-value, growing-perpetuity logic of the Gordon model carries over unchanged, just discounting expected free cash flow to equity instead of expected dividends.
4. So the constant-growth version looks exactly like the Gordon Growth Model with $FCFE_1$ in place of D_1 ; a full multi-stage FCFE model can likewise be built exactly like the multi-stage DDM, substituting FCFE flows for dividend flows.

 *Tip: Use FCFE-based valuation instead of DDM whenever a company pays low or no dividends but is clearly generating, and could pay out, real cash – DDM would understate its value there.*

4.17 FCFF-based Firm Valuation

★ Good to Know

$$\text{Value of Firm} = \frac{FCFF_1}{WACC - g}$$
$$\text{Value of Equity} = \text{Value of Firm} - \text{Market Value of Debt}$$

Where:

Value of Firm	total value of the company — the combined claims of both debt and equity holders
$FCFF_1$	expected Free Cash Flow to the Firm next year — cash available to all capital providers (debt and equity together) after operating expenses, tax and reinvestment needs, but before any interest or debt repayments
$WACC$	weighted average cost of capital — the blended required return of both debt and equity holders
g	constant expected growth rate of FCFF
Market Value of Debt	current market value of the company's outstanding debt, subtracted to isolate the equity portion

Build-up logic:

1. FCFF values only what belongs to equity holders, but it can be easier, or more reliable, to value the whole company first and back out the equity portion afterward — especially if capital structure or debt levels are expected to change.
2. FCFF is the cash generated by core operations that is available to pay everyone with a claim on the firm, lenders and shareholders alike, before any financing decisions such as interest or debt repayment are made.
3. Since FCFF belongs to both debt and equity holders together, it must be discounted at a rate reflecting both their required returns blended together — the WACC — not just the cost of equity.
4. Applying the same growing-perpetuity logic as before, but discounting FCFF at WACC instead of K_e , gives the total value of the firm (enterprise value).
5. Shareholders only own what's left after lenders are paid off, so subtracting the market value of outstanding debt from firm value isolates the value belonging to equity shareholders.

 *Tip: A common trap is forgetting to subtract debt after computing firm value via FCFF — the FCFF formula alone gives firm value, not equity value or share price.*

4.18 P/E Ratio-based Valuation

★★ Frequently Asked

$$P_0 = EPS \times \frac{P}{E}$$

Where:

P_0 estimated value per share

EPS expected earnings per share, usually next year's forecast figure

P/E the price-to-earnings multiple considered appropriate for this company – often an industry average, a comparable company's multiple, or a "justified" P/E derived from the Gordon model

Build-up logic:

1. Instead of forecasting distant dividends or cash flows, this approach asks a simpler, market-based question: what multiple of earnings do similar companies currently trade at?
2. If comparable companies of similar risk and growth trade at, say, 15 times earnings, this company's shares should also reasonably be worth around 15 times its own earnings, once those earnings are known or forecast.
3. So simply multiplying expected EPS by this benchmark P/E multiple gives an estimated fair value per share – a quick, market-relative cross-check rather than a from-scratch DCF valuation.
4. This "justified P/E" can itself be derived from the Gordon Growth Model – dividing both sides of $P_0 = D_1 / (K_e - g)$ by EPS gives $P/E = \text{Payout Ratio} / (K_e - g)$, tying relative valuation back to fundamentals.

 *Tip: A P/E-based value is only as good as the comparable chosen – always check the benchmark company or industry has genuinely similar risk and growth, or the answer misleads.*

4.19 PEG Ratio

★ Good to Know

$$PEG = \frac{P/E \text{ Ratio}}{\text{Expected Annual Growth Rate}}$$

Where:

P/E Ratio the company's current price-to-earnings multiple

Expected Annual Growth Rate expected growth rate of the company's earnings, entered as a plain number (20% growth is entered as 20, not 0.20)

Build-up logic:

1. A high P/E alone doesn't say whether a share is "expensive" — a fast-growing company deserves a higher P/E than a slow-growing one, so comparing raw P/E across companies with different growth rates can mislead.
2. The PEG ratio fixes this by scaling P/E down by the company's own expected growth rate — effectively asking "how much am I paying for each percentage point of growth?"
3. Dividing the P/E ratio by the expected growth rate, as a plain number, gives PEG.
4. As a rule of thumb, a PEG around 1 suggests fair pricing relative to growth; well below 1 hints the share may be undervalued relative to its growth prospects, and well above 1 hints it may be overvalued.

 *Tip: PEG is a rule-of-thumb screening tool, not a precise valuation model — treat "PEG below 1" as an indicative signal in an answer, not a guaranteed conclusion.*

4.20 Walter's Model

★★ Frequently Asked

$$P = \frac{D + \frac{r}{K_e}(E - D)}{K_e}$$

Where:

P market price per share

D dividend per share

E earnings per share

r firm's internal rate of return on its investments (return earned on retained earnings)

K_e cost of equity capital – shareholders' required rate of return

Build-up logic:

1. Walter's Model asks a specific question – does a company's dividend payout policy actually affect its share price? Its answer: yes, and it depends entirely on how r compares with K_e .
2. Total share value splits into two pieces: the dividend actually paid out, D , which shareholders receive directly and can value like a simple perpetuity, D/K_e ; plus the value created by whatever earnings are retained, $(E - D)$.
3. That retained portion is reinvested by the company at its own internal rate of return r , but shareholders still discount its eventual payoff at their own required rate K_e – so its contribution scales the retained amount by the ratio r/K_e , then capitalizes it as a perpetuity by dividing by K_e again.
4. Putting both pieces together and dividing by K_e overall gives Walter's formula, and it directly shows the payout-policy implication: if r is greater than K_e , retaining more (lower D) raises price; if r is less than K_e , paying out more (higher D) raises price; if r equals K_e , payout policy is irrelevant to price.

 Tip: Know the three cases cold – growth firm ($r > K_e$): optimal payout is 0%; declining firm ($r < K_e$): optimal payout is 100%; stable firm ($r = K_e$): optimal payout is any amount.

5

Portfolio Management

How to combine securities into a portfolio, price the risk that's left after diversifying, and judge whether a fund manager actually beat the market.

5.1 Expected Return and Variance of a Security's Return

★ Good to Know

$$E(R) = \sum_{i=1}^n p_i R_i$$

$$\sigma^2 = \sum_{i=1}^n p_i (R_i - E(R))^2$$

Where:

p_i probability of the i^{th} possible economic state/outcome

R_i return the security would give in that state

$E(R)$ expected (probability-weighted average) return of the security

σ^2 variance of the security's return; its square root σ is the standard deviation – the usual risk figure quoted

Build-up logic:

1. A security's future return isn't one fixed number – it depends on which economic scenario actually plays out, and each scenario has some probability attached.
2. To get one usable "average" figure, weight every possible return by how likely it is and add them up – that gives the expected return.
3. Risk is how far actual outcomes can wander from this average. Take each deviation ($R_i - E(R)$), square it so negative and positive swings don't cancel out and bigger swings count more, then weight by probability again.
4. Summing these weighted squared deviations gives variance; its square root brings the units back to plain "% return" – that's standard deviation, the number actually used to compare risk.

Tip: If the question gives equally-likely historical years instead of probabilities, just use $p_i = 1/n$ for each year – the logic is unchanged.

5.2 Expected Return of a Portfolio

★★ Frequently Asked

$$E(R_p) = \sum_{i=1}^n w_i E(R_i)$$

Where:

w_i weight — the proportion of total portfolio funds invested in security i ; all weights add up to 1

$E(R_i)$ expected return of security i

$E(R_p)$ expected return of the whole portfolio

Build-up logic:

1. A portfolio is nothing but several securities held together in fixed proportions — its return in any period is just the money-weighted average of what each security earned.
2. So the expected return of the whole portfolio must also be a weighted average — weight each security's own expected return by how much of your money sits in it.
3. Because the weights already represent the fraction of funds invested, simply summing $w_i E(R_i)$ across every holding gives the portfolio's expected return directly.
4. This is a straight-line (linear) relationship — unlike risk, expected return never gets a "diversification bonus"; only risk does, which is why the next formulas matter so much.

 *Tip: Return is a simple weighted average. Risk is not — that difference is the entire reason portfolio theory exists.*

5.3 Covariance between Two Securities' Returns

★★ Frequently Asked

$$\text{Cov}(R_A, R_B) = \sum_{i=1}^n p_i (R_{A,i} - E(R_A))(R_{B,i} - E(R_B))$$

Where:

$R_{A,i}, R_{B,i}$ returns of securities A and B in state i

$E(R_A), E(R_B)$ expected returns of A and B

$\text{Cov}(R_A, R_B)$ covariance – a measure of how the two securities' returns move together

Build-up logic:

1. The diversification benefit of holding two securities together depends entirely on whether they tend to move together or in opposite directions – we need a number that captures this.
2. In each state, check whether both securities sit above their own average together, or one is above while the other is below, by multiplying their deviations from their own means.
3. A positive product means both moved the same way that period; a negative product means they moved opposite ways; near-zero means little relationship that period.
4. Weight every state's product by its probability and sum them – an overall positive figure means the two generally move together, negative means they offset each other, and a value near zero means little linear relationship.

 *Tip: The sign of covariance matters more than its size – a negative covariance is what actually pulls portfolio risk down.*

5.4 Correlation Coefficient

★★ Frequently Asked

$$\rho_{AB} = \frac{\text{Cov}(R_A, R_B)}{\sigma_A \sigma_B}$$

Where:

$\text{Cov}(R_A, R_B)$ covariance between A and B's returns

σ_A, σ_B standard deviations (own risk) of A and B

ρ_{AB} correlation coefficient between A and B – always between -1 and $+1$

Build-up logic:

1. Covariance's numeric value depends on how volatile each security is, so a covariance of "150" on its own tells you nothing – it can't be compared across different pairs of securities.
2. To fix this, scale covariance down by each security's own standard deviation, which removes the effect of each security's individual volatility.
3. What remains – correlation – is a pure, unit-free number that must lie between -1 (perfectly opposite movement) and $+1$ (perfectly identical movement), with 0 meaning no linear relationship at all.
4. This makes different security pairs directly comparable: any correlation below $+1$ between two securities is what makes combining them reduce risk.

 Tip: Diversification benefit exists as long as $\rho < +1$; it is strongest – risk can fall dramatically – as ρ moves toward -1 .

5.5 Variance and Standard Deviation of a Two-Asset Portfolio

★ MUST KNOW

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \text{Cov}(R_A, R_B)$$

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho_{AB} \sigma_A \sigma_B$$

Where:

w_A, w_B	weights of A and B in the portfolio ($w_A + w_B = 1$)
σ_A^2, σ_B^2	variances of A and B
$\text{Cov}(R_A, R_B) / \rho_{AB}$	covariance or correlation between A and B — either form may be given, both formulas are equivalent
σ_p	portfolio standard deviation, the square root of σ_p^2

Build-up logic:

1. Unlike return, portfolio variance is *not* simply the weighted average of the two securities' own variances — the way the two securities move relative to each other also matters.
2. Each security still contributes its own risk in proportion to how big a share it is of the portfolio — squared, because variance is itself a squared measure: $w_A^2 \sigma_A^2$ and $w_B^2 \sigma_B^2$.
3. But a third piece is needed: how A and B move together. If they move together (positive covariance) total risk is pushed up further; if they offset each other (negative covariance) total risk is pulled down.
4. This interaction is counted twice — the "2" in the formula — because A's effect on B is exactly mirrored by B's effect on A; it is one symmetric cross-effect, fully captured by $2w_A w_B \text{Cov}(R_A, R_B)$.
5. Take the square root of the total variance to bring the risk figure back into the same "%" units as the individual securities' own standard deviations — this is σ_p .

Example: $w_A = 0.6$, $w_B = 0.4$, $\sigma_A = 20\%$, $\sigma_B = 15\%$, $\rho_{AB} = 0.5$. $\text{Cov} = 0.5 \times 20 \times 15 = 150$. $\sigma_p^2 = (0.36 \times 400) + (0.16 \times 225) + 2(0.6)(0.4)(150) = 144 + 36 + 72 = 252$, so $\sigma_p \approx 15.87\%$ — below the weighted-average risk of 18% ($= 0.6 \times 20 + 0.4 \times 15$), which is exactly the diversification benefit at work.

5.6 Portfolio Variance — General (n-Asset) Case

★ Good to Know

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(R_i, R_j)$$

Where:

w_i, w_j weights of securities i and j in the portfolio

$\text{Cov}(R_i, R_j)$ covariance between securities i and j ; when $i = j$, this is just σ_i^2 , the security's own variance

Build-up logic:

1. The two-asset formula had one variance term for each asset plus one covariance term for the single pair — with n assets, every possible pair now matters, not just one.
2. The clean way to capture "every asset's own variance plus every pair's covariance, correctly weighted" is a double summation running both i and j from 1 to n .
3. When $i = j$, the term $w_i w_i \text{Cov}(R_i, R_i)$ is just $w_i^2 \sigma_i^2$ — each security's own variance contribution, exactly as before.
4. When $i \neq j$, every pair is picked up twice in the double sum (once as i, j and once as j, i), which automatically supplies the "x2" that had to be written by hand in the two-asset version.
5. Practical takeaway: as n grows large, the number of covariance terms grows far faster than the number of variance terms — this is exactly why, in a large well-diversified portfolio, average covariance (not any one security's own variance) ends up driving most of total risk.

5.7 Minimum-Variance Portfolio Weights (Two-Asset Case)

★★ Frequently Asked

$$w_A^* = \frac{\sigma_B^2 - \text{Cov}(R_A, R_B)}{\sigma_A^2 + \sigma_B^2 - 2 \text{Cov}(R_A, R_B)}$$
$$w_B^* = 1 - w_A^*$$

Where:

σ_A^2, σ_B^2 variances of A and B

$\text{Cov}(R_A, R_B)$ covariance between A and B

w_A^*, w_B^* the specific weights that minimize portfolio variance — not the weights that maximize return, which is a separate objective

Build-up logic:

1. Combining two imperfectly correlated assets can reduce risk below either asset's own risk — that is diversification's core promise.
2. As weight is gradually shifted from one asset to the other, portfolio variance traces a U-shaped curve: it falls, hits a minimum, then rises again.
3. The bottom of that curve is the "minimum-variance portfolio" — the one specific mix where risk cannot be pushed any lower by shifting weight between just these two assets.
4. Working out exactly where that minimum occurs — by balancing each asset's own variance against its covariance with the other — reduces to the clean ratio shown, using only variances and covariance, with no need for expected returns at all.
5. As always with two assets, $w_B^* = 1 - w_A^*$, since the two weights must add up to the whole portfolio.

Tip: This weight-set minimizes risk only — it says nothing about return. Don't confuse it with the "optimal" portfolio on the efficient frontier, which also weighs expected return.

5.8 Beta of a Security

★ MUST KNOW

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2} = \frac{\rho_{im} \sigma_i}{\sigma_m}$$
$$R_i = \alpha_i + \beta_i R_m + e_i$$

Where:

$\text{Cov}(R_i, R_m)$ covariance between the security's return and the market's return

σ_m^2 variance of the market's return

$\rho_{im}, \sigma_i, \sigma_m$ correlation between security and market, and their respective standard deviations – an equivalent way to write the same beta

R_i, R_m period returns of the security and the market (this is the "characteristic line" regression)

α_i intercept – the return the security earns independent of the market

e_i residual/error term – firm-specific (unsystematic) return not explained by the market

Build-up logic:

1. Not all of a security's risk matters to a well-diversified investor – only the part that moves with the overall market survives after diversification, so a number is needed that isolates just that part.
2. Beta is built exactly like any sensitivity measure: covariance between the security and the market, scaled down by the market's own variance – this scaling turns raw covariance into "how many times more or less volatile than the market," which is easy to interpret.
3. The same idea appears if you plot the security's period returns (Y-axis) against the market's returns for the same periods (X-axis) and fit a straight line – the characteristic line. Its slope is precisely beta.
4. The intercept α_i of that line is the return earned even when the market's return is zero, and the scatter of points around the line (e_i) is firm-specific noise that diversification eventually cancels out.
5. So beta as "market sensitivity" can be computed two equivalent ways – directly from covariance/variance, or as the slope of a return-on-return regression – both give the identical number.

Tip: $\beta = 1$ moves with the market, $\beta > 1$ is more aggressive/volatile than the market, $\beta < 1$ is defensive, and $\beta = 0$ means no relationship with the market at all.

5.9 Beta of a Portfolio

★ Good to Know

$$\beta_p = \sum_{i=1}^n w_i \beta_i$$

Where:

w_i weight of security i in the portfolio

β_i beta of security i

β_p beta of the whole portfolio

Build-up logic:

1. A portfolio's systematic risk is nothing but the combined systematic risk of everything it holds.
2. Beta is already a linear sensitivity measure with no cross-terms to worry about (unlike variance), so the portfolio's beta is simply the weighted average of the individual securities' betas.
3. This is exactly why adding even one high-beta stock to an otherwise low-beta portfolio noticeably raises the whole portfolio's market sensitivity — beta gets no "diversification discount" the way variance does.

5.10 Capital Asset Pricing Model (CAPM)

★ MUST KNOW

$$K_e = R_f + \beta (R_m - R_f)$$

Where:

R_f	risk-free rate of return
R_m	expected return of the market portfolio
$(R_m - R_f)$	market risk premium – the extra return investors demand for bearing market risk
β	the security's systematic risk relative to the market
K_e	required/expected rate of return on the security

Build-up logic:

1. Every investor must be compensated at least at the risk-free rate simply for parking money over time – that is the base return, R_f .
2. On top of that, bearing market risk deserves extra reward – the market as a whole earns a premium of $(R_m - R_f)$ over the risk-free rate for carrying that risk.
3. But one specific security doesn't carry the "average" amount of market risk – it carries β times as much, so it should earn only β times the market risk premium, not the full premium.
4. Adding this scaled risk premium back onto the risk-free base gives the fair required return for that specific security – CAPM's central equation.
5. Crucially, unsystematic (firm-specific) risk earns no reward here – CAPM assumes it is fully diversified away, so only β (systematic risk) is priced into required return.

 Tip: CAPM prices only systematic risk (β). Total risk (σ) is irrelevant to the CAPM required return – a favourite trap in theory/MCQ questions.

Security Market Line (SML)

The SML is simply the CAPM equation drawn as a graph – beta (systematic risk) on the X-axis and required/expected return on the Y-axis, giving an upward-sloping straight line that starts at R_f (where beta is zero) and rises with a slope equal to the market risk premium $(R_m - R_f)$. In equilibrium every security, however risky or safe, should plot exactly on this line, because CAPM says required return depends only on beta. A security plotting above the SML is offering more return than its beta justifies – undervalued, a buy – while one plotting below is overpriced relative to the risk it carries – a sell. The SML applies equally to individual securities and to portfolios, since beta is defined for both.

5.11 Capital Market Line (CML)

★ Good to Know

$$E(R_p) = R_f + \frac{E(R_m) - R_f}{\sigma_m} \sigma_p$$

Where:

R_f risk-free rate

$E(R_m)$ expected return of the market portfolio

σ_m standard deviation (total risk) of the market portfolio

σ_p standard deviation (total risk) of the efficient portfolio being priced

$E(R_p)$ required/expected return of that efficient portfolio

Build-up logic:

1. Combine the risk-free asset with the market portfolio in different proportions, and every possible combination plots on one straight line running from R_f (all risk-free) through the market portfolio (all risky) – this line is the CML.
2. The line's slope, $\frac{E(R_m) - R_f}{\sigma_m}$, is the "market price of risk" – exactly how much extra expected return is rewarded for each extra unit of total risk (σ) taken on.
3. Since the relationship is a straight line, any efficient combination of R_f and the market portfolio must earn the risk-free rate plus that slope multiplied by however much total risk (σ_p) it carries.
4. Key restriction: this only holds for fully-diversified, efficient portfolios lying on this line – it cannot price a single security or an undiversified portfolio, since those still carry unsystematic risk that the market does not reward.

 Tip: CML uses total risk (σ) and applies only to efficient portfolios; SML uses systematic risk (β) and applies to any security or portfolio at all. This contrast is a classic theory question.

5.12 Sharpe's Single Index Model — Risk Decomposition

★★ Frequently Asked

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{ei}^2$$

Where:

- σ_i^2 total risk (variance) of the security's return
 $\beta_i^2 \sigma_m^2$ systematic risk — the portion of total risk caused by market-wide movements
 σ_{ei}^2 unsystematic (residual/firm-specific) risk — unrelated to the market

Build-up logic:

1. Sharpe's Single Index Model says a security's return is driven by one common factor, the market, plus a random firm-specific piece: $R_i = \alpha_i + \beta_i R_m + e_i$ — the same characteristic line seen earlier.
2. Because the market-driven piece and the firm-specific piece e_i are assumed statistically independent (the model's key simplifying assumption), the variance of their sum splits cleanly into the sum of their separate variances — no cross-term needed.
3. The market-driven piece has variance $\beta_i^2 \sigma_m^2$ — systematic risk, present in every security to some degree and impossible to diversify away.
4. The leftover piece, σ_{ei}^2 , is unsystematic risk caused by firm-specific events (a strike, a lawsuit, a product launch) unrelated to the market.
5. As more securities are added to a portfolio, their unsystematic pieces are unrelated to each other and tend to cancel out on average, so σ_{ei}^2 shrinks toward zero — but $\beta^2 \sigma_m^2$ never disappears, since every security is tied to the same market.

 *Tip: This is exactly why "total risk" and "priced (rewarded) risk" differ — the market only compensates the $\beta^2 \sigma_m^2$ part, never σ_{ei}^2 .*

5.13 Arbitrage Pricing Theory (APT)

★ Good to Know

$$E(R) = R_f + \beta_1(RP_1) + \beta_2(RP_2) + \dots + \beta_k(RP_k)$$

Where:

R_f	risk-free rate — the base return when all factor sensitivities are zero
$\beta_1, \beta_2, \dots, \beta_k$	the security's sensitivity to each of k macroeconomic risk factors (inflation, interest rates, GDP growth, etc.)
$RP_1, RP_2, \dots, RP_k$	the risk premium associated with each corresponding factor
$E(R)$	the security's expected/required return under APT

Build-up logic:

1. CAPM assumes only one systematic risk factor matters — the market as a whole — and packs everything into a single beta.
2. In reality, several distinct macroeconomic forces (inflation shocks, interest-rate changes, industrial growth, exchange-rate moves) can each systematically move returns, independently of one another.
3. APT simply generalizes CAPM's logic to multiple factors: a security earns a separate risk premium for its sensitivity to each factor, instead of just one market risk premium.
4. Add up the required compensation for every factor's exposure (that factor's sensitivity $\times$ that factor's own risk premium), and stack this on top of the risk-free base — exactly the way CAPM adds just one such term.
5. Unlike CAPM, APT never needs to identify "the market portfolio" or assume any particular return distribution — it only requires that returns be explainable by a set of common factors, making it more flexible (though the factors themselves must be identified empirically).

 *Tip: Think of APT as "CAPM with more than one beta" — set $k = 1$ with the market as the sole factor, and APT collapses back into CAPM.*

5.14 Sharpe Ratio

★ MUST KNOW

$$S_p = \frac{R_p - R_f}{\sigma_p}$$

Where:

R_p actual (realised) return of the portfolio

R_f risk-free rate

σ_p total risk (standard deviation) of the portfolio's return

S_p Sharpe Ratio of the portfolio

Build-up logic:

1. A manager's raw return means little without knowing how much risk was taken to earn it – two portfolios returning 15% are not equally good if one is twice as volatile.
2. Start by isolating the reward earned above the safe alternative: excess return $= R_p - R_f$.
3. Divide this excess return by the total risk taken to earn it, σ_p – this converts the reward into "excess return earned per unit of total risk carried."
4. Because it uses total risk, not just systematic risk, the Sharpe ratio is the right measure whenever the portfolio being judged represents an investor's *entire* wealth – since then all risk, systematic and unsystematic, matters to that investor.

5.15 Treynor Ratio

★ MUST KNOW

$$T_p = \frac{R_p - R_f}{\beta_p}$$

Where:

R_p actual (realised) return of the portfolio

R_f risk-free rate

β_p systematic risk (beta) of the portfolio

T_p Treynor Ratio of the portfolio

Build-up logic:

1. Treynor's ratio asks the same question as Sharpe's — reward per unit of risk — but assumes the investor already holds many other investments, so this one portfolio's unsystematic risk will be diversified away at the overall wealth level anyway.
2. So instead of dividing excess return by total risk (σ), divide it by only the risk that actually matters to a diversified investor: systematic risk, β_p .
3. The result again measures "excess return per unit of risk," but now per unit of the one risk that cannot be diversified away — market risk.
4. A higher Treynor ratio means the manager generated more return for each unit of unavoidable systematic risk taken on.

 *Tip: Use Sharpe when the portfolio is the investor's whole/only holding (total risk matters); use Treynor when it is just one of several holdings in an already-diversified wealth (only systematic risk matters).*

5.16 Jensen's Alpha

★★ Frequently Asked

$$\alpha_p = (R_p - R_f) - \beta_p (R_m - R_f)$$

Where:

R_p	actual (realised) return of the portfolio
R_f	risk-free rate
$\beta_p(R_m - R_f)$	the systematic-risk-adjusted excess return CAPM says the portfolio should have earned
α_p	Jensen's alpha — the manager's value added over and above what CAPM predicts

Build-up logic:

1. CAPM tells us exactly what excess return a portfolio with a given beta *should* have earned — the fair, risk-adjusted benchmark.
2. Compare the portfolio's actual excess return, $R_p - R_f$, against this CAPM-predicted excess return, $\beta_p(R_m - R_f)$.
3. The gap between the two is Jensen's alpha — positive means the manager beat the return that their systematic risk alone would justify (genuine skill), negative means they underperformed even after accounting for the risk taken.
4. Unlike Sharpe and Treynor, which are ratios ("per unit of risk"), alpha is an absolute number in % return — it directly answers "did the manager add value, and how much?"

 *Tip: Use Jensen's alpha to judge stock-selection/manager skill in absolute % terms; like Treynor, it uses β , so it too assumes a reasonably diversified portfolio.*

5.17 Constant Proportion Portfolio Insurance (CPPI)

★ Good to Know

$$\text{Exposure (Equity)} = \text{Multiplier} \times (\text{Portfolio Value} - \text{Floor Value})$$

Where:

Portfolio Value	current total value of the portfolio
Floor Value	the minimum value the investor will not allow the portfolio to fall below (often grows over time at the risk-free rate)
(Portfolio Value – Floor Value)	the "cushion" – how much value can be put at risk before hitting the floor
Multiplier	a fixed constant applied to the cushion, usually set as $1 \div$ (maximum one-period fall in the risky asset the strategy is designed to survive)
Exposure	the amount that should actually be invested in the risky asset (equity); the remainder goes into the safe asset

Build-up logic:

1. Portfolio insurance aims to protect against falling below some minimum acceptable value (the Floor) while still keeping upside from risky assets – CPPI does this by continuously adjusting the equity/safe-asset split.
2. Only the "cushion" – value sitting above the floor – is truly available to risk; the floor itself must stay protected in the safe asset.
3. Instead of investing the cushion rupee-for-rupee in equity, CPPI scales it up by a fixed Multiplier – a higher multiplier means a more aggressive strategy that raises equity exposure faster as the cushion grows.
4. This produces the strategy's defining, self-reinforcing behaviour: as the portfolio rises, the cushion grows, so more moves into equity (buy high); as the portfolio falls, the cushion shrinks, so equity is sold into the safe asset (sell low) – protecting the floor.
5. Because Exposure must be recalculated and rebalanced every time Portfolio Value changes, CPPI is a dynamically (actively) rebalanced strategy, not a fixed-weight one.

 *Tip: A higher multiplier chases more upside but hits the floor faster in a sharp fall – that's the direct trade-off in choosing it.*

Buy-and-Hold vs Constant-Mix Rebalancing

A Buy-and-Hold strategy fixes the rupee amount invested in the risky asset at the start and never rebalances – as equity rises, its share of the portfolio automatically grows, needs no action, but offers no active protection if equity falls sharply. A Constant-Mix strategy instead keeps the *proportion* in equity fixed – say, always 60% equity and 40% debt – which forces continuous rebalancing: selling equity after it rises and buying more after it falls, to restore the ratio. This makes Constant-Mix a contrarian, "buy low, sell high" approach that does well in flat, oscillating markets but underperforms Buy-and-Hold in a strongly trending market, up or down.

CPPI vs the Other Two Strategies

CPPI also rebalances actively, like Constant-Mix, but trades in the opposite direction. Since CPPI's equity exposure rises as the cushion (Portfolio Value minus Floor Value) grows and falls as the cushion shrinks, it buys equity after prices rise and sells after prices fall – a trend-following, "buy high, sell low" approach. This gives CPPI a payoff that participates strongly in a sustained bull market while still protecting the floor, but it can underperform in a flat, choppy market, where it keeps buying at local highs and selling at local lows (a repeated whipsaw cost sometimes called the "cost of insurance"). In short: Buy-and-Hold is passive, Constant-Mix is contrarian, and CPPI is dynamic and trend-following.

6

Securitization

How a pool of loans sitting on a lender's books gets converted into tradeable securities – and how investors are protected along the way.

Securitization — Meaning, Process and Parties Involved

Securitization converts a pool of illiquid, individually small receivables — housing loans, car loans, credit card dues — sitting on a lender's books into tradeable securities that can be sold to investors. The process starts with the Originator (the bank/NBFC that created these loans) selling or assigning the receivables pool to a Special Purpose Vehicle (SPV), a separate, bankruptcy-remote entity set up only to hold this pool and issue securities against it, so investors are insulated from the Originator's own credit risk. A Credit Rating Agency rates the securities issued by the SPV based on the quality of the underlying pool and any credit enhancement provided. Investors buy these rated securities, and as the underlying borrowers — the Obligors — repay their loans, those collections flow from the Obligors through the SPV (usually via a Servicer, often the Originator itself continuing to collect instalments) and out to the Investors as interest and principal.

Why Securitize? (Benefits)

For the Originator, securitization turns a slow-moving, illiquid loan book into immediate cash — improving liquidity, freeing up capital for fresh lending, and moving the asset (and its risk) off the balance sheet, which also improves capital adequacy for banks and NBFCs. It lets the Originator raise funds without taking on fresh debt or diluting equity. For Investors, it opens up a new asset class of asset-backed paper that can offer a better yield than similarly-rated bonds, and can be structured into tranches to suit different risk appetites. For the financial system as a whole, it spreads credit risk more widely instead of concentrating it inside a few lenders' books.

Pass-Through Certificates vs Pay-Through Securities vs Stripped Securities

In a Pass-Through Certificate (PTC), the SPV simply passes on the pool's cash flows — interest and principal, net of servicing fees — to investors exactly as collected, pro-rata; investors directly own a share of the underlying pool and bear its prepayment risk. A Pay-Through Security is different: it is a debt obligation of the SPV itself, backed by (not a direct ownership share of) the receivables pool, which lets the issuer restructure the pool's cash flows into multiple bonds or tranches of different maturities and coupons instead of passing them through untouched. Stripped securities go a step further by splitting the pool's interest and principal streams into two entirely separate instruments — an Interest-Only (IO) strip, which receives only the interest portion of every payment, and a Principal-Only (PO) strip, which receives only the principal portion — each attracting different investors depending on their view on prepayment speed and interest rates.

6.1 Weighted Average Coupon (WAC)

★★ Frequently Asked

$$WAC = \sum_{i=1}^n \left(\frac{\text{Principal}_i}{\text{Total Pool Principal}} \right) \times \text{Coupon Rate}_i$$

Where:

Principal_i outstanding principal of loan/receivable *i* in the pool

Total Pool Principal sum of outstanding principal across every loan in the pool

Coupon Rate_i interest rate charged on loan *i*

Build-up logic:

1. A securitized pool is made up of many individual loans, each carrying its own interest rate — there is no single "the" interest rate for the pool unless one is constructed.
2. The fair way to combine many different rates into one representative figure is to weight each loan's coupon by how much of the total pool it represents — a large loan should count for more than a tiny one.
3. So each loan's weight is simply its principal divided by the total pool principal, and WAC sums (weight × that loan's coupon rate) across every loan in the pool.
4. WAC tells investors the effective average interest income the pool generates — which is what ultimately funds the coupon paid out on the securities issued against it.

6.2 Weighted Average Maturity (WAM)

★★ Frequently Asked

$$WAM = \sum_{i=1}^n \left(\frac{\text{Principal}_i}{\text{Total Pool Principal}} \right) \times \text{Remaining Maturity}_i$$

Where:

Principal _{<i>i</i>}	outstanding principal of loan <i>i</i>
Total Pool Principal	sum of outstanding principal across the whole pool
Remaining Maturity _{<i>i</i>}	months/years left to maturity on loan <i>i</i>

Build-up logic:

1. Just as loans in the pool differ in coupon rate, they also differ in how many months or years of repayments remain – some are almost paid off, others were disbursed recently.
2. To get one representative maturity figure for the whole pool, again weight each loan's remaining maturity by its share of total pool principal, so bigger loans influence the average more.
3. Summing (weight × remaining maturity) across all loans gives WAM – the effective average time left for the pool's cash flows to be fully collected.
4. WAM matters to investors because it drives how long the securities backed by this pool are expected to keep paying – the securitization world's equivalent of a bond's average life.

Credit Enhancement — Internal vs External

Credit enhancement improves the credit quality of the securities issued by the SPV, giving investors extra comfort beyond the raw quality of the underlying pool. Internal credit enhancement is built into the structure itself – for example, over-collateralization, where the pool of receivables backing the securities is worth more than the securities issued, so the extra cushion absorbs early defaults first, or a cash collateral/reserve account funded upfront to cover shortfalls. External credit enhancement instead comes from a third party outside the pool, such as a bank guarantee, or insurance/a surety bond from an insurer, who agrees to make good any shortfall in collections up to an agreed limit. A higher degree of credit enhancement generally supports a higher rating (and lower yield) on the securities, since investors are better cushioned against the pool underperforming.

6.3 Degree of Credit Enhancement

★ Good to Know

$$\text{Degree of Credit Enhancement} = \frac{\text{Credit Enhancement Amount}}{\text{Total Value of the Securitized Pool}}$$

Where:

Credit Enhancement Amount rupee value of the protection provided — e.g. the over-collateralization amount, cash reserve, or guarantee amount

Total Value of the Securitized Pool total value of the receivables being securitized

Build-up logic:

1. Credit enhancement exists to absorb losses before investors take any hit — but "how much protection" only means something when expressed relative to the size of the pool being protected.
2. So express the rupee amount of credit enhancement as a percentage of the total pool value — this gives one comparable figure regardless of the deal's absolute size.
3. A higher degree of credit enhancement means investors are cushioned against a larger proportion of pool losses before bearing any loss themselves — exactly why it supports a higher credit rating on the securities issued.

How to value a mutual fund unit, see what an investor actually earned after costs, and judge a fund manager's performance.

7.1 Net Asset Value (NAV)

★ **MUST KNOW**

$$NAV = \frac{\text{Total Assets} - \text{Total Liabilities}}{\text{No. of Units Outstanding}}$$

Where:

Total Assets	market value of all securities held by the fund, plus cash/receivables
Total Liabilities	amounts owed by the fund — expenses payable, fees due, etc.
No. of Units Outstanding	total units issued to investors and not yet redeemed

Build-up logic:

1. A mutual fund pools money from many investors into one common kitty invested in securities, so first value everything the fund owns at current market prices — this gives Total Assets.
2. The fund also owes some amounts, like unpaid expenses and fees payable — subtract Total Liabilities to leave the net value that truly belongs to the unit-holders.
3. This net value belongs jointly to every unit outstanding, so divide it by the number of units outstanding to find the value of exactly one unit — the NAV.

 *Tip: NAV is always a per-unit figure — never confuse the size of the whole scheme (AUM) with NAV itself.*

7.2 Holding Period Return (HPR) on a Mutual Fund

★ MUST KNOW

$$HPR = \frac{(NAV_1 - NAV_0) + D}{NAV_0}$$

Where:

NAV_0 NAV at the start of the holding period (purchase NAV)

NAV_1 NAV at the end of the holding period (sale/current NAV)

D dividend/income distributed per unit during the holding period

Build-up logic:

1. An investor in a mutual fund unit earns in two ways — the unit's value can change (capital gain or loss), and the fund may pay out dividends along the way — both must be counted for total return.
2. Capital gain per unit is simply the change in NAV over the period: $NAV_1 - NAV_0$.
3. Add any dividend per unit, D , received during the holding period — this is income earned in addition to the price change.
4. Divide this total gain (price change plus dividend) by the NAV you started with, NAV_0 , to express the return as a percentage of the original investment — exactly like a holding period return on any other investment.

 *Tip: If NAV falls but the dividend paid more than compensates, HPR can still come out positive — always add both pieces before judging performance.*

7.3 Effect of Entry Load and Exit Load

★ Good to Know

$$\begin{aligned}\text{Purchase Price} &= NAV \times (1 + \text{Entry Load}) \\ \text{Redemption Price} &= NAV \times (1 - \text{Exit Load})\end{aligned}$$

Where:

NAV	the fund's Net Asset Value on the transaction date, before adjusting for load
Entry Load	the % charged by the fund at purchase, added on top of NAV
Exit Load	the % charged by the fund at redemption, deducted from NAV
Purchase Price	the effective price actually paid per unit when buying
Redemption Price	the effective price actually received per unit when selling

Build-up logic:

1. NAV alone tells you what a unit is intrinsically worth — but what an investor actually pays or receives can differ if the fund charges a load for entering or exiting.
2. An entry load is a charge added on top of NAV when buying, so the price actually paid per unit rises above NAV by that load percentage.
3. An exit load works the other way — a charge deducted from NAV at redemption, so the price actually received per unit falls below NAV by that load percentage.
4. Effect on return: an entry load raises the investor's effective cost base (use the loaded Purchase Price, not raw NAV, as NAV_0 in the HPR formula), while an exit load reduces what is actually received at the end (use Redemption Price as NAV_1) — both loads directly pull down the investor's real return compared to a no-load scheme.

 *Tip: SEBI abolished entry loads on Indian mutual funds with effect from August 2009 — if a question mentions only an exit load, do not assume an entry load also applies unless it is explicitly stated.*

Example: NAV = ₹50, Entry Load = 2%, Exit Load = 1%. Purchase Price = ₹50 × 1.02 = ₹51. If later redeemed at NAV = ₹55, Redemption Price = ₹55 × 0.99 = ₹54.45. The investor's return is based on the ₹51 actually paid and the ₹54.45 actually received — not on the raw NAVs of ₹50 and ₹55.

7.4 Expense Ratio

★ Good to Know

$$\text{Expense Ratio} = \frac{\text{Total Expenses of the Fund}}{\text{Average Net Assets of the Fund}} \times 100$$

Where:

Total Expenses all recurring costs of running the scheme over the period – management fees, admin costs, distribution expenses, etc.

Average Net Assets average value of the fund's net assets (AUM) over the same period

Build-up logic:

1. Running a mutual fund costs money – management fees, administrative costs, distribution expenses – and these are paid out of the fund's own assets, quietly reducing what investors actually earn.
2. To see how heavy this cost burden is relative to the fund's size, express total expenses as a percentage of the fund's average net assets rather than as a plain rupee figure.
3. This ratio lets investors compare cost-efficiency across funds of very different sizes – a lower ratio means more of the fund's returns reach the investor instead of being absorbed by running costs.
4. Expense ratio directly reduces investor return: a fund's return net of expenses is always lower than its gross portfolio return by roughly this ratio.

 *Tip: Unlike a load (a one-time charge), the expense ratio is charged every single year the investment is held – a small-looking difference compounds a lot over the long term.*

7.5 Sharpe Ratio (for Mutual Fund Evaluation)

★★ Frequently Asked

$$S_p = \frac{R_p - R_f}{\sigma_p}$$

Where:

R_p the fund's actual return over the period

R_f risk-free rate

σ_p standard deviation of the fund's returns (its total risk)

Build-up logic:

1. A fund's return alone doesn't say whether the manager took excessive risk to get there — return needs to be scaled by the volatility endured to earn it.
2. Subtract the risk-free rate from the fund's return to isolate the extra reward earned purely for investing in the fund rather than a safe asset: $R_p - R_f$.
3. Divide by the fund's own standard deviation, σ_p , its total risk — appropriate here because a retail investor typically holds this one fund as most or all of that money, not as one small slice of a much larger diversified book.
4. A higher Sharpe ratio means the fund delivered more return per unit of total risk taken — the natural first check when comparing funds with different risk profiles.

 *Tip: Sharpe is the go-to ratio for ranking ordinary mutual fund investments, since most investors don't hold a given fund as just one small piece of a much bigger diversified portfolio.*

7.6 Treynor Ratio (for Mutual Fund Evaluation)

★★ Frequently Asked

$$T_p = \frac{R_p - R_f}{\beta_p}$$

Where:

R_p the fund's actual return over the period

R_f risk-free rate

β_p beta of the fund relative to its benchmark/the market

Build-up logic:

1. If an investor already holds this fund as just one part of a larger, well-diversified portfolio, the fund's own unsystematic risk gets diversified away at the overall portfolio level, so it shouldn't be charged against this fund's score.
2. So instead of dividing excess return by total risk, divide by the fund's beta, β_p – the systematic risk that actually survives diversification.
3. This gives excess return earned per unit of market risk taken by the fund, letting investors compare funds purely on how well they used their exposure to market risk.
4. A manager who took on high beta merely to chase returns, without real skill, will show a weaker Treynor ratio than one who earned the same return with lower beta.

 *Tip: Use Treynor instead of Sharpe when the fund being judged is only one holding within a bigger, already-diversified investment portfolio.*

Tracking Error (Index Funds)

An index fund is designed simply to replicate a benchmark index, not beat it, so it needs its own way of being judged. Tracking error measures how closely the fund's actual returns follow its benchmark's returns – technically, the standard deviation of the difference between the fund's period returns and the index's period returns. A well-run index fund should show a very low tracking error, since large deviations mean it is failing at its one job of mirroring the index, due to reasons such as expenses, cash holdings, imperfect replication, or transaction costs. Unlike Sharpe, Treynor, or Jensen's alpha, tracking error is not about earning a superior risk-adjusted return – for an index fund, a lower tracking error is the sign of a better-managed fund, not a higher one.

8

Derivatives Analysis and Valuation

Pricing and valuing forwards, futures, options and swaps – nearly every formula here rests on one idea: two positions with identical future payoffs must cost exactly the same today.

Forwards & Futures

Forward vs Futures Contracts

A forward is a private, customised agreement between two parties to buy/sell an asset at a fixed price on a future date – negotiated directly (over-the-counter), so quantity, quality and delivery date can all be tailored, but it carries real counterparty (default) risk since no one guarantees performance. A futures contract is the same basic idea made standardised and exchange-traded – fixed lot sizes and expiry dates, with the exchange's clearing house stepping in as counterparty to every trade, which all but removes default risk. The key working difference is timing of settlement: a forward pays out its entire gain or loss only at maturity, while a futures position is "marked to market" and cash-settled every single day. Because futures are standardised and exchange-traded, they're also far easier to exit early – just take an opposite position – whereas a forward usually has to be held to maturity or unwound directly with the same counterparty.

Margins and Marking-to-Market

To open a futures position, both the buyer and the seller deposit an Initial Margin with the broker – a small percentage of contract value acting as a good-faith deposit against future losses, not a payment towards the asset itself. Every trading day, the exchange "marks to market": it revalues every open position at that day's closing futures price, credits the day's gain to whichever side made money, and debits the day's loss from the other side – settling profit/loss in cash daily instead of waiting for expiry. If losses pull an account's balance below a set Maintenance Margin level, the holder gets a margin call and must top it back up to the Initial Margin (paying "Variation Margin"), or the broker closes out the position. This daily cash settlement is exactly what makes futures far safer than forwards – losses can never quietly build up unnoticed over the life of the contract.

8.1 Cost of Carry Model (Forward/Futures Price)

★ MUST KNOW

$$F = S e^{rt} \quad F = S(1 + r)^t \quad (\text{continuous-compounding equivalent})$$

Where:

F fair/theoretical forward or futures price today

S current spot price of the underlying asset

r risk-free rate of interest per annum (as a decimal)

t time to maturity of the contract, in years

Build-up logic:

1. There are two ways to own the asset at time t : buy the forward/futures contract now and pay F only at maturity, or simply buy the asset today in the spot market for S and carry (hold) it until then.
2. Both routes must work out to the same cost today – if one were cheaper, everyone would rush to it, buying the cheap route and selling the expensive one until prices are forced back in line. This "no free lunch" idea is the entire basis of the model.
3. Buying spot today ties up S worth of cash right now; buying forward lets that same cash stay invested elsewhere earning the risk-free rate r until maturity – so the forward price must sit above spot by exactly the interest saved by delaying payment.
4. With no income or storage cost involved, that forgone/financing interest is the entire "cost of carrying" the asset – so S simply gets compounded forward at r for the holding period t , giving $F = S(1 + r)^t$.
5. If interest is instead compounded continuously rather than annually, the exact same idea rearranges to $F = S e^{rt}$ – same logic, different compounding convention.

 *Tip: Use whichever compounding the question specifies. ICAI numericals almost always intend plain annual/discrete compounding $(1 + r)^t$ unless "continuous compounding" is explicitly mentioned.*

8.2 Cost of Carry — Adjustment for Dividend Yield / Storage Cost

★★ Frequently Asked

$$F = S(1 + r + c - q)^t$$
$$F = (S - PV(D))(1 + r)^t$$

Where:

- S , r , t spot price, risk-free rate and time to maturity, as in 8.1
- c storage/insurance (carrying) cost of holding the asset, expressed as a % p.a. of asset value — relevant mainly for commodities
- q dividend yield or income earned while holding the asset, expressed as a % p.a. — relevant for shares/index
- $PV(D)$ present value of a known, specific dividend/income amount D expected to be received during the contract's life (used when a rupee figure is given instead of a yield %)

Build-up logic:

1. The basic model assumed holding the asset costs and earns nothing extra — but real assets often do one or the other: commodities need storage/insurance (an added cost), while shares and indices pay dividends (added income) during the wait.
2. An extra holding cost is one more expense the forward buyer avoids by not owning the asset yet, so it must be added on top of the financing cost — this is the "+ c " inside the rate, pushing F up.
3. Extra holding income works in reverse: the forward buyer misses out on dividends the actual owner collects while waiting, so the seller must give up part of that benefit by quoting a lower forward price — this is the " $-q$ ".
4. When the dividend is one specific rupee amount D rather than a yield %, the cleaner fix is to strip its present value straight out of today's spot price first — since that cash flow never belongs to the forward buyer — then compound only the remaining "ex-dividend" value forward at the plain rate r .
5. Both versions carry the same underlying idea: only compound forward the part of today's spot value that the forward buyer actually gets full exposure to.

 *Tip: The most common slip here is a sign error — storage cost ADDS to the cost of carry (raises F), dividend/income yield SUBTRACTS from it (lowers F). Mixing the two up is the #1 mistake examiners catch.*

8.3 Value of an Existing Forward Contract

★★ Frequently Asked

$$V_{long} = \frac{F_t - K}{(1 + r)^{T-t}}$$
$$V_{short} = \frac{K - F_t}{(1 + r)^{T-t}} = -V_{long}$$

Where:

V_{long} mark-to-market value today of the contract to the party who is long (agreed to buy)

F_t the fair forward price today, for a brand-new contract maturing on the same date T – recomputed using the cost-of-carry model with today's spot price

K the delivery price locked in when this contract was originally entered into – fixed for life, never changes

r risk-free rate

$T - t$ remaining time to maturity, in years, from today onward

V_{short} mark-to-market value today to the party who is short (agreed to sell) – always the exact negative of V_{long}

Build-up logic:

1. At the moment a forward is entered into, it's fair to both sides by construction – its value is ₹0, and the agreed delivery price K exactly equals the fair forward price at that time.
2. Time passes and the spot price moves, so a brand-new forward for the same remaining maturity would now trade at a new fair price F_t , generally different from the old locked-in K .
3. The long side is still contractually entitled to buy at the old K , while the "going rate" for the same future purchase is now F_t – if $F_t > K$, the long holder is locked into buying below the going rate, a paper gain worth $F_t - K$.
4. That gain is only realised at maturity, so it must be discounted back to today at the risk-free rate over the remaining life $T - t$.
5. The short side's position is the exact mirror image – whatever the long side gains, the short side loses by the same amount – so its value is simply the negative of the long's.
6. Sanity check: right at maturity $T - t = 0$ and F_t collapses to the spot price then, so the formula correctly reduces to the familiar final payoff, $S_T - K$, for the long holder.

 *Tip: If a question asks for the "gain/loss on an existing forward position" before maturity, this is exactly what it wants – don't confuse it with the final payoff $S_T - K$, which only applies once $T - t = 0$.*

8.4 Basis and Convergence

★ Good to Know

$$\text{Basis} = S - F$$

Where:

S current spot price of the underlying

F current futures price of the same underlying, for the relevant contract

Basis the gap between the two – can be positive or negative

Build-up logic:

1. A futures price is really just spot price plus the cost of carry for the remaining time to maturity – so the size of the gap between them (the basis) is driven mainly by how much time is left and the cost-of-carry rate.
2. As the contract's remaining life shrinks day by day, there is less and less time left for cost of carry to build up, so the gap between futures and spot must keep shrinking too.
3. Right at expiry there is no time left at all – cost of carry is zero – so the futures price must exactly equal the spot price. This forced equality is called convergence, and basis becomes zero.
4. If the two were still apart at expiry, a costless arbitrage would exist (buy the cheaper one, sell the dearer one, settle instantly) – so convergence isn't a coincidence, it's enforced by the same no-arbitrage logic behind the cost-of-carry model itself.
5. Before expiry, though, basis can move somewhat unpredictably as time passes – this residual uncertainty is called basis risk, and it's exactly why even a well-calculated hedge can leave a small profit or loss if unwound before the futures contract's own expiry.

 *Tip: A hedge removes price risk but not basis risk – some basis risk always remains unless the hedge is held right until the futures contract expires.*

8.5 Hedge Ratio — Number of Futures Contracts

★★ Frequently Asked

$$N = \frac{\beta_p \times V_p}{V_F} \quad (\text{to fully hedge: target beta of zero})$$
$$N = \frac{(\beta_{\text{target}} - \beta_p) \times V_p}{V_F} \quad (\text{to reach any target beta})$$

Where:

- N number of index futures contracts to buy (+) or sell (-)
- β_p current beta (systematic risk) of the portfolio
- V_p current market value of the portfolio
- V_F value of one futures contract = futures index price $\times$ contract multiplier (lot size)
- β_{target} the beta the portfolio manager wants the portfolio to have after the futures position is added

Build-up logic:

1. Beta measures how much a portfolio moves for every 1% move in the market index — futures on that same index move almost exactly with the index too, making them a natural tool to offset (or amplify) market-linked risk.
2. To fully neutralise market risk, the rupee amount of futures exposure taken must be scaled so its gain/loss exactly cancels the portfolio's own beta-driven gain/loss — that needs "beta times portfolio value" worth of offsetting exposure, not just an equal rupee amount.
3. Dividing this required rupee exposure by the value of one futures contract converts it into a whole number of contracts to actually transact — going short (selling) this many contracts drags portfolio beta down towards zero.
4. The same logic generalises: a manager might want to only partially hedge, or even raise market exposure — either way, what really needs to change is the beta gap, $(\beta_{\text{target}} - \beta_p)$, so using this gap instead of the full β_p gives the number of contracts to buy (positive) or sell (negative).
5. Setting $\beta_{\text{target}} = 0$ in this general formula collapses it straight back to the simple full-hedge version above, confirming both are really the same idea.

 *Tip: Watch the sign — to REDUCE beta, sell (short) futures; to INCREASE beta above its current level, buy (go long) futures.*

Options

8.6 Intrinsic Value and Time Value of an Option

★★ Frequently Asked

$$IV_{call} = \max(S - X, 0) \quad IV_{put} = \max(X - S, 0)$$
$$\text{Time Value} = \text{Premium} - IV$$

Where:

S	current market price of the underlying
X	exercise (strike) price of the option
IV	intrinsic value – the payoff if the option were exercised right now
Premium	the market price actually paid for the option
Time Value	the extra amount the market pays above intrinsic value, for the chance the option becomes more valuable before expiry

Build-up logic:

1. An option's premium bundles two layers together: what it would be worth if exercised this instant, plus a cushion for whatever could still happen before expiry.
2. A call is only worth exercising if the market price is above the strike (buy cheap via the option, worth more in the market), so its intrinsic value is $S - X$, floored at zero – you'd never exercise an option that loses you money.
3. A put works the opposite way – worth exercising only if the market price is below the strike (sell high via the option instead of low in the market) – so its intrinsic value is $X - S$, again floored at zero.
4. Whatever premium is left after stripping out intrinsic value is time value – the market's price for remaining uncertainty and remaining life, which is why it's largest for at-the-money options with long life left and shrinks as expiry nears.
5. At expiry there's no time left for anything to change, so time value must fall to exactly zero and the option's price collapses to pure intrinsic value.

 *Tip: An out-of-the-money option always has $IV = 0$ – its entire premium, however small, is pure time value.*

8.7 Put-Call Parity

★ MUST KNOW

$$\begin{aligned} C + PV(X) &= P + S && \text{(no dividend)} \\ C + Xe^{-rt} &= P + Se^{-qt} && \text{(continuous dividend yield } q) \end{aligned}$$

Where:

- C price/premium of a European call option
- $PV(X)$ present value of the common strike price X , discounted at the risk-free rate over the option's life (either $X/(1+r)^t$ or Xe^{-rt} , as the question's compounding convention requires)
- P price/premium of a European put option – same underlying, same strike X , same expiry
- S current spot price of the underlying
- r risk-free rate; t = time to expiry
- q continuous dividend yield paid by the underlying, if any

Build-up logic:

1. Build two portfolios that must produce exactly the same payoff at expiry no matter what the stock does – if two things always pay off identically, they must cost the same today too, or arbitrage would be free money.
2. Portfolio A: one call plus cash today equal to $PV(X)$, enough to grow to exactly X by expiry. Portfolio B: one put plus one share of the stock.
3. If the stock ends above X : A's call is exercised (worth $S_T - X$) and the cash has grown to X , total = S_T . B's put expires worthless and the share is worth S_T , total = S_T . Equal.
4. If the stock ends at or below X : A's call expires worthless and the cash is worth X , total = X . B's put is exercised (worth $X - S_T$) plus the share worth S_T , total = X . Equal again.
5. Since both portfolios always match regardless of which way the stock moves, they must cost the same today:
 $C + PV(X) = P + S$.
6. If the stock pays a continuous dividend yield q while held, owning the actual share (Portfolio B) also earns that dividend along the way – so the share must be counted at its dividend-stripped value Se^{-qt} instead of S for the two portfolios to still match exactly.

 Tip: Put-call parity is the fastest way to find any one of C , P , S or X when the other three are known – a common exam shortcut instead of running a full Black-Scholes/Binomial calculation.

8.8 Binomial Option Pricing Model (One Period)

★ MUST KNOW

$$p = \frac{(1+r) - d}{u - d}$$

$$V_0 = \frac{pC_u + (1-p)C_d}{1+r}$$

Where:

- p risk-neutral ("hedge") probability of an up-move – a pricing convenience, not the real-world probability
- r risk-free rate of interest for one period
- d down-factor – the multiple by which the stock price falls in the down state ($d < 1$)
- u up-factor – the multiple by which the stock price rises in the up state ($u > 1$); no-arbitrage requires $(d < 1+r < u)$
- V_0 fair value of the option today
- C_u, C_d the option's known payoff at expiry in the up-state (stock = $S \times u$) and down-state (stock = $S \times d$) – e.g. for a call, $C_u = \max(Su - X, 0)$ and $C_d = \max(Sd - X, 0)$

Build-up logic:

1. Assume the stock can only do one of two things over the period – go up to Su or down to Sd – a simplification, but it makes the option's payoff at the end completely known in just two scenarios, C_u or C_d .
2. Instead of guessing real-world probabilities (subjective, and they don't even matter for pricing), build a portfolio of the stock plus borrowing/lending that exactly replicates the option's payoff in both states – with only two assets and two future states, this can always be done.
3. Because this replicating portfolio matches the option's payoff in every possible outcome, it must be worth exactly the same as the option today too – the same no-arbitrage logic used in put-call parity, just applied to a stock instead of another option.
4. Working through that replication algebraically produces a clean shortcut: pretend the up-move happens with a fictitious "risk-neutral" probability $p = \frac{(1+r) - d}{u - d}$ instead of the true probability, and take the option's expected payoff under this fictitious probability.
5. Discount that expected payoff back one period at the risk-free rate – not some risky required return, because using p has already "neutralised away" the risk – giving $V_0 = \frac{pC_u + (1-p)C_d}{1+r}$.
6. This only works because u, d and r are internally consistent ($d < 1+r < u$)

Tip: For a two-period tree, apply this exact one-step formula twice: first find the payoffs at the three possible year-2 nodes (up-up, up-down, down-down), value the two year-1 nodes by discounting back from year 2 using the same p , then value today by discounting back once more from those two year-1 values.

Example: $S = 100$, $u = 1.10$, $d = 0.90$, $r = 8\%$, $X = 100$ (call). $C_u = \max(110 - 100, 0) = 10$, $C_d = \max(90 - 100, 0) = 0$. $p = \frac{1.08 - 0.90}{1.10 - 0.90} = 0.90$. $V_0 =$

$$\frac{0.90(10) + 0.10(0)}{1.08} = \frac{9}{1.08} \approx ₹8.33.$$

8.9 Black-Scholes Option Pricing Model (BSM)

★ MUST KNOW

$$C = S N(d_1) - X e^{-rt} N(d_2)$$

$$d_1 = \frac{\ln(S/X) + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} \quad d_2 = d_1 - \sigma\sqrt{t}$$

$$P = X e^{-rt} N(-d_2) - S N(-d_1)$$

Where:

C	fair value/premium of a European call option
S	current market price of the underlying share
$N(d_1)$	cumulative probability under the standard normal distribution up to d_1 (read off a standard normal table)
X	exercise (strike) price of the option
r	continuously-compounded risk-free interest rate per annum
$N(d_2)$	cumulative standard normal probability up to d_2
$\ln$	natural logarithm
σ	annualised standard deviation (volatility) of the underlying's returns
t	time to expiry, in years
P	fair value/premium of the corresponding European put (same S, X, r, t); $N(-d) = 1 - N(d)$

Build-up logic:

1. Black-Scholes is really the Binomial model pushed to its limit — chop the option's life into infinitely many, infinitesimally small up/down steps, so the stock price follows a smooth continuous path instead of a few discrete jumps.
2. Just like Binomial, the call price is a discounted, risk-neutral expected payoff — structured as (value of what you'd receive) minus (value of what you'd have to pay), each weighted by how likely that outcome is.
3. $N(d_2)$ is the risk-neutral probability the option finishes in-the-money and actually gets exercised, so $X e^{-rt} N(d_2)$ is today's value of the strike price you'd expect to have to pay, weighted by the chance you'll actually pay it.
4. $N(d_1)$ plays a related but distinct role — it's the probability-weighted factor for actually receiving the stock upon exercise, so $S \cdot N(d_1)$ is today's value of "getting the stock if exercised."
5. The call price is simply what you'd receive minus what you'd have to pay: $C = S N(d_1) - X e^{-rt} N(d_2)$.
6. d_1 and d_2 are standardised measures of "how far in-the-money the option is expected to land" — built from the log of the moneyness ratio S/X , adjusted for drift (r) and total variance built up over time ($\sigma^2 t$), then scaled to standard-normal (z-score) units by $\sigma\sqrt{t}$; $d_2 = d_1 - \sigma\sqrt{t}$ simply strips out one extra dose of volatility, since "will it be exercised" and "how much will it be worth if exercised" are two subtly different questions. The put price then follows directly from put-call parity (8.7).

Tip: If the stock pays a continuous dividend yield q , replace S with $S e^{-qt}$ everywhere above (including inside d_1) and use the dividend-adjusted parity from 8.7 — the rest of the formula is unchanged.

Example: $S = X = 100$, $r = 10\%$, $\sigma = 20\%$, $t = 1$. $d_1 = \frac{0 + (0.10 + 0.02)(1)}{0.20} = 0.60$; $d_2 = 0.60 - 0.20 = 0.40$. Using $N(0.60) = 0.7257$, $N(0.40) = 0.6554$: $C = 100(0.7257) - 100e^{-0.10}(0.6554) = 72.57 - 59.30 \approx ₹13.27$. Put (via parity) $= C - S + Xe^{-rt} \approx ₹3.75$.

8.10 Option Delta (Δ)

★★ Frequently Asked

$$\begin{aligned}\Delta_{call} &= N(d_1) \\ \Delta_{put} &= N(d_1) - 1\end{aligned}$$

Where:

- Δ_{call} call option's delta – how much the call's price moves for a ₹1 move in the underlying
- $N(d_1)$ the same cumulative standard normal value computed for d_1 in the Black-Scholes formula (8.9)
- Δ_{put} put option's delta – always negative, between -1 and 0

Build-up logic:

1. Delta answers a practical question: if the underlying moves up by ₹1, roughly how much does the option's price move? It's simply the slope of the option price against the stock price.
2. In the Black-Scholes call formula $C = S N(d_1) - Xe^{-rt} N(d_2)$, the entire "size" of exposure to the stock sits in the coefficient in front of S – so it's no surprise that this same $N(d_1)$ is exactly the call's delta.
3. Since $N(d_1)$ is a cumulative probability, it always lies between 0 and 1 – a deep out-of-the-money call barely moves when the stock moves (delta near 0), while a deep in-the-money call moves almost ₹1-for-₹1 with the stock (delta near 1).
4. A put's delta must equal the call's delta minus 1: put-call parity says $C - P = S - PV(X)$ at every stock price, so the two sides' sensitivities to a change in S must also match – and the right side's sensitivity to S is exactly 1.
5. Delta doubles up as a hedge ratio – to hedge one written (short) call, hold Δ shares of the stock, so the combined position barely moves for small stock price changes – the same replicating-portfolio idea as the Binomial model, applied instant by instant.

 *Tip: Don't confuse delta with the probability of finishing in-the-money – that's $N(d_2)$, not $N(d_1)$, even though the two look similar and are easy to mix up under exam pressure.*

The Other Greeks — Gamma, Theta, Vega, Rho

Beyond Delta, four other Greeks describe how an option's price reacts to other changes. Gamma measures how fast Delta itself changes as the stock price moves — it tells you how often a delta-hedge needs rebalancing, and is highest for at-the-money options close to expiry. Theta measures the option's daily "time decay" — how much value it loses purely from one more day passing, all else unchanged (usually negative for a long option position, since running out of time hurts the holder). Vega measures sensitivity to a change in the underlying's expected volatility — higher expected volatility raises both call and put premiums, since it raises the chance of a large favourable move. Rho measures sensitivity to a change in the risk-free interest rate — usually the smallest, least-watched Greek in exam problems, since short-term rate moves rarely swing option prices much.

Swaps

Interest Rate Swap (IRS) — Concept

An Interest Rate Swap is an agreement between two parties to exchange interest payment streams on a common notional principal, over a fixed period — one side pays a fixed rate, the other pays a floating (market-linked) rate, usually referenced to a benchmark like MIBOR. Only the interest differential is actually settled in cash each period; the notional principal itself is never exchanged, it exists only on paper to calculate the interest amounts. Firms use swaps to convert an existing floating-rate loan into an effectively fixed-rate one (or vice versa) without touching the underlying loan itself — useful for managing interest rate risk or acting on a view that rates will move a certain way. Because no principal changes hands and usually only the net interest is settled, credit risk on a swap is far smaller than on an equivalent loan of the same notional value.

8.11 Valuation of an Interest Rate Swap

★★ Frequently Asked

$$\begin{aligned}V_{\text{fixed receiver}} &= B_{\text{fixed}} - B_{\text{floating}} \\V_{\text{floating receiver}} &= B_{\text{floating}} - B_{\text{fixed}}\end{aligned}$$

Where:

- V value of the swap today, to the stated party
- B_{fixed} value of a hypothetical bond paying the swap's fixed coupon each period plus the notional at maturity, discounted at today's market interest rates
- B_{floating} value of a hypothetical bond paying the swap's floating (market-linked) coupon each period plus the notional at maturity

Build-up logic:

1. Strip the swap apart into two imaginary bonds sitting on the same notional principal: one paying the fixed swap rate every period, the other paying whatever the floating reference rate turns out to be.
2. Whoever receives fixed and pays floating on the swap has exactly the cash flows of someone who owns the fixed-rate bond and owes the floating-rate bond — so the swap's value to them must equal that combined bond position: $B_{\text{fixed}} - B_{\text{floating}}$.
3. B_{fixed} is valued exactly like any ordinary coupon bond — discount every remaining fixed coupon and the final notional back to today at current market interest rates.
4. B_{floating} has a neat shortcut: a floating-rate bond's coupon always resets to the current market rate, so right after each reset date the bond is worth exactly its face value (par) — no more, no less — because it's now offering exactly the return the market demands going forward.
5. So at any point between reset dates, B_{floating} is simply the present value of (notional + the next floating coupon, already fixed at the last reset), discounted for the short time remaining — usually close to the notional itself.
6. The value to the other party (paying fixed, receiving floating) is the mirror image, $B_{\text{floating}} - B_{\text{fixed}}$ — a zero-sum position between the two counterparties, just like a forward contract.

 *Tip: At the moment a swap is first set up, the fixed rate is deliberately chosen so $B_{\text{fixed}} = B_{\text{floating}}$ — making the swap worth zero to both sides at inception, exactly like a forward contract.*

Currency Swap — Concept

A Currency Swap works like an interest rate swap, but the two payment streams are in two different currencies, and — unlike an IRS — the notional principal amounts are actually exchanged, both at the start of the swap and again at maturity, at rates agreed in advance. One party pays interest (fixed or floating) in one currency while receiving interest in another currency, on their respective notional amounts. Firms use currency swaps to convert a loan raised in one currency into an effective loan in another — for example, a company that has borrowed in US Dollars but earns mostly in Rupees can swap its Dollar interest and principal obligations into Rupee ones, removing its foreign exchange risk on that borrowing.

8.12 Valuation of a Currency Swap

★ Good to Know

$$V_{\text{receive foreign}} = S_0 \times B_F - B_D$$
$$V_{\text{receive domestic}} = B_D - S_0 \times B_F$$

Where:

V value of the swap today, in domestic currency, to the stated party

S_0 current spot exchange rate, expressed as units of domestic currency per unit of foreign currency

B_F value of the notional foreign-currency bond underlying the swap (remaining foreign coupons + foreign principal, discounted at the foreign interest rate), in foreign currency units

B_D value of the notional domestic-currency bond underlying the swap (remaining domestic coupons + domestic principal, discounted at the domestic interest rate)

Build-up logic:

1. Just like an IRS, split the currency swap into two notional bonds — but here each is denominated in its own currency: a foreign bond (coupon = foreign rate, principal = foreign notional) and a domestic bond (coupon = domestic rate, principal = domestic notional).
2. Whoever is contracted to receive foreign-currency payments and pay domestic-currency payments effectively holds the foreign bond and owes the domestic bond — mirroring exactly how an IRS fixed-receiver holds one bond and owes the other.
3. Value each bond separately, in its own currency, first — discount the foreign bond's remaining cash flows at the foreign interest rate to get B_F , and the domestic bond's remaining cash flows at the domestic rate to get B_D .
4. Since the final answer needs one common currency, convert the foreign bond's value into domestic terms using today's spot rate — multiplying B_F by S_0 .
5. With both legs now in the same currency, the swap's value is simply what's held minus what's owed: $S_0 B_F - B_D$ — exactly parallel to the IRS's $B_{\text{fixed}} - B_{\text{floating}}$.

 *Tip: Unlike an IRS, a currency swap's value can shift sharply from exchange rate movements alone (through S_0) — even if neither country's interest rates move at all.*

How Swap Rates Are Quoted

In practice, a swap dealer quotes the fixed rate of an interest rate swap against a standard floating benchmark "flat" — for example, "5 years at 7.50% against 6-month MIBOR" means the fixed-rate payer pays a flat 7.50% while the floating-rate payer pays the prevailing 6-month MIBOR each reset, with no extra spread added on the floating leg. Dealers typically quote a two-way price — a slightly lower fixed rate they will pay (bid) and a slightly higher fixed rate they will receive (ask/offer) — earning the difference as their margin for making a market in the swap. Fixed swap rates are also often expressed as a spread over the government security (G-Sec) yield of matching maturity, since G-Sec yields serve as the domestic risk-free benchmark.

How to read a two-way exchange-rate quote, predict where it should move using PPP/IRP/IFE, spot a risk-free arbitrage when parity breaks down, and lock in a rate using forwards, money markets, futures or options.

9.1 Direct Quote and Indirect Quote

★ Good to Know

$$\text{Indirect Quote} = \frac{1}{\text{Direct Quote}}$$

Where:

Direct Quote units of home currency per 1 unit of foreign currency (e.g. in India, ₹83 = \$1) — the standard convention used through this chapter unless stated otherwise

Indirect Quote units of foreign currency per 1 unit of home currency (e.g. \$0.01205 = ₹1)

Build-up logic:

1. An exchange rate is just the price of one currency stated in terms of another, so it can always be expressed from either side of the pair.
2. A direct quote states this the way a local resident naturally thinks about it — "how many rupees do I pay for one unit of foreign currency" — home currency is the changing amount, foreign currency is fixed at one unit.
3. An indirect quote flips the roles — "how many units of foreign currency does one unit of my home currency buy" — home currency is now the one fixed unit.
4. Since both describe the exact same pair of currencies, one must be the precise mathematical reciprocal of the other — a direct quote multiplied by its indirect quote always equals exactly 1.
5. India (like most countries) has followed the direct quote system since August 1993, so unless a question says otherwise, treat any bare rate given as a direct quote.

Tip: Look for the phrase "per rupee" or "per dollar" in the question — it tells you instantly whether you've been handed a direct or indirect quote. Taking the reciprocal of the wrong one is a common, entirely avoidable mistake.

9.2 Bid Rate, Ask Rate and Spread

★★ Frequently Asked

$$\% \text{ Spread} = \frac{\text{Ask Rate} - \text{Bid Rate}}{\text{Ask Rate}} \times 100$$
$$\text{New Bid} = \frac{1}{\text{Old Ask}}, \quad \text{New Ask} = \frac{1}{\text{Old Bid}}$$

Where:

Ask Rate (Offer Rate)	the rate at which the dealer/bank sells the foreign currency to you — always the higher of the two rates
Bid Rate	the rate at which the dealer/bank buys the foreign currency from you — always the lower of the two rates
Old Bid / Old Ask	the two-way quote as originally given (say, ₹ per \$)
New Bid / New Ask	the two-way quote for the reciprocal pair (say, \$ per ₹), found by inverting and swapping sides

Build-up logic:

1. A bank quoting currency is really running a small trading desk — it must buy low and sell high on every trade to earn a margin and cover its risk, exactly like a moneychanger.
2. So it never quotes one single rate — it quotes two, a Bid (what it pays you for foreign currency) and an Ask (what it charges you for foreign currency) — and Bid is always lower than Ask.
3. The gap, Ask – Bid, is the bank's compensation for market-making risk and cost; scaling it by the Ask rate turns it into one comparable percentage figure across currency pairs of very different absolute value.
4. When the very same pair is quoted the other way round, whichever side previously let the bank buy the original currency at the cheapest terms must now become the new Ask, and vice versa — so inverting a quote also swaps which number is bid and which is ask, never just inverts each number in place.
5. Getting this swap right matters hugely in cross-rate and arbitrage problems, because using the wrong side of a two-way quote silently manufactures a profit — or hides a loss — that isn't actually there.

 *Tip: Bid is always less than Ask, full stop. When you flip a quote to its reciprocal, remember it's New Bid = 1/Old Ask (not 1/Old Bid) — the labels swap along with the numbers.*

9.3 Cross Rate

★ Good to Know

$$\frac{A}{C} = \frac{A}{B} \times \frac{B}{C}$$

Where:

A/B given quote of currency A in terms of currency B (units of B per unit of A)

B/C given quote of currency B in terms of currency C (units of C per unit of B)

A/C the cross rate being found – units of C per unit of A

Build-up logic:

1. Two currencies are sometimes never quoted directly against each other at all – international markets quote the pound in dollars (\$ per £), and an Indian bank quotes the dollar in rupees (₹ per \$), but nobody publishes a direct £/₹ rate.
2. Both available quotes share one common "vehicle" currency (here, the dollar) – this is the bridge linking the two separate markets.
3. Treat each quote like a unit of measurement and let the common currency cancel algebraically, exactly as you'd chain km → miles → feet using two conversion factors.
4. Multiplying (\$ per £) × (₹ per \$) cancels the "\$" sitting in the middle, leaving a rate purely between £ and ₹ – the cross rate, in rupees per pound.
5. If instead both given quotes happened to have the vehicle currency in the same position (say, both expressed as "amount per dollar"), multiplying them directly would leave the dollar squared, not cancelled – invert one of them first so the vehicle currency genuinely cancels out.

 *Tip: Write both given quotes with the common currency on the same "level" (both as numerator or both as denominator relative to it) before multiplying – if they don't line up, invert one first.*

9.4 Forward Premium / Discount (Annualized)

★★ Frequently Asked

$$\text{Forward Premium/Discount} = \frac{\text{Forward Rate} - \text{Spot Rate}}{\text{Spot Rate}} \times \frac{12}{\text{Months Forward}} \times 100$$

Where:

Forward Rate the contracted rate for delivery at a future date

Spot Rate today's rate for immediate delivery

Months Forward the tenor of the forward contract, in months

Build-up logic:

1. A forward rate being simply "higher" or "lower" than spot doesn't say much on its own — you need to know by how much, and over what period, before judging whether the gap is large or small.
2. Start with the plain percentage change from spot to forward, $(\text{Forward} - \text{Spot})/\text{Spot}$ — positive means the foreign currency costs more forward (a premium on it); negative means a discount.
3. This change happened only over the forward's own tenor (say 3 months), so it isn't yet comparable to annual figures like interest rates.
4. Multiply by $(12 \div \text{number of months})$ to scale this partial-period change up to what it would be if the same pace continued for a full year — a simple linear annualization, the standard convention here.
5. Multiply by 100 to express the final answer as a percentage per annum.

 *Tip: State clearly which currency the premium/discount belongs to — a positive answer means the foreign (quoted) currency is at a forward premium; a negative answer means it's at a forward discount.*

9.5 Purchasing Power Parity (PPP)

★ MUST KNOW

$$S_1 = S_0 \times \frac{1 + \pi_h}{1 + \pi_f}$$

Where:

S_0 current spot rate (home currency per unit of foreign currency)

S_1 expected spot rate at the end of the period, as predicted by PPP

π_h expected inflation rate in the home country over the period

π_f expected inflation rate in the foreign country over the period

Build-up logic:

1. PPP starts from a simple idea: exchange rates exist to keep the real purchasing power of money roughly equal across countries – a basket of goods should cost about “the same” everywhere once converted to a common currency.
2. If home inflation (π_h) runs higher than foreign inflation (π_f), the home currency's domestic purchasing power is falling faster, so its purchasing power abroad – its exchange value – must fall too, to keep things in balance; the home currency should depreciate.
3. The ratio $(1 + \pi_h)/(1 + \pi_f)$ captures exactly how much faster prices are rising at home than abroad over the period.
4. Multiplying today's spot rate by this ratio scales it in the same proportion: since the quote is home-currency-per-unit-of-foreign-currency, a ratio above 1 (home inflation higher) correctly pushes the rate up – more home currency needed per unit of foreign currency, i.e. home-currency depreciation.
5. This gives S_1 , the rate PPP predicts will prevail once both countries' inflation has worked through – its main use in AFM is forecasting future exchange rates for international capital budgeting.

 *Tip: PPP forecasts the expected future spot rate, not the forward rate (that comes from IRP, using interest rates, not inflation) – the two formulas share the identical structure and often coincide, but they are conceptually different tools.*

9.6 Interest Rate Parity (IRP)

★ MUST KNOW

$$F = S \times \frac{1 + i_c}{1 + i_b}$$

Where:

S spot rate, quoted as units of the counter currency per 1 unit of the base currency (in a direct home quote: counter = home currency, base = foreign currency)

F forward rate for the same pair, same quote convention

i_c interest rate (per annum, for the period concerned) in the counter-currency country – the home country, in a direct quote

i_b interest rate (per annum, for the period concerned) in the base-currency country – the foreign country, in a direct quote

Build-up logic:

1. IRP rests on a no-arbitrage idea: an investor should end up equally well off whether they invest at home for the period, or convert to the foreign currency, invest abroad, and convert back at a forward rate agreed today – both routes are equally riskless once forward-covered.
2. Investing 1 unit of counter (home) currency at home for the period grows it to $(1 + i_c)$.
3. Instead, convert that 1 unit to base (foreign) currency at spot – getting $1/S$ units – invest it abroad at i_b , and after the period it becomes $(1/S)(1 + i_b)$ units of base currency.
4. To compare fairly, convert this future foreign amount back to counter currency using the forward rate F fixed today: value = $(1/S)(1 + i_b) \times F$.
5. For neither route to beat the other, the two must be equal: $(1 + i_c) = (1/S)(1 + i_b) \times F$; rearranging for F gives $F = S \times (1 + i_c)/(1 + i_b)$.
6. Net result: whichever currency carries the higher interest rate must trade at a forward discount, and the lower-interest currency at a forward premium – the interest differential and the forward differential always offset each other under IRP.

 *Tip: The currency with the HIGHER interest rate always moves to a forward DISCOUNT (and vice versa). If your computed F looks the wrong way round, check it against this rule of thumb before finalizing an answer.*

9.7 International Fisher Effect (IFE)

★ Good to Know

$$\frac{S_1 - S_0}{S_0} = \frac{i_c - i_b}{1 + i_b}$$

Where:

S_0, S_1 current spot rate and expected future spot rate (counter currency per unit of base currency)

i_c, i_b nominal interest rates in the counter-currency (home) and base-currency (foreign) countries — same meaning as in IRP

Build-up logic:

1. Fisher's original idea: a nominal interest rate is roughly the real interest rate plus expected inflation; if real returns tend to equalize across countries as capital chases the best deal, differences in nominal rates mostly just reflect differences in expected inflation.
2. Relative PPP separately says a currency's expected depreciation should track the inflation differential between the two countries.
3. Chain these two ideas: the interest-rate differential proxies the inflation differential (step 1), and the inflation differential predicts currency depreciation (step 2) — so the interest-rate differential itself should predict the expected % change in the exchange rate. That chained conclusion is IFE.
4. Algebraically it has exactly the same shape as the IRP derivation — just replace the contracted forward rate F with the market's unbiased *expectation* of the future spot rate, S_1 , since IFE assumes forward rates are, on average, unbiased predictors of future spot rates.
5. So the currency of the higher-interest country ($i_c > i_b$) is expected to depreciate — offsetting its nominal-rate advantage so the real, currency-adjusted return ends up similar in both countries.

 *Tip: IFE explains why covered interest arbitrage profits don't persist even without hedging — the very depreciation IFE predicts quietly erodes the higher-interest currency's advantage on an uncovered basis too.*

9.8 Covered Interest Arbitrage (CIA)

★★ Frequently Asked

$$\text{Gain per unit invested} = \left[\frac{F}{S} \times (1 + i_b) \right] - (1 + i_c)$$

Where:

F, S the actual market-quoted forward and spot rates (counter currency per unit of base currency)

i_b interest rate available on an investment in the base (foreign) currency

i_c interest rate available on an investment in the counter (home) currency

Build-up logic:

1. IRP tells you what the forward rate should theoretically be for both routes to earn exactly the same; in practice, actual market forward rates can drift away from this IRP-implied level, at least briefly.
2. Whenever that happens, one route genuinely earns more than the other with zero exchange-rate risk, because spot, forward, and both interest rates are all locked in today – hence "covered".
3. To spot it: compute the covered foreign return, $\frac{F}{S} \times (1 + i_b)$, and compare it to the plain domestic return, $(1 + i_c)$ – a positive "Gain" above means going abroad wins.
4. If Gain is positive: borrow in the home (counter) currency, convert the borrowed sum to foreign currency at spot, invest it abroad at i_b , and simultaneously sell the maturity proceeds forward at F – the forward-converted proceeds exceed what's needed to repay the home loan with interest, and the excess is the riskless profit.
5. If Gain is negative instead, run it in reverse: borrow abroad in the foreign currency, convert to home currency at spot, invest at home, and buy the foreign currency needed for repayment forward today.
6. This arbitrage is self-correcting – the extra spot and forward buying/selling it triggers nudges rates back toward the IRP-consistent level, which is why such gaps rarely persist for long in an efficient market.

 *Tip: The golden rule – borrow in the currency that's "cheap" relative to what covered investment abroad can earn, and invest in whichever side gives the better covered (forward-locked) return. Never leave the foreign leg unhedged, or it stops being riskless arbitrage.*

9.9 Forward Market Hedge

★★ Frequently Asked

$$\begin{aligned}\text{Hedged Value (Receivable)} &= \text{FC Receivable} \times \text{Forward Bid Rate} \\ \text{Hedged Value (Payable)} &= \text{FC Payable} \times \text{Forward Ask Rate}\end{aligned}$$

Where:

FC Receivable	foreign-currency amount due to be received on a future date (e.g. export proceeds)
FC Payable	foreign-currency amount due to be paid on a future date (e.g. an import bill)
Forward Bid Rate	the bank's forward rate for buying foreign currency from you – relevant when you'll be selling FC forward
Forward Ask Rate	the bank's forward rate for selling foreign currency to you – relevant when you'll be buying FC forward

Build-up logic:

1. Transaction exposure exists from the moment a foreign-currency receivable or payable is contracted but not yet settled – between now and settlement, the spot rate can move against you.
2. A forward hedge removes this uncertainty the simplest way possible: lock in today, via a forward contract with your bank, the exact rate at which the future foreign currency will be converted, so the home-currency value is known right now.
3. If you're due to receive foreign currency (an exporter), you will be the one selling it to the bank on the future date – so the relevant rate is the bank's forward bid (buying) rate.
4. If you owe foreign currency (an importer), you will be buying it from the bank on the future date – so the relevant rate is the bank's forward ask (selling) rate.
5. Multiplying the exposure by the correct side of the forward quote turns an uncertain future home-currency amount into a fixed, known-today figure – exposure eliminated, at the cost of giving up any chance of a favourable rate move.

 *Tip: Receivable → use the bank's forward BID; Payable → use the bank's forward ASK. Using the wrong side of the quote is one of the most common marks lost on this topic.*

9.10 Money Market Hedge (MMH)

★ MUST KNOW

$$\begin{aligned}\text{FC to Borrow Today (hedging a Receivable)} &= \frac{\text{FC Receivable}}{1 + i_f} \\ \text{FC to Invest Today (hedging a Payable)} &= \frac{\text{FC Payable}}{1 + i_f}\end{aligned}$$

Where:

FC Receivable / FC Payable the foreign-currency amount due to be received/paid at a future date

i_f the relevant foreign-currency interest rate for the period – the *borrowing* rate abroad when hedging a receivable; the *deposit/investment* rate abroad when hedging a payable

Build-up logic:

1. A money market hedge replaces "wait and convert at an unknown future spot rate" with "convert at today's known spot rate", by artificially creating an offsetting foreign-currency cash flow today instead of waiting for the real one to arrive later.
2. For a receivable: you don't have the foreign currency in hand yet, but you can borrow it now – so borrow just enough FC today that, growing at the foreign borrowing rate i_f , the loan exactly equals the receivable on the due date. That's why the amount borrowed is the receivable discounted back by $(1 + i_f)$.
3. Convert this borrowed FC into home currency immediately at today's known spot rate – this locks in a certain home-currency sum right now. When the real receivable arrives later, it repays the foreign loan exactly, principal and interest.
4. For a payable: instead of waiting to buy FC later at an unknown rate, place FC on deposit today that will grow, at the foreign deposit rate i_f , to exactly the payable amount by the due date – hence the amount invested today is the payable discounted back by $(1 + i_f)$.
5. Fund that deposit by converting home currency to FC at today's spot rate (borrowing the home currency now if it isn't already on hand) – again locking in a known home-currency cost today instead of an uncertain one later.
6. Either way, the future foreign-currency flow – the real receivable, or the maturing deposit – is used purely to settle the loan/payable; the hedge's whole effect is to move the currency conversion from an uncertain future date to a known rate today.

 *Tip: A money market hedge and a forward hedge should give an identical home-currency result whenever IRP holds exactly. If a problem's two methods give different answers, that gap is itself evidence that IRP does not hold in the given data.*

Example: An Indian importer owes \$100,000 in 6 months; the \$ deposit rate is 4% p.a. (2% for 6 months).

Invest $\$100,000 \div 1.02 \approx \$98,039$ today. Convert this to ₹ at spot to lock in today's certain ₹ outflow, then place the \$98,039 on deposit – in 6 months it grows to exactly \$100,000, used to pay the supplier.

Currency Futures and Currency Options — Alternative Hedges

Currency futures are exchange-traded, standardized versions of a forward contract — fixed contract size, fixed maturity dates, traded on an exchange rather than negotiated bank-to-bank — that hedge FX exposure the same basic way a forward does, but add daily mark-to-market, margin requirements, and the ability to close out the position early with an offsetting trade. This flexibility is useful when an exact-amount OTC forward isn't easily available, though standardized sizes and dates rarely match an exposure exactly, leaving some residual "basis" risk. Currency options instead give the holder the right, but not the obligation, to buy (call) or sell (put) a currency at a fixed strike rate — an exporter worried about the foreign currency weakening would buy a put option, while an importer worried about it strengthening would buy a call option. Unlike a forward or futures hedge, an option lets the firm walk away and deal at the better spot rate if the market moves favourably; that flexibility costs an upfront, non-refundable premium, which is the most the firm can lose if the option goes unused.

Translation Exposure — Current Rate Method and Monetary/Non-Monetary Method

Translation (or accounting) exposure arises purely from consolidation — when a parent restates a foreign subsidiary's financial statements into the parent's reporting currency at each balance sheet date, exchange-rate moves create paper gains or losses even though no actual cash has moved. Under the Current Rate Method, every asset and liability on the subsidiary's balance sheet is translated at the closing (year-end) rate, so the entire net worth is exposed to translation movement, while income-statement items are usually translated at an average rate for the period. Under the Monetary/Non-Monetary Method, monetary items — cash, receivables, payables, debt, anything fixed in currency terms — are translated at the current/closing rate, while non-monetary items — inventory, fixed assets, other cost-based items — are translated at the historical rate that applied when they were originally acquired, reflecting the idea that only monetary items truly carry exchange risk. Because it is a reporting effect rather than a real cash-flow effect, most firms treat translation exposure as a lower hedging priority than transaction or economic exposure.

Economic Exposure

Economic (or operating) exposure is the broadest and, in the long run, the most important currency exposure a firm faces — the extent to which unexpected exchange-rate changes affect the present value of its future operating cash flows, competitive position, and overall value, well beyond any single contracted transaction. It hits an exporter who competes on price against foreign rivals (home-currency appreciation makes its exports costlier abroad, hurting volumes and margins) just as much as a purely domestic firm that competes with imports or sources inputs from abroad, even if it never itself transacts in foreign currency. Unlike transaction exposure, it can't be hedged away with one forward contract or money market position, because it isn't a fixed, known amount at a known date. Firms instead manage it strategically — diversifying markets, suppliers and production across countries and currencies, building pricing flexibility into contracts, and matching the currency mix of costs to the currency mix of revenues.

Exposure Netting (Multinational Groups)

A multinational group with many subsidiaries constantly generates offsetting currency exposures — one unit may owe another unit money in Currency X while a third unit is separately due to receive Currency X from someone else — so hedging every single exposure individually with an outside bank wastes money once the group looks at itself as a whole. Exposure netting consolidates intra-group receivables and payables in the same (or closely correlated) currencies and settles only the net difference — bilaterally, where two units settle just their mutual net balance, or multilaterally, where a central treasury/netting centre nets all group-wide intercompany balances in a currency and settles one net figure with each unit. This sharply cuts the number and total value of cross-border transactions that need actual external hedging, saving on transaction costs, bid-ask spreads and bank charges — only the group's genuine residual exposure to the outside world still needs to be hedged externally.

How to value a project once a foreign currency, a foreign cost of capital, or subsidized foreign financing enters the picture – and how a multinational manages the working-capital and political risks that come with operating across borders.

10.1 Foreign Project Appraisal — Home Currency Approach

★ **MUST KNOW**

$$NPV_h = \sum_{t=1}^n \frac{CF_t^{FC} \times S_t}{(1 + k_h)^t} - I_0$$

Where:

- CF_t^{FC} the project's expected cash flow in the foreign/local currency for year t
- S_t expected spot rate at year t (home currency per unit of foreign currency) – usually forecast using PPP off the current spot rate and the expected inflation differential
- k_h discount rate appropriate to the project's risk, expressed in home-currency terms – the parent's required return
- I_0 initial investment, expressed in home currency (convert at today's spot rate first, if given in foreign currency)

Build-up logic:

1. A parent company ultimately cares about value created in its own currency, since that's what it reports to shareholders and pays dividends in – so the most direct approach converts everything to home currency as early as possible.
2. Each year's foreign-currency cash flow is converted separately, using that specific year's expected spot rate S_t – not today's spot rate – because the home-currency value of a cash flow received three years out depends on the exchange rate expected three years out, not today's.
3. These future spot rates are usually not handed to you outright; they're forecast by escalating today's spot rate year by year using PPP and the expected inflation differential between the two countries.
4. Once every year's cash flow is in home-currency terms, discount each one at k_h – the risk-adjusted rate you'd use for a similarly risky domestic project, since currency conversion has already been handled separately, year by year.
5. Subtracting the home-currency initial investment from the sum of these discounted values gives the project's NPV in the currency the parent's shareholders actually care about.

 *Tip: A common mistake is converting only the final total at today's spot rate. Always convert year-by-year, each year at that year's own expected rate, before discounting in this approach.*

10.2 Foreign Project Appraisal — Local Currency Approach

★ Good to Know

$$NPV_{FC} = \sum_{t=1}^n \frac{CF_t^{FC}}{(1 + k_f)^t} - I_0^{FC} ; \quad NPV_h = NPV_{FC} \times S_0$$

Where:

CF_t^{FC}, I_0^{FC}	the project's cash flows and initial investment, kept in local/foreign currency throughout
k_f	discount rate appropriate to the project's risk, expressed in foreign-currency terms — built from foreign risk-free rate and market premium, not the home rate re-used
NPV_{FC}	the project's NPV, still expressed in foreign currency
S_0	today's spot rate (home currency per unit of foreign currency), used to convert only the final NPV figure
NPV_h	the project's NPV converted into home currency

Build-up logic:

1. Instead of converting every year's cash flow, this approach keeps the entire calculation in the foreign currency the project actually operates in — cash flows, discount rate and initial outlay all stay local.
2. Discount the foreign-currency cash flows at a rate that itself reflects foreign-currency risk and return, k_f — this must be a genuinely foreign rate, not the home rate borrowed for convenience.
3. This produces NPV_{FC} , which is already a present ("today") value in foreign-currency terms — so only today's spot rate is needed to translate it, unlike operating cash flows, which needed year-specific future rates.
4. Multiply NPV_{FC} by today's spot rate S_0 , once, at the very end, to state the answer in home currency for comparison with the parent's other opportunities.
5. Since conversion happens only once, this route is quicker when a reliable foreign discount rate is available — but it leans entirely on correctly estimating k_f , which is often the harder number to pin down.

 *Tip: Never mix the two approaches halfway. Either convert cash flows every year and discount at k_h , or discount in foreign currency at k_f and convert only the final NPV at S_0 — converting foreign cash flows at S_0 but discounting at k_h is a common, marks-losing muddle.*

Why the Two Approaches Must Reconcile (PPP-Consistency)

Both routes are meant to answer exactly the same question — what is this foreign project worth to the home-currency parent — so, done correctly, they must land on the identical NPV. This reconciliation holds specifically when the exchange-rate forecasts used to convert cash flows year-by-year in the home-currency approach are consistent with the gap between k_h and k_f — in practice, this means the future spot rates must be generated using Purchasing Power Parity (or an equivalent parity-consistent forecast) off the very same inflation/interest differential that explains why k_h and k_f differ in the first place. If a question instead hands you an exchange-rate forecast that doesn't actually match the implied gap between k_h and k_f — an arbitrary analyst forecast, say, assuming more or less currency movement than PPP would predict — the two approaches will genuinely give two different NPVs. Neither is "wrong" in that case; they simply rest on different, inconsistent assumptions about the exchange rate, so always use whichever approach the question's data actually supports.

10.3 Adjusted Present Value (APV) for Cross-Border Projects

★★ Frequently Asked

$$APV = NPV_{\text{Base Case}} + PV(\text{Financing Side Effects})$$

$$PV(\text{Subsidy Benefit}) = (\text{Normal Rate} - \text{Concessional Rate}) \times \text{Loan} \times (1 - t) \times PVIFA(\text{Normal Rate}, n)$$

Where:

$NPV_{\text{Base Case}}$	the project's NPV assuming it's financed entirely by equity, discounted at the ungeared required return for the project's business risk – financing ignored altogether
$PV(\text{Financing Side Effects})$	the present value of benefits/costs that arise specifically from <i>how</i> the project is actually financed, over and above its pure operating economics
Normal Rate	the market/commercial interest rate the borrower would otherwise pay
Concessional Rate	the actual, lower rate charged on subsidized/soft financing (e.g. from a host government or development agency)
Loan	principal amount of the subsidized loan
t	the borrower's tax rate
$PVIFA(\text{Normal Rate}, n)$	present value annuity factor for n years at the normal market rate

Build-up logic:

1. Ordinary NPV bundles the operating decision (is the project worth building?) and the financing decision (how should it be paid for?) into one discount rate – APV deliberately separates the two, which matters most for cross-border projects, where financing terms are often unusual and don't apply generically to every project the firm undertakes.
2. First, value the project purely on its commercial merits, as if funded 100% by equity – this base-case NPV, discounted at the required return for the project's business risk alone, captures pure operating value with no financing effects mixed in.
3. Then separately value every distinct side effect arising purely from the financing method chosen – the most common one in cross-border projects is a concessional loan from a host government keen to attract the investment, charging less than the market rate.
4. The benefit each year is the interest saved, $(\text{Normal Rate} - \text{Concessional Rate}) \times \text{Loan}$, tax-adjusted: had the firm paid full market interest, that interest would have been tax-deductible, so the true net benefit of paying less interest is this saving net of the tax shield it forgoes – multiply by $(1 - t)$.
5. This net annual benefit repeats every year the concessional loan runs, so its present value uses an annuity factor over the loan's tenor, discounted at the normal market rate – the rate that reflects the loan's own risk.
6. Adding this (and any other financing side effect, such as issue/arrangement costs, entered as a negative) to the base-case NPV gives the APV – a total that transparently shows how much of the project's worth comes from the business itself versus from favourable financing.

 *Tip: APV suits cross-border capital budgeting precisely because it isolates one-off financing perks – like a subsidized loan or a host-government tax concession – that don't recur for the firm's other, normally-*

financed projects. Bundling them into one blended WACC-based discount rate would misstate the project's true underlying operating worth.

Example: A ₹10,00,000 loan is available at a concessional 6% instead of the market 10% for 5 years; tax rate 30%. Annual after-tax saving = $(10\% - 6\%) \times 10,00,000 \times (1 - 0.30) = ₹28,000$. PV of this saving at 10% for 5 years (PVIFA ≈ 3.791) $\approx ₹1,06,148$ — this is added to the base-case NPV to get APV.

10.4 Cost of Capital for a Foreign Project

★★ Frequently Asked

$$k_{\text{foreign project}} = k_{\text{equivalent domestic project}} + \text{Country/Political Risk Premium}$$

Where:

$k_{\text{equivalent domestic project}}$ the discount rate that would apply to a project of similar business/operating risk located at home — e.g. via CAPM, $R_f + \beta(R_m - R_f)$, or the firm's normal risk-adjusted required return

Country/Political Risk Premium additional return investors demand for risks specific to the host country over and above ordinary business risk — expropriation, currency inconvertibility, political instability, capital controls, etc.

Build-up logic:

1. CAPM-style discount rates capture business and market risk, but say nothing about risk that's specific to a country rather than to the project's industry — a factory in a politically unstable country isn't equally risky as an identical factory at home, even though the "business" is the same.
2. To reflect this extra layer, add a country/political risk premium on top of the rate that would normally apply to a similarly-risky project, raising the hurdle the foreign project's cash flows must clear.
3. This premium is built up from the host country's risk profile — political stability, expropriation risk, currency-convertibility restrictions, sovereign credit standing — and a common practical shortcut is to use the extra yield a host country's government bond pays over a risk-free benchmark of similar maturity, since that market-observed spread already reflects investors' collective view of the country's risk.
4. A higher discount rate automatically penalizes distant cash flows more heavily, since discounting compounds — which fits the intuition that political risk tends to bite harder the longer money stays committed in that country.
5. Some AFM problems instead load the risk into the numerator — cutting expected cash flows for the probability of adverse political events and discounting at a normal rate — rather than the denominator; both approaches build the same risk into the appraisal, just at a different stage of the calculation.

 **Tip:** Don't double-count. If the cash flow forecasts already carry a haircut for political risk, don't also add a country risk premium to the discount rate — that penalizes the same risk twice.

International Working Capital Management — Netting and Leading & Lagging

Multinational groups manage working capital tied up in cross-border intercompany trade using two classic techniques. Netting — the same idea used for currency exposure — consolidates the many intercompany payables and receivables between group units in different currencies and settles only the net balances, either bilaterally between two units or multilaterally through a central treasury/netting centre for the whole group, cutting the volume, cost and float of cross-border currency transfers along with the bid-ask spreads and bank charges each transfer would otherwise attract. Leading and lagging means deliberately timing intercompany (and sometimes third-party) payments earlier ("leading") or later ("lagging") than their normal due date, based on the expected direction of exchange-rate movement: a subsidiary paying a bill in a currency expected to appreciate should lead — pay early, before it gets costlier — while one due to pay in a currency expected to depreciate should lag, since settling will get cheaper. Both techniques let a multinational actively optimize its group-wide currency and cash position without buying any external hedging instrument, though leading and lagging between group units is closely watched by tax and exchange-control authorities, since it can shift value between jurisdictions.

Political Risk and How Firms Mitigate It

Political risk is the risk that actions of a host government — expropriation or nationalization of assets, sudden changes in tax or regulatory policy, restrictions on repatriating profits or capital, exchange controls, or outright political instability — reduce or wipe out the value of a firm's foreign investment, over and above ordinary commercial risk. Since it can't be hedged with a simple forward contract the way currency risk can, firms instead rely on structural safeguards: structuring the investment as a joint venture with a well-connected local partner (or even government participation), so the host country has its own stake in the venture's survival; deliberately keeping the subsidiary dependent on the parent for a critical input — technology, key components, or export markets — so that expropriating it destroys much of its value to the host country; financing the project heavily with local or multilateral debt, so local banks or agencies like the World Bank/IFC would also lose out from expropriation; and taking political risk insurance from agencies such as India's ECGC or the multilateral MIGA (Multilateral Investment Guarantee Agency), which compensate the firm if political events cause a loss. Many firms also simply diversify across several countries, so a political shock in any one location can't seriously damage the group as a whole.

11

Interest Rate Risk Management

How a borrower, lender or bond portfolio manager locks in today's interest rate for a future period, or protects a portfolio's value against rate moves, using FRAs, futures, options and duration-based hedges.

Interest Rate Risk — Meaning and Sources

Interest rate risk is the risk that a change in market interest rates hurts a firm or investor before the exposure has been settled — it cuts both ways, unlike most risks which only threaten one side of a transaction. A borrower with a future loan to take, or a floating-rate loan already running, is hurt if rates rise, since borrowing costs go up. A lender, a fixed-income investor, or anyone about to invest surplus cash is hurt if rates fall, since the return they can lock in drops. Bond and bond-portfolio holders face a related but distinct risk — a bond already held loses market value when yields rise, because its fixed coupons become less attractive relative to new, higher-yielding issues. Because a single firm can be a borrower on one leg and a lender or bond-holder on another, well-run treasuries measure and hedge interest rate risk actively rather than simply hoping rates move in their favour.

11.1 Forward Rate Agreement (FRA) — Settlement Amount

★ MUST KNOW

$$\text{Settlement Amount} = \frac{(R - FRA) \times N \times \frac{d}{B}}{1 + \left(R \times \frac{d}{B}\right)}$$

Where:

- R the actual reference/benchmark rate (e.g. MIBOR) prevailing on the FRA's fixing/settlement date
- FRA the fixed rate agreed in the FRA contract at inception
- N notional principal — the amount on which the interest differential is calculated (no actual principal ever changes hands)
- d number of days in the FRA's contract period (the forward period being fixed)
- B day-count basis used for the currency concerned (360 or 365, per market convention)

Build-up logic:

1. An FRA fixes the interest rate on a notional loan/deposit for a future period starting at a future date — but unlike a real loan, no principal is ever exchanged, so the parties only need to settle the interest-rate difference, not the whole loan.
2. If the reference rate on the fixing date turns out higher than the FRA rate, the party who "bought" the FRA — protecting against rising rates, as a would-be borrower — is compensated for the extra interest they'd otherwise pay; that compensation is the rate difference, $R - FRA$, applied to the notional over the period's length as a fraction of a year, d/B .
3. In a real loan, that interest difference would only actually fall due at the END of the period (in arrears) — but an FRA settles immediately, right at the START of the contract period, so the raw amount from step 2 isn't yet due; it must be brought back to today's (the settlement date's) value first.
4. Discount that raw amount by dividing by $[1 + (R \times d/B)]$, using the reference rate itself as the discounting rate — that's the genuine cost of money for that period, as observed on the fixing date.
5. This discounted figure is the actual net cash settlement paid on the FRA's settlement date — to the FRA buyer if the reference rate rose above the FRA rate, or to the seller if it fell below.

 *Tip: The single most commonly missed step in FRA numericals is the discounting — students correctly compute the interest differential but forget that FRA settlement happens upfront, not at period-end, and skip dividing by the discount factor.*

Example: A ₹1,00,00,000 loan starting in 3 months is hedged with a 3×6 FRA at 8%. If the actual 3-month reference rate on settlement day is 9% (90 days, 365-day basis): raw differential = $(9\% - 8\%) \times 1,00,00,000 \times 90/365 \approx ₹24,658$. Discounted at the reference rate: $24,658 \div [1 + (0.09 \times 90/365)] \approx ₹24,122$ — received by the borrower, since actual rates rose above the FRA rate.

Interest Rate Futures — Hedging Logic

Interest rate futures let a firm lock in today the rate/price for a transaction that will actually happen later, exactly like an index future locks in the level of an equity index. The key mechanical fact: futures prices on interest-rate instruments move opposite to interest rates, since the futures contract is priced off an underlying debt instrument — when rates rise, its price falls, and vice versa. A borrower who fears rates will rise before they actually borrow should therefore SELL interest rate futures now: if rates do rise, futures prices fall and the short position profits, offsetting the higher borrowing cost paid in the cash market. A lender or bond investor who fears rates will fall — cutting future reinvestment income or the yield they can lock in — should instead BUY interest rate futures: if rates fall, futures prices rise and the long position profits, compensating for the lower rate now available. This is the exact mirror image of hedging with equity index futures, where a manager who fears the market will fall sells index futures to profit as they fall alongside it — here, the "falling asset" being protected against is future interest income or rates, not portfolio value.

Interest Rate Cap, Floor and Collar

An Interest Rate Cap is a series of call options on an interest rate, bought by a borrower with floating-rate debt to guard against rates rising — the writer pays the buyer whenever the reference rate exceeds the agreed strike, capping the effective borrowing cost at that strike plus the premium paid, while still letting the borrower benefit fully if rates fall below it. An Interest Rate Floor is the mirror image — a series of put options on an interest rate, bought by a lender or investor with floating-rate income to guard against rates falling — the writer pays the buyer whenever the reference rate drops below the agreed strike, guaranteeing a minimum return while still letting the investor gain if rates rise above it. An Interest Rate Collar combines both: a borrower who has bought a cap can partly or fully fund its premium by simultaneously selling (writing) a floor at a lower strike, collecting premium from a counterparty willing to bet rates won't fall too far. This locks the borrower's effective rate into a band between the floor and cap strikes — cheaper than buying a cap alone, potentially even zero-cost if the strikes are chosen right — at the cost of giving up any benefit if rates fall below the floor strike.

11.2 Duration-Based Hedge Ratio (Number of Futures Contracts)

★ MUST KNOW

$$N = \frac{D_P \times V_P}{D_F \times P_F}$$

Where:

D_P (modified) duration of the portfolio being hedged

V_P market value of the portfolio being hedged

D_F duration of the instrument underlying the futures contract (e.g. the cheapest-to-deliver bond)

P_F price of one futures contract – the futures price $\times$ contract size/multiplier

N number of futures contracts to buy or sell

Build-up logic:

1. Duration measures how sensitive a bond or bond portfolio's price is to a small change in yields – a higher duration means a bigger price swing for the same rate move; to hedge, the futures position's own price sensitivity must offset the portfolio's.
2. The portfolio's rupee sensitivity to a rate change is proportional to $D_P \times V_P$ – bigger portfolios and higher-duration portfolios both swing more in rupee terms for the same move in yields.
3. Each futures contract's own rupee sensitivity to that same rate change is proportional to $D_F \times P_F$ – the duration of the instrument underlying the future, times the rupee value one contract represents.
4. Dividing the portfolio's sensitivity by one contract's sensitivity gives N , the number of contracts whose combined price movement should offset the portfolio's for a given change in yields.
5. Direction: a manager hedging against rising rates (which hurt a bond portfolio's value) sells N contracts, so a rate rise that hurts the cash portfolio produces an offsetting gain on the short futures position – exactly the beta-hedge logic used for equity portfolios, just with duration as the sensitivity measure instead of beta.

 *Tip: This is structurally identical to the equity beta-hedge formula, Number of Contracts = (Portfolio Beta $\times$ Portfolio Value) $\div$ Futures Price – just swap "beta" for "duration". If you already know one, you already know the other.*

Basis Point Value (BPV) / Price Value of a Basis Point (PVBP)

Basis Point Value (BPV), also called Price Value of a Basis Point (PVBP) or DV01, measures how much a bond or bond portfolio's price changes for a one-basis-point (0.01%) change in yield – it turns the abstract idea of duration into a concrete rupee figure that traders and risk managers can use directly. It can be quickly estimated as $\text{Modified Duration} \times \text{Price} \times 0.0001$, since duration already gives the approximate percentage price change for a small yield move, and multiplying by 0.0001 simply scales that sensitivity down to the effect of just one basis point instead of a full 1% move. BPV is especially useful for sizing hedges across instruments of very different duration and price – to hedge a bond position using a different bond or a futures contract, a trader can match the two positions' BPVs directly, rather than juggling duration and value separately, since BPV already blends both into one rupee-sensitivity number. A higher BPV means a position's value swings more for every basis point the market moves, which is why trading books and portfolios are often given BPV limits as a simple, practical cap on interest-rate risk.

How to put a rupee value on a whole business — using the cash it will generate, what similar listed companies trade at, or what its net assets are worth.

12.1 Free Cash Flow to Firm (FCFF)

★ **MUST KNOW**

$$FCFF = EBIT(1 - t) + \text{Depreciation} - \text{Capital Expenditure} - \Delta \text{Working Capital}$$

Where:

EBIT	earnings before interest and tax — the firm's operating profit
t	applicable tax rate
Depreciation	non-cash charge added back (include amortization and other non-cash items too)
Capital Expenditure	cash spent on acquiring/replacing fixed assets to sustain or grow operations
Δ Working Capital	increase in net working capital during the period

Build-up logic:

1. FCFF is the cash available to ALL capital providers together — both lenders and shareholders — so start from a profit figure measured before any financing decision: EBIT, which is untouched by how the firm is funded.
2. Tax is a real cash outflow regardless of financing, so convert EBIT to an after-tax basis, $EBIT(1-t)$, giving the firm's after-tax operating cash-generating power.
3. Add back depreciation because it reduced accounting profit but never actually left the firm's bank account — it must be restored to get to real cash.
4. Subtract capital expenditure, since the firm must keep reinvesting in fixed assets just to sustain and grow the business — that cash never becomes "free."
5. Subtract the increase in working capital too — growing sales usually locks up more cash in inventory and receivables, so that cash is tied up in the business, not free to distribute.
6. Whatever survives all these operating and reinvestment needs is the cash genuinely free to be claimed by lenders and shareholders combined — Free Cash Flow to the Firm.

 *Tip: Never deduct interest while computing FCFF — it belongs to the whole firm, before any financing effect, which is exactly why it gets discounted at WACC later.*

12.2 Free Cash Flow to Equity (FCFE)

★ **MUST KNOW**

$$FCFE = FCFF - \text{Interest}(1 - t) + \text{Net Borrowing}$$
$$FCFE = \text{Net Income} + \text{Depreciation} - \text{Capital Expenditure} - \Delta \text{Working Capital} +$$

Where:

FCFF	free cash flow to the firm, as computed above
Interest(1-t)	after-tax interest actually paid to lenders during the period
Net Borrowing	fresh debt raised minus debt repaid in the period (can be negative)
Net Income	profit after tax and after interest – PAT belonging to equity shareholders
Depreciation, Capital Expenditure, Δ Working Capital	same meaning as in the FCFF formula above

Build-up logic:

1. FCFF belongs jointly to lenders and shareholders, so to isolate what's left specifically for equity, first strip out what lenders are owed: the after-tax interest actually paid to them, $\text{Interest}(1-t)$ – after-tax to stay consistent with FCFF's own after-tax basis.
2. Equity shareholders also gain whenever the firm raises fresh debt (that cash substitutes for money equity would otherwise have had to put in) and lose out whenever debt is repaid (equity funds effectively cover it) – so add back Net Borrowing.
3. Equivalently, FCFE can be built straight from accounting Net Income, which is already after interest and tax and so already "belongs" to equity: just add back depreciation, subtract capex and the rise in working capital exactly like FCFF, then add Net Borrowing for the same debt-financing effect.
4. Both routes land on the same number, since Net Income is essentially $\text{EBIT}(1-t)$ minus after-tax interest – they are just two different starting points reaching the identical equity cash flow.

 *Tip: FCFE is what's left ONLY for equity shareholders after lenders are serviced – that is exactly why it gets discounted at Cost of Equity, never at WACC.*

12.3 Terminal Value — Gordon Growth Model

★ MUST KNOW

$$TV_n = \frac{CF_{n+1}}{k - g} = \frac{CF_n(1 + g)}{k - g}$$

Where:

- TV_n terminal value as at the end of year n , i.e. end of the explicit forecast period
- CF_{n+1} the cash flow expected in the first year after the explicit forecast (year $n+1$)
- CF_n the last explicit forecast year's cash flow (FCFF or FCFE, whichever is being valued)
- g constant growth rate the cash flow is assumed to grow at forever after year n
- k discount rate matching the cash flow — WACC for FCFF, Cost of Equity for FCFE

Build-up logic:

1. No DCF can forecast individual cash flows forever, so after n years of detailed explicit projections, assume the firm settles into a stable, mature phase growing at one constant rate g forever — a realistic simplification for the distant future.
2. A cash flow stream growing at a constant rate forever is exactly a growing perpetuity, and its present value, one period before the first cash flow, is simply (first cash flow) $\div$ (discount rate – growth rate).
3. Here the perpetuity's "first cash flow" is next year's figure, $CF_{n+1} = CF_n(1 + g)$ — the last explicit year's cash flow grown one more year.
4. Applying the growing-perpetuity formula values this entire post-year- n stream as of the end of year n — one period before CF_{n+1} arrives — and that lump-sum value at year n is the Terminal Value.
5. Being a value as at year n , TV is itself still a future amount — it must be discounted back to today at the same rate k , for the same n years, before it can be added to the explicit-period cash flows' present values.

 *Tip: TV is the value AT the end of year n , not at time 0 — always discount it back n years along with the explicit-period cash flows; adding it in undiscounted is one of the most common exam mistakes.*

12.4 Enterprise Value & Equity Value — FCFF / WACC Approach

★ MUST KNOW

$$EV = \sum_{t=1}^n \frac{FCFF_t}{(1+WACC)^t} + \frac{TV_n}{(1+WACC)^n}$$

Equity Value = EV – Market Value of Debt + Cash and Non-operating Investments

Where:

$FCFF_t$	free cash flow to the firm forecast for year t
WACC	weighted average cost of capital — the blended required return of lenders and shareholders together
TV_n	terminal value at end of year n (Gordon growth, using WACC since it's still FCFF-based)
Market Value of Debt	value of interest-bearing debt to be netted off, since EV belongs to debt and equity together
Cash and Non-operating Investments	surplus cash/investments not needed for operations, since FCFF excludes any return they generate

Build-up logic:

1. FCFF is shared by everyone who financed the firm — lenders and shareholders together — so discounting it at their blended required return, WACC, values the whole operating business: the Enterprise Value.
2. Add the present value of each explicit-period FCFF to the present value of the Terminal Value (itself discounted back n years at WACC), to capture both the near-term forecast and everything beyond it.
3. Enterprise Value belongs to lenders and shareholders combined, so to isolate what's left for equity, subtract the market value of debt — lenders' prior claim, which must be repaid first.
4. Add back any surplus cash or non-operating investments on the balance sheet — rightly excluded from operating FCFF, but still assets that equity shareholders own on top of the operating business.
5. What remains belongs purely to equity shareholders — divide by the number of equity shares if a per-share value is needed.

Tip: Stopping at Enterprise Value and calling it the equity value is a very common mistake — always run the debt-and-cash bridge to reach the actual value of equity shares.

Example: FCFF for years 1-3 = ₹100, ₹110, ₹120 lakh; WACC = 12%; terminal growth $g = 5\%$. $TV_3 = \frac{120(1.05)}{0.12 - 0.05} = \frac{126}{0.07} = ₹1,800$ lakh. Discount $FCFF_1$, $FCFF_2$, $FCFF_3$ and TV_3 — all at 12%, TV_3 over 3 years like $FCFF_3$ — and sum for Enterprise Value; then subtract debt and add cash to reach Equity Value.

12.5 Equity Value Directly — FCFE / Cost of Equity Approach

★★ Frequently Asked

$$\text{Equity Value} = \sum_{t=1}^n \frac{FCFE_t}{(1 + K_e)^t} + \frac{TV_n}{(1 + K_e)^n}$$

Where:

$FCFE_t$ free cash flow to equity forecast for year t

K_e cost of equity — equity shareholders' own required rate of return (e.g. via CAPM)

TV_n terminal value of the FCFE stream at end of year n , computed the same Gordon-growth way but using $FCFE_n$ and K_e in place of $FCFF_n$ and WACC

Build-up logic:

1. FCFE is already the cash flow left exclusively for equity shareholders, so it must be discounted at the rate equity itself requires, K_e — using WACC here would understate the risk equity alone bears.
2. Forecast FCFE explicitly for n years, just like FCFF, and find the present value of each year's FCFE at K_e .
3. Build a Terminal Value for the post-year- n FCFE stream with the same Gordon-growth logic, and discount it back n years at K_e as well.
4. Adding these present values gives Equity Value directly, in one step — with no separate debt/cash bridge needed, since FCFE already reflects the cash effects of interest and net borrowing inside it.

 *Tip: Mixing up discount rates — using WACC on FCFE, or K_e on FCFF — is one of the most common mark-losing errors in this topic; match the cash flow to its own rate.*

12.6 P/E Multiple Valuation

★★ Frequently Asked

$$\text{Value per Share} = EPS \times \text{P/E Multiple (Comparable Peer)}$$

Where:

EPS the company's own earnings per share

P/E Multiple price-to-earnings ratio of a comparable listed peer, or an industry average

Build-up logic:

1. Relative valuation assumes similar companies should trade at similar multiples of a common financial metric — here, earnings.
2. Find the P/E the market currently places on a genuinely comparable company or industry average — this shows how many rupees investors pay per rupee of that peer's earnings.
3. Apply this same multiple to the company being valued: multiply its own EPS by the peer P/E to estimate what the market would similarly pay for it.
4. This is quick and market-grounded, but only as reliable as the comparability of the peer chosen — different growth, risk, or capital structure will distort the result.

 *Tip: P/E only works on genuinely comparable, profit-making companies — it breaks down for a loss-making firm with negative EPS.*

12.7 EV/EBITDA Multiple Valuation

★★ Frequently Asked

$$EV = EBITDA \times \text{EV/EBITDA Multiple (Comparable Peer)}$$
$$\text{Equity Value} = EV - \text{Debt} + \text{Cash}$$

Where:

EBITDA	earnings before interest, tax, depreciation & amortization — a pure operating cash-profit proxy
EV/EBITDA Multiple	the ratio observed for comparable peers or the industry
Debt	market value of interest-bearing debt, netted off since EV belongs to debt and equity together
Cash	surplus cash/non-operating investments, added back since EBITDA excludes any return they generate

Build-up logic:

1. EBITDA sits above both interest and depreciation, so it is neutral to capital structure and depreciation policy — allowing fairer comparison across firms with different debt levels or accounting choices than PAT/EPS would.
2. Just as with P/E, find what multiple of EBITDA the market pays for comparable firms — their EV ÷ their EBITDA.
3. Apply that multiple to the target's own EBITDA to estimate its Enterprise Value.
4. Since EV includes both debt and equity claims, bridge back to Equity Value exactly as in the DCF approach — subtract debt, add back cash.

 *Tip: Prefer EV/EBITDA over P/E when comparing firms with very different leverage or depreciation policies, since EBITDA sits above both.*

12.8 Price to Book Value (P/BV) Multiple

★★ Frequently Asked

$$\text{Value per Share} = \text{Book Value per Share} \times \text{P/BV Multiple (Comparable Peer)}$$

Where:

Book Value per Share net worth (equity share capital + reserves) ÷ number of equity shares

P/BV Multiple price-to-book ratio observed for comparable peers

Build-up logic:

1. Book value per share reflects the accounting net worth backing each share — a stable reference point especially for asset-heavy businesses like banks and NBFCs where earnings can swing but the balance sheet doesn't.
2. Identify the P/BV multiple the market assigns to comparable companies — how many times their book value their shares trade at.
3. Multiply the target's own book value per share by this peer multiple to estimate a market-consistent value per share.
4. P/BV is most meaningful when book values genuinely reflect economic worth (financial companies); it is far weaker for asset-light, intangible-heavy businesses.

12.9 PEG Ratio

★ Good to Know

$$PEG = \frac{\text{P/E Ratio}}{\text{Expected Annual EPS Growth Rate}}$$

Where:

P/E Ratio the stock's current price-to-earnings multiple

Expected Annual EPS Growth Rate projected earnings growth, used as a plain number (e.g. 15, for 15% growth)

Build-up logic:

1. A high P/E alone doesn't prove a stock is expensive — a fast-growing company deserves a higher P/E than a slow one, so P/E must be judged relative to growth, not in isolation.
2. Divide P/E by the expected earnings growth rate to see how much an investor pays, in P/E terms, for each percentage point of expected growth.
3. A PEG near 1 is generally read as fairly valued; well below 1 suggests the stock may be cheap relative to its growth, and well above 1 suggests it may be overpriced relative to growth.

 Tip: Keep the growth rate as a plain number (20, not 0.20) — PEG is conventionally expressed this way.

12.10 Net Asset Value (NAV) / Adjusted Book Value Method

★★ Frequently Asked

$$\text{Value per Share} = \frac{\text{Net Assets Available to Equity Shareholders (at Fair Value)}}{\text{No. of Equity Shares}}$$
$$\text{Net Assets Available to Equity} = \text{Fair Value of Total Assets} - \text{Fair Value of Outside Liabilities} - \text{Preference Share Capital}$$

Where:

No. of Equity Shares	number of equity shares currently outstanding
Fair Value of Total Assets	every asset restated at current realisable/replacement value, not historical book value
Fair Value of Outside Liabilities	all amounts owed to outsiders — debt, creditors, provisions — at settlement value
Preference Share Capital	preference capital, which ranks ahead of equity and is deducted before arriving at what belongs to equity

Build-up logic:

1. This method values a company the way an asset-based buyer or liquidator would: what would be left for equity shareholders if every asset were sold at fair value and every outside liability paid off?
2. Start from the fair value of all assets, not book value — historical cost less depreciation rarely reflects what land, buildings, or investments are really worth today.
3. Subtract fair value of outside liabilities, since these must be settled before shareholders get anything, and subtract preference capital too, since preference shareholders rank ahead of equity in a wind-up.
4. What remains is the net worth genuinely available to equity, at current values — divide by the number of equity shares to get a per-share figure.
5. This method ignores future earning capacity entirely, so it is normally used as a floor value or cross-check alongside earnings-based methods, not as a stand-alone value for a healthy going concern.

 *Tip: ICAI numericals that ask you to value a company "by multiple methods" almost always expect NAV as the asset-backed cross-check alongside DCF and relative multiples.*

12.11 Economic Value Added (EVA) & EVA-based Valuation

★★ Frequently Asked

$$EVA = NOPAT - (WACC \times \text{Capital Employed})$$
$$\text{Value of Firm} = \text{Capital Employed} + \text{Present Value of all Future EVAs}$$

Where:

NOPAT	net operating profit after tax = $EBIT(1-t)$
WACC	the minimum return the firm must earn to satisfy all its capital providers
Capital Employed	total long-term capital invested in the business, usually taken at book value at the start of the period
Present Value of all Future EVAs	every future year's EVA discounted back to today at WACC

Build-up logic:

1. EVA asks a sharper question than plain profit: did the firm earn more than the minimum return its capital providers demand? $WACC \times \text{Capital Employed}$ is exactly that minimum acceptable profit – the "charge" for using the capital.
2. Subtracting this capital charge from the actual after-tax operating profit, NOPAT, leaves the true economic surplus generated over and above what investors required – the EVA for one period.
3. A firm's total worth then splits into two pieces: the capital already invested (already reflected in Capital Employed) plus the extra value the firm is expected to keep creating from that capital in future – the stream of future EVAs.
4. So firm value builds up as Capital Employed (what's already there) plus the present value of every future EVA, each discounted at WACC exactly like any other cash flow stream.
5. Sanity check: if a firm is expected to earn exactly its WACC forever, every future EVA is zero, and the formula correctly collapses to $\text{Value of Firm} = \text{Capital Employed}$ – confirming that value is created only when returns exceed the cost of capital, not merely by being profitable.

 *Tip: A firm can show a healthy positive PAT and still be destroying value (negative EVA) if it isn't even covering its own cost of capital – profit and value creation are not the same thing.*

Sum-of-the-Parts (SOTP) Valuation

When a company runs several distinct businesses with very different risk-return profiles — say, a stable FMCG arm alongside a high-growth tech arm — valuing it as one blended whole with a single multiple or discount rate can badly mis-value it. Sum-of-the-Parts fixes this by valuing each segment separately, using whichever method and multiple/discount rate genuinely fits that segment's own industry and risk (a DCF for one division, a peer P/E or EV/EBITDA multiple for another), then simply adding up all the segment values. Corporate-level items that don't belong to any one segment — head-office costs, unallocated debt, cross-holdings, contingent liabilities — are adjusted centrally after summing the parts, to arrive at the overall enterprise or equity value. SOTP is especially useful for conglomerates, diversified holding companies, and businesses being considered for a demerger, since it reveals whether the market is valuing the combined entity at a discount to what its individual pieces would be worth apart — the so-called "conglomerate discount."

The maths behind how many new shares get issued in a merger, whether the deal actually creates value, and how goodwill gets recorded on the acquirer's books.

13.1 Swap Ratio (Exchange Ratio) — General Logic

★★ Frequently Asked

$$\text{Swap Ratio} = \frac{\text{Target's Relevant Value per Share}}{\text{Acquirer's Relevant Value per Share}} \times (1 + \text{Control Premium, if any})$$

Where:

Target's Relevant Value per Share	the chosen basis — EPS, market price, or book value — for the company being acquired
Acquirer's Relevant Value per Share	the same basis, computed for the acquiring company
Control Premium	extra % sometimes layered on to compensate target shareholders for giving up control, if the deal specifies one

Build-up logic:

1. In a share-swap merger, target shareholders aren't paid cash — they're paid in the acquirer's own shares, so the core question is: how many acquirer shares is one target share genuinely worth?
2. The fairest starting point is to compare both companies on one common per-share yardstick — earnings, market price, or book value — since the swap should broadly preserve the economic value each shareholder held before the deal.
3. Dividing the target's per-share value by the acquirer's per-share value on that yardstick gives exactly the number of acquirer shares carrying the same value as one target share — the swap ratio.
4. Target shareholders are often also being asked to give up control or independence, so a control premium may be layered on top of the plain ratio to make the deal acceptable to them — pushing the ratio, and the number of acquirer shares issued, a little higher.
5. This same logic mechanically repeats whichever basis is chosen next — only the specific per-share figure being compared changes.

 *Tip: The swap ratio always answers "how many acquirer shares per one target share" — double-check which company sits in the numerator; inverting it is a very common exam slip. Unless a question explicitly gives a control premium, work with the plain ratio.*

13.2 Swap Ratio — Earnings (EPS) Basis

★ **MUST KNOW**

$$\text{Swap Ratio (EPS basis)} = \frac{EPS_{\text{Target}}}{EPS_{\text{Acquirer}}}$$

Where:

EPS_{Target} earnings per share of the target company (latest or maintainable profits)

EPS_{Acquirer} earnings per share of the acquiring company

Build-up logic:

1. On the EPS basis, "relevant value per share" is simply the earnings each share generates — shareholders are compensated in proportion to the earning power of the share they're giving up.
2. Divide the target's EPS by the acquirer's EPS to find how many acquirer shares — each carrying the acquirer's own EPS — together match the earning power of one target share.
3. This is the most commonly used basis in practice and in ICAI numericals, since earnings are the most direct measure of what a share is fundamentally producing for its owner.

 *Tip: The EPS-basis ratio is purely earnings-driven — it ignores market price and asset backing entirely, so exam problems often ask you to compute all three bases and then compare or negotiate between them.*

13.3 Swap Ratio — Market Price Basis

★★ **Frequently Asked**

$$\text{Swap Ratio (Market Price basis)} = \frac{MP_{\text{Target}}}{MP_{\text{Acquirer}}}$$

Where:

MP_{Target} current market price per share of the target company

MP_{Acquirer} current market price per share of the acquiring company

Build-up logic:

1. Here "relevant value per share" is simply what the market is currently paying for each share — the most direct measure of value when both shares are actively and fairly traded.
2. Dividing the target's market price by the acquirer's market price gives the number of acquirer shares whose combined market value equals one target share's market value at prevailing prices.
3. This basis works well only when both companies are actively traded with reliable prices — it is unreliable if a stock is thinly traded, or if prices are distorted by speculation right around the announcement.

13.4 Swap Ratio — Book Value Basis

★ Good to Know

$$\text{Swap Ratio (Book Value basis)} = \frac{BV_{\text{Target}}}{BV_{\text{Acquirer}}}$$

Where:

BV_{Target} book value per share of the target = net worth ÷ number of shares

BV_{Acquirer} book value per share of the acquirer, on the same basis

Build-up logic:

1. Here "relevant value per share" is the accounting net worth backing each share, so the logic values shareholders in proportion to the balance-sheet assets standing behind their shares, not earnings or market sentiment.
2. Divide the target's book value per share by the acquirer's book value per share to get the swap ratio on this basis.
3. Book value is the least-used of the three bases in practice, since accounting values can be quite stale versus real economic worth, but it remains a standard basis in ICAI problems, especially when a reliable market price isn't available for an unlisted company.

13.5 Post-Merger EPS of Combined Entity

★★ Frequently Asked

$$\text{Combined EPS} = \frac{PAT_{\text{Acquirer}} + PAT_{\text{Target}} \pm \text{Adjustments}}{\text{No. of Acquirer's Shares} + \text{New Shares Issued to Target}}$$

Where:

PAT_{Acquirer} , PAT_{Target} profit after tax of each company, pre-merger (or as adjusted for expected synergies if given)

Adjustments any post-merger change to combined profit caused directly by the deal – extra interest if debt-funded, interest saved if debt is repaid, incremental depreciation on revalued assets, synergy gains, etc.

No. of Acquirer's Shares the acquirer's own equity shares already outstanding before the merger

New Shares Issued to Target number of target's shares $\times$ the swap ratio used

Build-up logic:

1. After the merger there is only one surviving reporting entity, so combined earnings available to its shareholders is simply the sum of both companies' profits, adjusted for anything that genuinely changes because of the deal itself.
2. The share count also changes: the acquirer's own shares remain outstanding, and new shares are issued to the target's shareholders in exchange for their now-cancelled shares, based on the agreed swap ratio.
3. Dividing the new combined profit by the new combined share count gives post-merger EPS – the number the acquirer's own shareholders watch most closely, since it shows whether the deal helped or hurt their per-share earnings.

 *Tip: Always use the NEW total share count – acquirer's original shares plus the newly issued swap shares – forgetting the new shares is a very common numerical slip.*

13.6 Post-Merger Share Price

★ Good to Know

$$\text{Post-merger Market Price} = \text{Post-merger EPS} \times \text{P/E Ratio (Combined/Acquirer)}$$

Where:

Post-merger EPS as computed above

P/E Ratio (Combined/Acquirer) the multiple the market is expected to apply to the combined entity – usually taken as the acquirer's own pre-merger P/E unless the question states otherwise

Build-up logic:

1. Once post-merger EPS is known, the market price the combined share should command follows the same basic relationship that links price, earnings, and multiple for any listed share: $\text{Price} = \text{EPS} \times \text{P/E}$.
2. The P/E applied is the one the market is expected to place on the combined, larger entity – commonly the acquirer's own P/E as the surviving listed entity, unless the problem specifies the market will re-rate it to a different multiple.
3. Comparing this new price to the acquirer's pre-merger price directly shows whether the deal has helped or hurt the acquirer's original shareholders on a market-value basis.

13.7 Synergy from Merger

★★ Frequently Asked

$$\text{Synergy} = \text{Value of Combined Firm} - (\text{Standalone Value of Acquirer} + \text{Standalone Value of Target})$$

Where:

Value of Combined Firm	value of the merged entity as one going concern after the merger, capturing any real benefits the deal is expected to create
Standalone Value of Acquirer/Target	what each company would be worth valued independently, as if the merger were never happening

Build-up logic:

1. A merger is worth pursuing, from a pure value-creation standpoint, only if the combined entity is worth more together than the two firms were worth apart — otherwise shareholders are better off with no deal at all.
2. Value each company's standalone worth first, exactly as if no merger were happening, and add them together — this is the "no synergy" benchmark.
3. Separately estimate what the combined entity is actually expected to be worth after the merger, capturing real operational or financial benefits — cost savings, cross-selling, tax benefits, higher combined debt capacity, and so on.
4. The difference between these two figures is the pure value created (or destroyed) purely by combining the two firms — the real economic justification for paying any premium to the target's shareholders.

 *Tip: If the premium paid to the target exceeds the synergy actually generated, the acquirer's own shareholders end up worse off even though the target's shareholders did well — always compare premium paid against synergy created, not just against the target's standalone value.*

13.8 Purchase Consideration — Net Assets Method

★ MUST KNOW

$$\text{Purchase Consideration} = \text{Agreed Value of Assets Taken Over} - \text{Agreed Value of Liabilities Taken Over}$$

Where:

Agreed Value of Assets Taken Over	fair/agreed values of only those assets the acquirer has actually agreed to take over — some assets may be excluded from the deal
Agreed Value of Liabilities Taken Over	fair/agreed values of only those liabilities the acquirer specifically agrees to assume — some may be left with the vendor

Build-up logic:

1. Under this method, purchase consideration is built directly from the balance sheet being acquired — think of it as what the acquirer ends up net-owning once the deal completes.
2. Total up the agreed value of every asset the acquirer is actually taking over, using the values both parties have agreed for the deal, not necessarily the vendor's book values.
3. Subtract the agreed value of every liability the acquirer specifically agrees to assume along with those assets — the acquirer is only "paying" for the net position, since it also inherits these obligations.
4. What's left is the net value being acquired, and that net amount is exactly what the acquirer must pay the vendor — in cash, shares, or a mix — as purchase consideration, since the vendor is handing over precisely this net value.

 *Tip: Include only assets/liabilities actually agreed to be taken over — if a question says a specific asset (e.g. cash) is excluded from the deal, or a liability is retained by the vendor, leave its value out entirely.*

13.9 Purchase Consideration — Net Payment Method

★ Good to Know

$$\text{Purchase Consideration} = (\text{Equity Shares Issued} \times \text{Agreed Value per Share}) + (\text{Pref. Shares Issued} \times \text{Agreed Value per Share}) + \text{Cash Paid} + \text{Value of Debentures/Other Securities Issued}$$

Where:

Equity Shares Issued	number of the acquirer's equity shares given to the vendor company/its shareholders
Agreed Value per Share	the value per share both parties have agreed to use for this issue
Pref. Shares Issued	number of preference shares issued, if any, as part of the consideration
Cash Paid	any cash component paid directly to the vendor
Value of Debentures/Other Securities Issued	value of any debentures or other instruments given as part of the deal

Build-up logic:

1. Under this method, forget the target's balance sheet entirely — instead, simply add up everything of value the acquirer actually hands over to the vendor company or its shareholders as payment.
2. This can take several forms in the same deal — equity shares at an agreed price, preference shares, cash, debentures, even deferred payments — every form given must be valued and included.
3. Summing all these individual payments together, from the payer's side, gives the purchase consideration under this method.
4. Both methods should normally arrive at the same figure when full information is available — the difference is only which side of the deal you calculate from; use whichever the question gives you enough data for.

 *Tip: Net Payment is usually easier when the question directly states the payment package; Net Assets is needed when the question instead gives the target's balance sheet with agreed/revalued figures.*

13.10 Goodwill / Capital Reserve on Acquisition

★ MUST KNOW

$$\text{Goodwill (or Capital Reserve)} = \text{Purchase Consideration} - \text{Net Assets Acquired (at Fair Value)}$$

Where:

Purchase Consideration

total amount paid, from either method above

Net Assets Acquired (at Fair Value)

fair value of assets taken over minus fair value of liabilities taken over – the same net figure as under the Net Assets Method

Build-up logic:

1. When one company buys another, it is really paying for the net assets received – so compare what was actually paid against the fair value of what was actually received to see if a premium was paid or a bargain was struck.
2. If purchase consideration is MORE than the fair value of net assets acquired, the acquirer has paid extra for things not sitting on the balance sheet – brand value, customer relationships, expected synergies – and this excess is recorded as Goodwill, an intangible asset.
3. If purchase consideration is LESS than the fair value of net assets acquired, the acquirer got a bargain – net assets worth more than what was paid – and this shortfall is recorded as Capital Reserve, credited straight to reserves.
4. This one formula tells you which case you're in purely from the sign of the result – positive means Goodwill, negative means Capital Reserve.

 *Tip: Don't confuse this with any goodwill already sitting in the target's own pre-merger books – only fair values of identifiable net assets are used here; any pre-existing goodwill on the target's books is ignored, not carried forward.*

Example: Purchase Consideration = ₹80 lakh, Net Assets Acquired (fair value) = ₹65 lakh → Goodwill = ₹15 lakh (an asset, since PC > Net Assets). If Net Assets Acquired were instead ₹95 lakh, the ₹15 lakh shortfall (₹80 – ₹95) would be Capital Reserve, not Goodwill.

EPS Accretion vs EPS Dilution for the Acquirer

A merger is EPS-accretive for the acquirer's original shareholders when the post-merger EPS of the combined entity comes out higher than the acquirer's own pre-merger EPS — each of their shares now earns more than before. It is EPS-dilutive when post-merger EPS comes out lower. As a quick rule of thumb, a share-swap deal tends to be accretive when the acquirer buys a target at a P/E lower than its own (paying a relatively cheap price for the target's earnings), and dilutive when it pays a P/E higher than its own, because a rich swap ratio then issues more new shares than the target's earnings alone can support at the acquirer's own earnings yield. Accretion or dilution is purely an EPS-mechanics outcome, though, not the same thing as value creation — a deal can be EPS-accretive in year one and still destroy shareholder value if no real synergy exists (and vice versa), so exam problems often expect both an EPS check and a separate synergy/value check on the same deal.

Leveraged Buyout (LBO)

A Leveraged Buyout is the acquisition of a company funded mostly with borrowed money and only a thin slice of the sponsor's (typically a private equity fund's) own equity — the target's own assets and future cash flows are pledged as security for, and used to service and repay, this acquisition debt. The appeal is leverage itself: since only a small amount of equity is put in, even a modest rise in the company's value translates into a much larger percentage return on that thin equity slice, and operating cash flows used to pay down acquisition debt over the holding period increase the equity portion of value even if enterprise value itself doesn't move at all. A quick way to estimate the sponsor's likely equity IRR is the entry/exit multiple method: Entry Equity Value = (Entry EV/EBITDA multiple × Entry-year EBITDA) – Entry Debt, and Exit Equity Value = (Exit EV/EBITDA multiple × Exit-year EBITDA) – Debt Remaining at Exit, giving $IRR \approx \left(\frac{\text{Exit Equity Value}}{\text{Entry Equity Value}} \right)^{1/n} - 1$, where n is the number of years held. The sponsor's return is thus driven by three levers together — EBITDA growth during the holding period, debt pay-down from operating cash flows, and any multiple expansion from exiting at a richer multiple than the entry multiple.

How investors and founders agree on what a pre-revenue startup is worth round after round, and the basic cash-survival numbers every founder tracks.

14.1 Pre-Money & Post-Money Valuation

★ **MUST KNOW**

$$\text{Post-money Valuation} = \text{Pre-money Valuation} + \text{New Investment}$$

$$\text{Investor's Ownership \%} = \frac{\text{New Investment}}{\text{Post-money Valuation}}$$

Where:

Pre-money Valuation	the agreed value of the startup immediately before the new investment comes in
New Investment	the fresh capital being invested in this funding round
Post-money Valuation	the value of the startup immediately after the new money is added
Investor's Ownership %	the % of the company the new investor receives in exchange for the investment, on a post-money basis

Build-up logic:

1. Before a funding round, the startup already has some agreed worth based on its business, team, and prospects — this is the Pre-money Valuation, negotiated between founders and investor.
2. The moment the investor's cash actually comes in, that cash itself becomes part of the company's value — so simply add the New Investment to Pre-money Valuation to get what the company is worth right after the round: Post-money Valuation.
3. The investor is buying a slice of this newly-enlarged company with their investment, so their ownership share must be measured against the POST-money value — the value that already includes their own money — not the smaller pre-money value.
4. Dividing the investment by the post-money valuation therefore gives exactly the percentage of the now-bigger company the investor is entitled to in exchange for the money put in.

 *Tip: A very common error is dividing investment by Pre-money instead of Post-money — always use Post-money in the denominator for ownership %, since the investor's own money is part of what they now own a share of.*

Example: Pre-money = ₹8 crore, New Investment = ₹2 crore → Post-money = ₹10 crore, and the investor's ownership = $2/10 = 20\%$.

14.2 Venture Capital (VC) Method

★ MUST KNOW

$$\begin{aligned}\text{Post-money Valuation} &= \frac{\text{Exit (Terminal) Value}}{\text{Expected/Required ROI}} = \frac{\text{Exit Value}}{(1 + r)^n} \\ \text{Required Ownership \%} &= \frac{\text{Investment}}{\text{Post-money Valuation}} \\ \text{Required Pre-money Valuation} &= \text{Post-money Valuation} - \text{Investment}\end{aligned}$$

Where:

Exit (Terminal) Value	the value the VC expects the startup to be worth at exit (IPO/sale), usually a projected exit-year earnings or revenue figure times a comparable multiple
Expected/Required ROI	the money-multiple the VC needs to earn to compensate for the very high risk of startup investing, equivalently $(1 + r)^n$ for a required annual return r over the holding period
n	expected number of years to exit
Investment	the amount this VC is putting into the current round
Required Ownership %	the stake the VC must hold today to be assured of its required share of the future exit value
Required Pre-money Valuation	the value implied for the startup today, before this round's money, consistent with the VC's required return

Build-up logic:

1. A VC investing in an early-stage startup faces very high risk — most startups fail — so instead of valuing the company on today's negligible or unreliable earnings, the VC works backward from what the company could realistically be worth at a future exit: the Exit/Terminal Value.
2. The VC then asks: what value would I need to place on the company TODAY so that my investment compounds, at my required rate of return, into my required share of that future exit value — found by discounting the exit value at the required return over the holding period, i.e. dividing by $(1 + r)^n$, or equivalently by the target money-multiple wanted.
3. This discounted figure represents the value of the ENTIRE company today under the VC's required-return assumption, and since the investment being made is already part of what will grow into the exit value, this figure is exactly the Post-money Valuation the VC is implicitly working with.
4. Once Post-money Valuation is known, Required Ownership % is simply the investment as a fraction of it — the same ownership logic as the basic post-money formula.
5. Required Pre-money Valuation then follows immediately as Post-money Valuation minus the investment itself — the maximum pre-money value the VC can justify offering, given its required return and view of the likely exit.

 *Tip: The "required ROI" here is usually a steep target multiple (5x-10x or more), precisely because it must compensate for the many other portfolio companies that will return little or nothing — don't mistake it for an ordinary listed-market cost of equity.*

Example: Expected exit value in 5 years = ₹100 crore; VC requires a 10× return. Post-money Valuation today = $100/10 = ₹10$ crore. If the VC invests ₹2 crore, Required Ownership % = $2/10 = 20\%$, and Required Pre-money = $10 - 2 = ₹8$ crore.

14.3 First Chicago Method

★★ Frequently Asked

$$\text{Expected Value} = (P_{\text{Success}} \times V_{\text{Success}}) + (P_{\text{Sideways}} \times V_{\text{Sideways}}) + (P_{\text{Failure}} \times V_{\text{Failure}})$$

Where:

$P_{\text{Success}}, P_{\text{Sideways}}, P_{\text{Failure}}$ probability assigned to each scenario (must add up to 100%)

$V_{\text{Success}}, V_{\text{Sideways}}, V_{\text{Failure}}$ the estimated (present) value of the startup under each of those scenarios

Build-up logic:

1. A single-point valuation forces you to pick one story about the startup's future, but its real outcomes are spread across a wide range — from a runaway success to a modest survivor to outright failure — so it makes more sense to value all three and blend them.
2. Under each scenario, separately project the startup's likely cash flows or exit outcome and value it (often with a scenario-specific DCF or multiple), then discount each scenario's value back to the present.
3. Assign a realistic probability to each of the three scenarios based on the startup's stage, sector, team, and traction — these probabilities must add up to 100%.
4. The overall value is the probability-weighted average of the three scenario values — multiply each value by its probability and add them up, exactly like computing an expected value in statistics.

Tip: First Chicago is considered more realistic than a single-scenario DCF for early-stage startups precisely because it explicitly builds in the chance of failure — always confirm the given probabilities sum to 100% before using them.

14.4 Cap Table Dilution Across Funding Rounds

★★ Frequently Asked

$$\text{Ownership \%}_{\text{After}} = \text{Ownership \%}_{\text{Before}} \times \frac{\text{Pre-money Valuation}_{\text{New Round}}}{\text{Post-money Valuation}_{\text{New Round}}}$$

Where:

Ownership % _{Before}	the existing investor's ownership % just before the new round is raised
Pre-money Valuation _{New Round}	the startup's agreed value just before the new round's fresh investment
Post-money Valuation _{New Round}	pre-money of the new round plus the new investment raised in it

Build-up logic:

1. Every time a startup raises a fresh round, it issues brand-new shares to the new investor — the total share count rises while an existing investor's own share count stays the same, so their percentage ownership must fall, even though the rupee value of their stake may still be rising.
2. The new investor's slice of the enlarged company equals its investment ÷ the new round's post-money valuation, so the fraction of the company that everyone who was already a shareholder collectively retains works out to exactly (pre-money of the new round) ÷ (post-money of the new round).
3. Since this dilution factor applies equally to every existing shareholder's slice — in the absence of anti-dilution protection or pro-rata top-up rights — multiply any one existing investor's prior ownership % by this same ratio to get their new, diluted ownership %.
4. This compounds across multiple rounds — an early investor's percentage stake gets diluted a little further at every subsequent round, which is exactly why cap tables are tracked carefully round after round.

 *Tip: A falling ownership % doesn't necessarily mean an investor has lost money — if the valuation is rising fast enough, a smaller % of a much bigger company can still be worth far more in rupee terms than a larger % of a smaller company earlier on.*

14.5 Burn Rate & Runway

★ Good to Know

$$\text{Runway (months)} = \frac{\text{Cash Balance}}{\text{Average Monthly Burn Rate}}$$

Where:

Cash Balance cash and cash equivalents currently available with the startup

Average Monthly Burn Rate average net cash the startup spends each month to keep operating (cash outflows minus cash inflows/revenue)

Build-up logic:

1. A pre-profit startup spends more cash than it earns every month — that monthly net cash outflow is the Burn Rate, and it steadily eats into the cash sitting in the bank.
2. If the startup keeps burning cash at roughly this same average pace with no further funding, dividing the cash currently on hand by the monthly burn tells you exactly how many months of cash remain.
3. This gives the Runway — the number of months the startup can survive before running out of cash — a critical planning number that tells founders exactly how urgently they need to cut burn or close their next fundraise.

 *Tip: Runway assumes the current burn rate stays constant — flag that a rising burn rate, as the team and spending scale up, shortens the runway faster than a simple straight-line calculation suggests.*

ESOP Pool Sizing and its Dilution Effect

An Employee Stock Ownership Plan (ESOP) pool is a block of shares, or options over shares, set aside specifically to be granted to employees over time, as a talent-retention and incentive tool rather than a cash cost.

Investors typically insist that this pool be created (or topped up) BEFORE their investment comes in, and that it be carved out of the pre-money valuation — which means the ESOP pool dilutes the founders' ownership, not the incoming investor's, since the investor's ownership is calculated after the pool already exists as part of the pre-money share count. A larger ESOP pool therefore directly reduces the effective pre-money value the founders are really getting, even though the headline pre-money valuation number looks unchanged — a classic point of negotiation between founders and VCs at every round. As the company raises further rounds, the pool itself gets diluted along with everyone else's holdings, so investors commonly ask for it to be topped back up — again out of the pre-money, meaning founder dilution — at each subsequent round to keep enough unallocated options available for future hiring.

Other Startup Valuation Approaches

Because most early-stage startups have little or no revenue or earnings history, several qualitative, scorecard-style approaches are used alongside the VC and First Chicago methods. The Berkus Method assigns a specific value to each of a handful of qualitative success factors — a sound idea, quality of the management team, existence of a working prototype, strategic relationships, and initial sales or roll-out — and adds them up to reach a pre-revenue valuation. The Scorecard Method starts from the average pre-money valuation of comparable, recently-funded startups in the same region and sector, then adjusts it up or down based on how the target compares on factors like team strength, market size, product, and competition. The Cost-to-Duplicate Method estimates value as simply the cost someone would incur to build an identical business from scratch — technology, product development, IP — useful as a rough floor value, though it ignores future earning potential and intangibles like brand or market position entirely. The Risk Factor Summation Method starts from an average valuation and adjusts it up or down against a checklist of specific risk categories — management risk, technology risk, funding risk, competition risk, and so on — adding for each favourable factor and subtracting for each unfavourable one.